

Research Article

Online time-optimal trajectory planning along parametric toolpaths with strict constraint satisfaction and certifiable feasibility guarantee

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ABSTRACT

As a fundamental technique in computer numerical control (CNC) machining, time-optimal trajectory planning (TOTP) is crucial for enhancing machining efficiency, geometric accuracy, and surface quality. However, it remains challenging to reliably generate smooth time-optimal trajectories online, particularly under complex constraints. This study aims to address the above longstanding challenges which limit the reliability and efficiency of online optimization-based TOTP approaches, thereby unlocking their practical advantages in real-world machining. A TOTP method along parametric toolpaths, namely TOTP-SPLP, is developed. The proposed piecewise linear objective function and sequential procedure improve time-optimality and avoid singularities at zero feedrate compared with existing linear formulations. Complex constraints are strictly satisfied at grid points over the entire toolpath without relying on static approximations, where exceedance between grid points is negligible and theoretically bounded by the grid density based on the developed analytical interpolation. To certifiably guarantee the feasibility during online TOTP, a hierarchical look-ahead windowing (HLAW) framework is established, which achieves a 100% success rate of optimization feasibility for long toolpaths. Simulation and real-world experiments demonstrate that the proposed TOTP-SPLP significantly outperforms existing online TOTP baselines in terms of time-optimality and numerical stability within limited computational cost. Specifically, TOTP-SPLP reduces terminal time by 15.6%–56.0% compared with baselines of lower computational cost, and outperforms more computationally expensive baselines by 9.5%. Comparable performance to our TOTP-SPLP is achieved by offline baselines only when consuming tenfold computational time. For a long toolpath with millions of grid points, the developed HLAW reduces the thousands of infeasibility cases encountered by existing online framework to zero. The strict satisfaction of complex constraints helps suppress vibrations and improve surface quality in real-world machining. In summary, this work fundamentally advances online TOTP by achieving offline-level time-optimality with certifiable feasibility under strict complex constraints, making reliable online optimization-based TOTP possible and practical in CNC machining.

1. Introduction

1.1. Background and related work

In computer numerical control (CNC) machining, the trajectories of cutting tools have a great impact on machining efficiency [1,2], geometric accuracy [3,4], and surface quality [5,6]. As a fundamental technique in CNC machining, time-optimal trajectory planning (TOTP) [7–9] aims at minimizing the machining time along a given geometric toolpath while satisfying kinematic and dynamic constraints, such as velocity, acceleration, and jerk. Over the past few decades,

TOTP has received growing interest in both academic and industrial communities [10]. However, it remains challenging to stably generate smooth time-optimal trajectories online, particularly when facing complex high-order constraints. In other words, the three requirements — (a) time-optimality, (b) strict satisfaction of complex constraints, and (c) reliable online planning — form an impossible trinity, where existing works typically fulfill at most two and inevitably compromise the third.

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For example, toolpaths are usually represented by G01 blocks in the workpiece coordinate system [11], while it is the axial kinematics in the tool coordinate system, corresponding to the driving axes, that directly affects the machine tool vibration [12]. In this case, a time-optimal trajectory satisfying axial constraints in another coordinate system is necessary, where the online and stable requirements further increase the difficulty of the TOTP process. In practice, axial constraints are indirectly addressed by tangential and centripetal ones in workpiece coordinates. As a result, either constraints at points of high curvature are easily violated, which is harmful to tool vibrations [13], or more conservative constraints are applied, leading to a significant reduction in machining efficiency [1]. Therefore, a TOTP approach that can handle complex constraints is urgently needed in CNC machining. Furthermore, a reliable online performance is critical for embedding TOTP into commercial CNC systems.

One of the most universally applied online TOTP methods is keypoint-based planning due to its high computational efficiency and numerical stability [13]. Keypoints are defined as points on the toolpath with locally maximal curvature, at which the feedrate is specified to reach a local minimum, i.e., tangential accelerations is zero. In keypoint-based methods, tangential velocity, acceleration, and jerk can be constrained along the entire toolpath based on one-axis kinematics [14]. Normal kinematic constraints are approximated by centripetal acceleration and jerk [15]. Some other constraints, such as the chord error in the interpolation process [16] and the contour error in the servo control process [17], can also be represented as the feedrate constraint at keypoints. After determining the feedrate at keypoints, the feedrate profiles between keypoints can be connected by S-shaped [13], sine-shaped [18], or other template trajectories to guarantee the tangential kinematic smoothness. However, these constraints can only be indirectly addressed or approximated as the so-called static constraints [7] at keypoints, while the varying geometric features and kinematic parameters of the entire toolpath are not considered. As a result, the final interpolated trajectories might significantly violate the constraints, particularly the jerk constraint.

Inspired by the G01 structure of toolpaths in CNC machining, one-step methods have been widely studied where toolpath smoothing and TOTP are conducted together. In this way, the toolpath geometries can be designed to facilitate constraint handling. A typical approach is kinematic smoothing, where acceleration and jerk are predefined within the transitional segments. Zhang et al. [19] considered the constraints on tangential and normal acceleration without relying on centripetal approximations. Wang et al. [20] further considered the constraints on tangential and normal jerk. More recently, a novel filtering-based approach was developed where the total acceleration and jerk are constrained. Tajima et al. investigated the contour error when filtering long [21] and short [1] G01 segments. Fang et al. [22] pre-discretized G01 segments to improve the feedrate flexibility, thereby improving machining efficiency. Within limited computational cost, some one-step methods can be conducted online, capable of handling acceleration and jerk constraints with simple forms. However, the acceleration and deceleration strategies are inflexible, which hinders time-optimality when facing complex toolpaths. For example, filtering-based methods require the same transition time for each acceleration and deceleration, regardless of the magnitude of the velocity change. As a result, time-optimality is compromised since the dynamic performance of the machine tool is not fully exploited.

Convex optimization-based TOTP methods, which were originally developed in the robotics domain [23,24] and subsequently introduced to CNC applications, have demonstrated great advantages in handling complex constraints. While third-order methods for robotic manipulators [24] can be directly applied to CNC machining, and some studies explored nonlinear and even nonconvex optimization [7,25], their high computational cost limits the online application. Inspired by the initial solution of [24], Erkorkmaz et al. [8] planned sub-optimal trajectories online by solving a linear programming (LP) problem and connecting

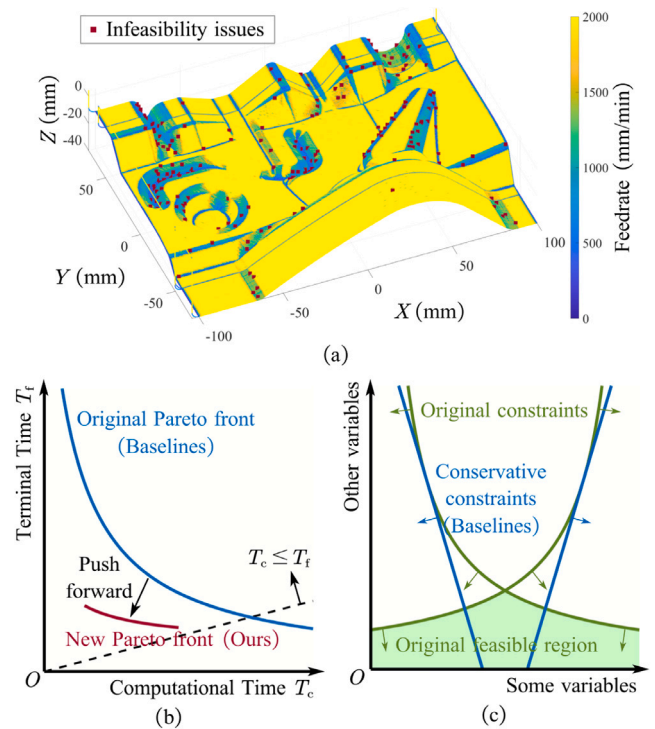


Fig. 1. Challenges of existing online optimization-based TOTP methods. (a) Example of online TOTP failure in finishing machining of a complex part. This example corresponds to SLP-PW in Section 6.3.2, where only one of ten infeasibility cases is shown. (b) Trade-off between time-optimality and computational cost. Pushing forward the Pareto front is critical for online TOTP. (c) Mechanism of artificial infeasibility in online TOTP. An improper linearization point may lead to infeasibility in an originally feasible problem.

profiles through a parallel windowing scheme. However, due to the objective linearization and conservative constraint scaling, most existing LP-based methods fail to achieve near-optimality in practice. To achieve a trade-off between time-optimality and computational cost, DiCola et al. [26] further optimized the trajectory generated by [8] through a nonlinear programming (NLP) problem, where the window in the NLP stage is much shorter than that in the LP stage. However, the feasibility of the windowing method in the LP stage is not guaranteed. In other words, infeasibility may arise when some constraints contradict others. Furthermore, numerical singularities at zero feedrate are inevitable, and feedrate oscillations remain unavoidable due to toolpath discretization. These longstanding issues and the impractical trade-off between time-optimality and computational cost have hindered the real-world application of reliable online optimization-based TOTP methods in CNC machining.

1.2. Problem statement

This paper aims to address the longstanding challenges which limit the reliability and efficiency of online optimization-based TOTP approaches mentioned in Section 1.1, thereby unlocking their practical advantages in real-world CNC machining. Notably, most optimization-based works evaluated their methods on relatively short toolpaths, typically within manufacturing time of only several minutes [7,26,27]. In contrast, the finishing machining of complex parts usually involves long toolpaths, where the machining time might be over tens of minutes or even hours. In this case, online TOTP should be sufficiently reliable and numerically stable in the entire machining process.

An example is shown in Fig. 1a, where a baseline is applied to a Mercedes-Benz mold. On the one hand, a trade-off exists between

time-optimality and computational cost, as shown in Fig. 1b. A Pareto front can be constructed based on existing optimization-based TOTP methods, and those methods with shorter computational time than terminal time are potential candidates for online application. However, as will be demonstrated in Section 6.3.1, existing online methods fail to achieve offline-level time-optimality within limited computational resources, which results in significant machining time loss for hour-level machining. This paper significantly pushes the Pareto front forward by developing a novel TOTP method, in which the novel objective function and constraint linearization jointly improve time-optimality with limited computational cost.

On the other hand, the feasibility requirement in online TOTP has been long overlooked in existing literature. As shown in Fig. 1c, artificial infeasibility may arise in an originally feasible problem due to the improper linearization point. Online frameworks divide the entire toolpath into multiple windows with overlaps to reduce computational cost and avoid full stops. However, the artificial infeasibility issue is not avoided when connecting two windows, as the linearization points at the adjacent windows are not consistent. For the example in Fig. 1a, thousands of infeasibility cases occur in a finishing machining process, while even one infeasibility case is intolerable in real-world machining. To address the above issue, this paper establishes an online framework with a novel topological structure, where the feasibility of each window is certifiably guaranteed during online TOTP.

Besides the above two challenges, this paper also addresses the numerical singularity at zero feedrate and the feedrate oscillations induced by toolpath discretization. In summary, this paper aims to develop an online TOTP framework for CNC machining with strict constraint satisfaction and certifiable feasibility guarantee.

1.3. Paper organization and contributions

The organization of this paper is as follows. Section 2 formulates the TOTP optimal control problem and introduces the overall framework of the proposed approach. A real-time analytical interpolation method is developed in Section 3, which achieves a lossless discretization of the toolpath. In Section 4, the TOTP problem is solved by a novel sequential piecewise linear programming approach. To certifiably guarantee the feasibility of online TOTP, a hierarchical look-ahead windowing framework is established in Section 5. Simulation and real-world experiments are conducted in Section 6. For completeness, a procedure for implementing the proposed method is summarized in Appendix A. The main contributions of this paper are summarized as follows:

- This paper proposes a TOTP method based on sequential piecewise linear programming (TOTP-SPLP) for CNC machining. Similar to existing optimization-based TOTP approaches [8,24], TOTP-SPLP can address a broad range of constraints in the literature, such as axial, tangential, and normal kinematic constraints. Asymmetric and parameter-varying constraints can be simultaneously handled in both the tool and workpiece coordinate systems. Unlike most keypoint-based methods [13,17], TOTP-SPLP strictly respect complex constraints at grid points along the entire toolpath without relying on static approximations, where exceedance between grid points is negligible and theoretically bounded by the grid density. TOTP-SPLP is applicable to arbitrary piecewise C^3 -continuous toolpaths, noting that any G01 blocks can be smoothed to achieve the required continuity [3].
- In TOTP-SPLP, the nonlinear objective function is novelly approximated by a piecewise linear one. In this way, near-lossless time-optimality is achieved within 90.6% shorter computational time than nonlinear optimization approaches [24]. Compared to existing linear programming (LP)-based methods [8], the developed piecewise linear structure not only achieves better time-optimality but also mitigates the risk of numerical singularities at zero feedrate. In this context, the singularity refers to the

violation of the nonnegativity of the squared feedrate, which makes the interpolation process infeasible. Experiments demonstrate the practicality of TOTP-SPLP under different constraints and computational resources. Specifically, TOTP-SPLP reduces terminal time by 15.6%–56.0% compared with baselines of lower computational cost, and still outperforms baselines by 9.5% even when they consume twice the computational time.

- To enable online TOTP for long toolpaths with high robustness and numerical stability, this paper establishes a hierarchical look-ahead windowing (HLAW) framework that certifiably guarantees the feasibility of all optimization problems throughout the online planning process. The infeasibility issue of online TOTP is formally identified and analyzed in CNC machining for the first time. Within the scope of the experiments, the developed HLAW framework achieves a 100% success rate of optimization feasibility in online planning for complex toolpaths, whereas artificial infeasibility remains a challenge in existing online optimization-based frameworks [8,26]. Owing to the consistency of linearization points among different windows, the feasibility in HLAW is guaranteed theoretically.
- An analytical interpolation is proposed where constraint errors between grid points are bounded. By enforcing constraints at multiple evaluation points within each interval, TOTP-SPLP can reduce feedrate oscillations that might degrade surface quality, which is a common issue in previous works on optimization-based TOTP. Furthermore, the strict satisfaction of other constraints in TOTP-SPLP enables vibration reduction and improved surface quality in real-world machining. In summary, this work fundamentally advances online TOTP by achieving offline-level time-optimality with certifiable feasibility under strict complex constraints, making reliable online optimization-based TOTP possible and practical in CNC machining.

2. Preliminaries

This section introduces some basic concepts and notations that will be used throughout this paper. In Section 2.1, the optimal control problem of TOTP is generally formulated, and assumptions are made for rigorous analysis. The overall framework of the proposed approach is introduced in Section 2.2. For readability, some examples of application scenarios are provided in Appendix A.2, and a toy example is provided in Appendix B.1.

2.1. Problem formulation and assumptions

Consider an n -axis machine tool with a given parametric toolpath $\mathcal{P} : \mathbf{q} = \mathbf{q}(u)$, $u \in [u_s, u_f]$, where u is the path parameter and $\mathbf{q} \in \mathbb{R}^n$ is the tool configuration. Note that u is not necessarily the arc length of the toolpath. TOTP aims to find a parameterization $u = u(t)$, $t \in [t_s, t_f]$, s.t. the trajectory $\mathbf{q}(u(t))$ satisfies some constraints and the motion duration $(t_f - t_s)$ is minimized.

Denote $\dot{\bullet}$, $\ddot{\bullet}$, $\dddot{\bullet}$ as the first, second, third derivative of a variable $\bullet$ with respect to (w.r.t.) t , respectively. Denote $\bullet'$, $\bullet''$, $\bullet'''$ as the first, second, third derivative of a variable $\bullet$ w.r.t. u , respectively. An assumption on the toolpath is as follows.

Assumption 1. The parametric toolpath $\mathcal{P}$ is C^2 - and piecewise C^3 -continuous. $\mathcal{P}$ is also regular, i.e., $\forall u \in (u_s, u_f)$, $\mathbf{q}'(u) \neq \mathbf{0}$.

Generally, the TOTP problem can be formulated as the following optimal control problem:

$$\min_{(u, t_f)} J = t_f - t_s = \int_{t_s}^{t_f} dt, \quad (1a)$$

$$\text{s.t. } \dot{u} \geq 0, \forall u \in [u_s, u_f], \quad (1b)$$

$$\dot{u} \leq v_{\alpha}(u), \forall \alpha \in \mathcal{A}, u \in (u_{\alpha 1}, u_{\alpha 2}), \quad (1c)$$

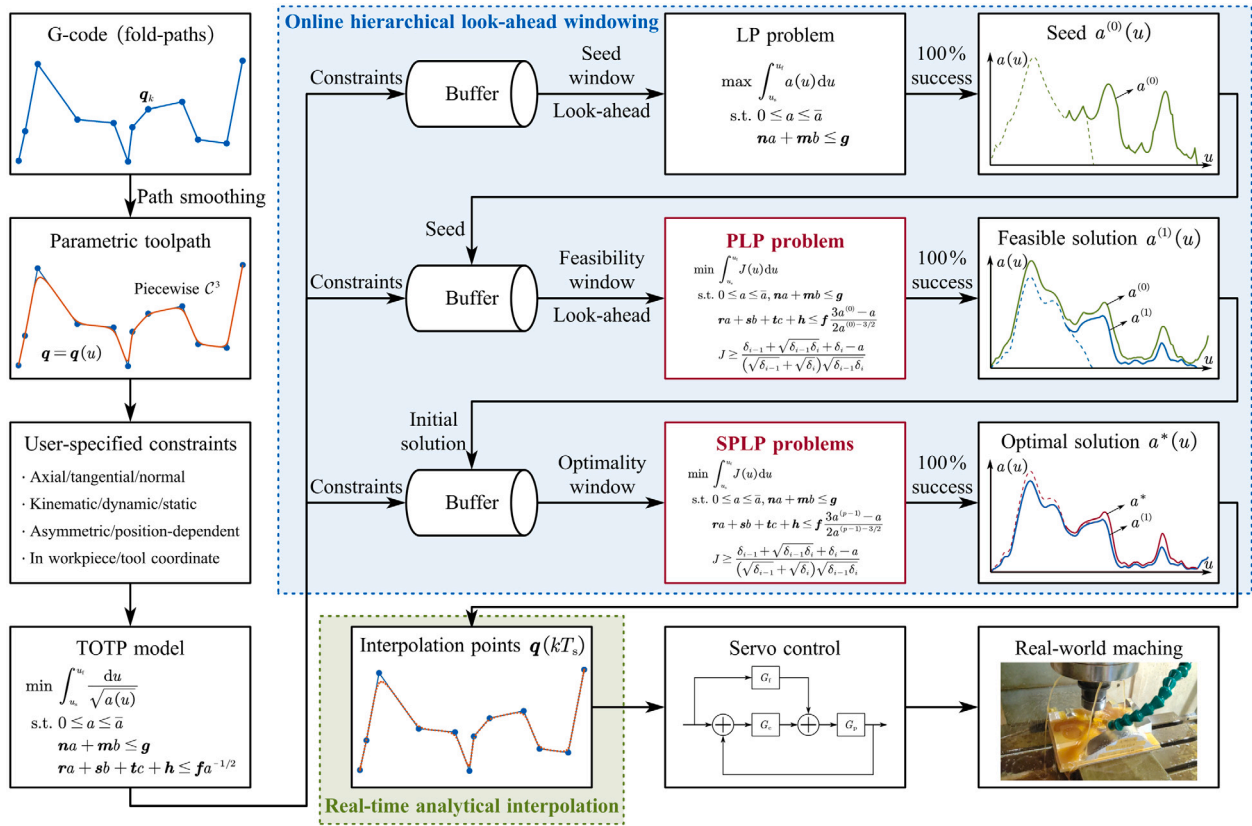


Fig. 2. The framework of online TOTP for CNC machining in this paper.

$$n_{\beta}(u)\dot{u}^2 + m_{\beta}(u)\ddot{u} \leq g_{\beta}(u), \forall \beta \in \mathcal{B}, u \in (u_{\beta 1}, u_{\beta 2}), \quad (1d)$$

$$r_{\gamma}(u)\dot{u}^3 + s_{\gamma}(u)\ddot{u} + t_{\gamma}(u)\ddot{\ddot{u}} + h_{\gamma}(u)\dot{u} \leq f_{\gamma}(u), \quad (1e)$$

$$\forall \gamma \in \mathcal{G}, u \in (u_{\gamma 1}, u_{\gamma 2}),$$

$$u(t_s) = u_s, \dot{u}(t_s) = \dot{u}_s, \ddot{u}(t_s) = \ddot{u}_s, \quad (1f)$$

$$u(t_f) = u_f, \dot{u}(t_f) = \dot{u}_f, \ddot{u}(t_f) = \ddot{u}_f. \quad (1g)$$

In problem (1), constraints (1c), (1d), and (1e) are called the 1st-, 2nd-, and 3rd-order constraints, respectively. α , β , and γ are the indexes of constraints. $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{G}$ are the finite index sets. The above constraints are dependent on the toolpath $\mathcal{P}$ and some upper bounds specified by the user. (1f) and (1g) are the initial and terminal conditions, respectively. The specific forms of the constraints are determined by the specific requirements of the manufacturing process, where some examples are provided in Appendix A.2 and Section 6.2.

Remark. In problem (1), $\dot{u}(t)$, $\ddot{u}(t)$, and $\ddot{\ddot{u}}(t)$ should be functions of time t , while constraints (1c), (1d), and (1e) are functions of the path parameter u . Along a time-optimal trajectory, q is not permitted to remain stationary over a period; hence $\dot{u}(t) > 0$ holds almost everywhere (a.e.) in $[t_s, t_f]$. In other words, $u(t)$ increases strictly monotonically in $[t_s, t_f]$. Therefore, the map $t \mapsto u$ is invertible, and the formulation of problem (1) is well-defined.

Some assumptions and discussions are given as follows:

Assumption 2. All constraints in problem (1) are bounded and C^2 -continuous w.r.t. u . In other words, $\forall \alpha \in \mathcal{A}$, $v_{\alpha}(u)$ is bounded and C^2 -continuous in $(u_{\alpha 1}, u_{\alpha 2})$. $\forall \beta \in \mathcal{B}$, $n_{\beta}(u)$, $m_{\beta}(u)$, and $g_{\beta}(u)$ are bounded and C^2 -continuous in $(u_{\beta 1}, u_{\beta 2})$. $\forall \gamma \in \mathcal{G}$, $r_{\gamma}(u)$, $s_{\gamma}(u)$, $t_{\gamma}(u)$, $h_{\gamma}(u)$, and $f_{\gamma}(u)$ are bounded and C^2 -continuous in $(u_{\gamma 1}, u_{\gamma 2})$.

Remark. Constraints in problem (1) are allowed to be piecewise continuous along the toolpath. By Assumption 2, constraints in different

processes can be formulated by different indexes α , β , and γ , where each process corresponds to a smooth segment of the constraints. Additionally, problem (1) can cover board constraints in existing literature on manufacturing, as will be shown in Appendix A.2.

Assumption 3. A near-stationary trajectory is uniformly feasible pointwise in problem (1). In other words, $\exists \delta^* > 0$ is a small positive number, s.t. $\forall u \in (u_s, u_f)$, if

$$0 \leq \dot{u} \leq \delta^*, |\ddot{u}| \leq \delta^*, |\ddot{\ddot{u}}| \leq \delta^* \quad (2)$$

hold at u , then all constraints in (1) are satisfied at u . Equivalently, it holds that $v_{\alpha}(u) > 0$, $g_{\beta}(u) > 0$, and $f_{\gamma}(u) > 0$.

Remark. Assumption 3 holds in common manufacturing scenarios. For example, when the machining process is sufficiently slow, then constraints on velocity, acceleration, jerk, and torque are satisfied along the entire toolpath.

Remark. An implicit corollary of Assumption 3 is that the representation of the toolpath $q = q(u)$ should be well-defined w.r.t. the constraints. For example, consider a toolpath of a 5-axis machine tool which passes through a singular configuration at u_1 . Through inverse kinematics, the toolpath in the tool coordinate system can be expressed as $\bar{q} = \bar{q}(u)$. If $\bar{q}$ is discontinuous at u_1 , then full stops will unavoidably occur at u_1 for any feasible trajectory with limited jerk. In this case, Assumption 3 is violated, and $q = q(u)$ is regarded as ill-defined. In fact, how to avoid or smoothly pass through singularities is the focus of the preprocessing stage before TOTP, whereas the toolpath must be well-defined as the input to the TOTP stage. Therefore, Assumption 3 is valid for the TOTP stage.

The following assumption requires the parametric jerk $\ddot{\ddot{u}}$ is bounded a.e. in $[u_s, u_f]$. In other words, the trajectory should be 3rd-order smoothness.

Assumption 4. Define the feasible set of $\ddot{u}$ at u as $\mathcal{F}(u) = \{\ddot{u} \in \mathbb{R} : \exists (\dot{u}, \ddot{u}) \text{ satisfies (1c) and (1d), s.t. (1e) holds}\}$. Then, $\mathcal{F}(u)$ is bounded in $[u_s, u_f]$ a.e. In other words, $\exists M > 0$ large enough, s.t. if $u = u(t)$ is a feasible solution of problem (1), then $|\ddot{u}| \leq M$ holds a.e. in $[u_s, u_f]$.

The goal of this paper is to solve problem (1) online with limited computational cost, while strictly satisfying all constraints along the entire toolpath.

2.2. Overall framework

The overall framework of the proposed online TOTP approach for CNC machining is shown in Fig. 2. Prior to the TOTP process, the input G-code is first smoothed into piecewise C^3 -continuous toolpaths using existing techniques like [3]. Based on the user-specified constraints, the TOTP optimal control model (1) is formulated as in Section 2.1. Assuming that a feasible solution of problem (1) is obtained, the planned trajectory can be analytically interpolated in real-time using the method developed in Section 3. Finally, the interpolation points can be sent to the low-level servo controller for execution.

The remaining work lies in efficiently solving problem (1) online. To this end, this paper develops a novel optimization-based TOTP method and a hierarchical look-ahead windowing framework to achieve reliable online TOTP, which will be presented in Sections 4 and 5, respectively. Specifically, the feasibility window operates after the seed window, followed by the optimality window. In the seed window, a second-order LP problem is solved without considering third-order constraints. Linearized at the second-order solution, the feasibility window solves the third-order problem, i.e., the PLP problem in Fig. 2, to generate a feasible initial trajectory. If the computational resource allows, the optimality window further optimizes the feasible solution to improve time-optimality, where the SPLP problems in Fig. 2 are solved sequentially.

3. Lossless discretization and real-time analytical interpolation with bounded constraint error

This section introduces how to discretize problem (1) and interpolate a planned feasible trajectory in real-time. Problem (1) is discretized in Section 3.1, and a real-time analytical interpolation approach is presented in Section 3.2. Combined with the proposed analytical interpolation, the discretization of problem (1) is lossless, i.e., an analytical smooth trajectory in continuous time domain can be obtained by the discretized numerical solution. The discretization error is analyzed in Section 3.3, where the constraints are theoretically guaranteed within a bounded error in the interpolation process.

3.1. State transformation and problem discretization

3.1.1. State transformation into the parameter domain

The developed TOTP problem (1) is inherently nonlinear and non-convex. Note that in problem (1), a bijection $t \mapsto u$ is implicitly introduced by the system dynamics $\frac{d}{dt}(u, \dot{u}, \ddot{u}) = (\dot{u}, \ddot{u}, \ddot{\ddot{u}})$. The constraints w.r.t. $(u, \dot{u}, \ddot{u}, \ddot{\ddot{u}})$ in (1c), (1d), and (1e) are dependent on the path parameter u , which is significantly complex under a nonlinear toolpath $\mathcal{P}$ and flexible user-specified constraints. Inspired by [24], the state vector $(a(u), b(u), c(u))$ is constructed as follows:

$$a(u) = \dot{u}^2(t(u)), \quad b(u) = \ddot{u}(t(u)), \quad c(u) = \frac{\ddot{\ddot{u}}(t(u))}{\ddot{u}(t(u))}. \quad (3)$$

Then, the system dynamic equation is

$$a'(u) = \frac{da}{dt} \frac{dt}{du} = 2b(u), \quad b'(u) = \frac{db}{dt} \frac{dt}{du} = c(u). \quad (4)$$

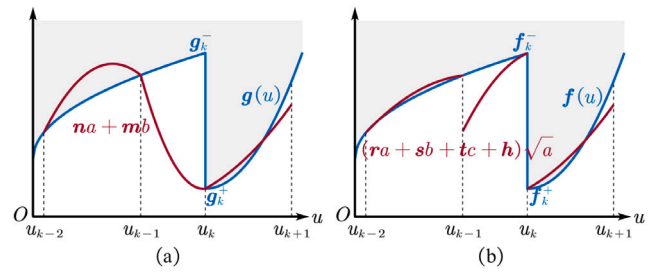


Fig. 3. Constraint discretization. (a) 2nd-order constraints. (b) 3rd-order constraints. Constraints can be discontinuous at the grid points. The constraints are strictly satisfied at the grid points, while constraint violations within intervals are bounded in Theorem 1.

The objective function (1a) can be reformulated as

$$J = \int_{u_s}^{u_f} \frac{du}{\dot{u}(t(u))} = \int_{u_s}^{u_f} \frac{du}{\sqrt{a(u)}}. \quad (5)$$

Therefore, the considered TOTP problem can be equivalently reformulated in a parameter domain as follows:

$$\min_{(a,b,c)} J = \int_{u_s}^{u_f} \frac{du}{\sqrt{a(u)}}, \quad (6a)$$

$$\text{s.t. } a'(u) = 2b(u), \quad b'(u) = c(u), \quad \forall u \in [u_s, u_f], \quad (6b)$$

$$a \geq 0, \quad \forall u \in [u_s, u_f], \quad (6c)$$

$$a \leq \bar{a}_\alpha \triangleq \bar{v}_\alpha^2, \quad \forall \alpha \in \mathcal{A}, \quad u \in (u_{\alpha 1}, u_{\alpha 2}), \quad (6d)$$

$$n_\beta a + m_\beta b \leq g_\beta, \quad \forall \beta \in \mathcal{B}, \quad u \in (u_{\beta 1}, u_{\beta 2}), \quad (6e)$$

$$r_\gamma a + s_\gamma b + t_\gamma c + h_\gamma \leq f_\gamma a^{-1/2}, \quad \forall \gamma \in \mathcal{G}, \quad u \in (u_{\gamma 1}, u_{\gamma 2}), \quad (6f)$$

$$a(u_s) = a_s \triangleq \dot{u}_s^2, \quad b(u_s) = b_s \triangleq \ddot{u}_s, \quad (6g)$$

$$a(u_f) = a_f \triangleq \dot{u}_f^2, \quad b(u_f) = b_f \triangleq \ddot{u}_f. \quad (6h)$$

Compared to problem (1), the constraints in (6) are linear except the term $a^{-1/2}$ in (6f), while the nonlinearity in problem (1) is eliminated. Furthermore, the parameter domain $[u_s, u_f]$ is fixed, which is more convenient for discretization compared to the time domain $[t_s, t_f]$ with free terminal time t_f .

Remark. In (3), the state $c(u)$ is well-defined if and only if $a(u) > 0$. If $a(u) = 0$ holds at some u , then the objective function (5) is a generalized integral. Similar conclusions hold for the 3rd-order constraints (6f). The above issues will be addressed in Sections 3.2 and 4.

3.1.2. Problem discretization in the parameter domain

Consider a discretization grid $\{u_k\}_{k=0}^N$ with $u_s = u_0 < u_1 < \dots < u_N = u_f$. To guarantee the smoothness of constraints in each interval, the grid should include boundaries of constraints' domains. In other words, $\forall \alpha \in \mathcal{A}$, $\beta \in \mathcal{B}$, and $\gamma \in \mathcal{G}$, the boundaries $u_{\alpha 1}$, $u_{\alpha 2}$, $u_{\beta 1}$, $u_{\beta 2}$, $u_{\gamma 1}$, $u_{\gamma 2} \in \{u_k\}_{k=0}^N$. Note that the grid is not required to be uniform. A toy example is provided in Appendix B.1.

The trajectory should satisfy the constraints at the grid points $u_k^\pm$, while the constraints in the interval are limited in Section 3.3. In the above notations, $u_k^\pm$ denote the right and left limits of u_k . As shown in Fig. 3, the constraints and the control $c(u)$ can be discontinuous at u_k . Denote $a_k = a(u_k)$ and $b_k = b(u_k)$. The discretized constraints in problem (6) are expressed as follows:

$$0 \leq a_k \leq \bar{a}_k, \quad \forall 0 < k < N, \quad (7a)$$

$$n_k a_k + m_k b_k \leq g_k, \quad \forall 0 < k < N, \quad (7b)$$

$$r_k^+ a_k + s_k^+ b_k + t_k^+ c(u_k^+) + h_k^+ \leq f_k^+ a_k^{-1/2}, \quad \forall 0 \leq k < N, \quad (7c)$$

$$r_k^- a_k + s_k^- b_k + t_k^- c(u_k^-) + h_k^- \leq f_k^- a_k^{-1/2}, \quad \forall 0 < k \leq N, \quad (7d)$$

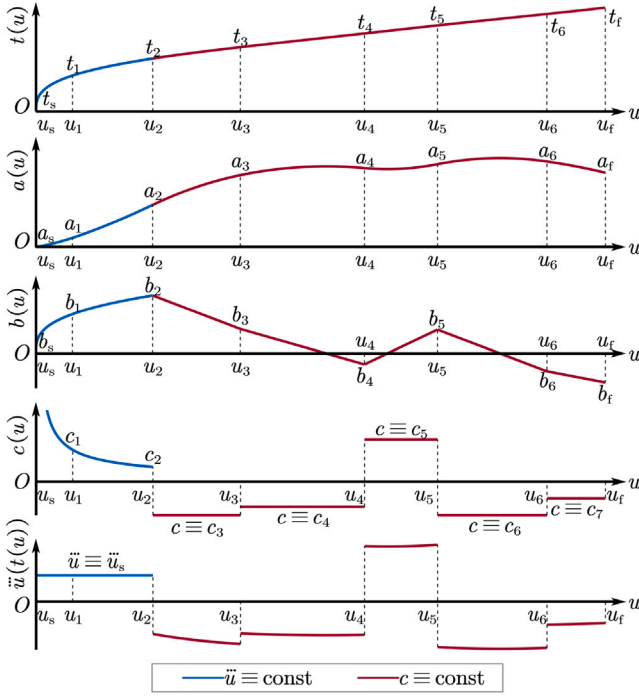


Fig. 4. The analytical interpolation of a feasible trajectory in intervals. A stationary initial condition and a non-stationary terminal condition are assumed, i.e., $a_s = b_s = 0$, $a_f \neq 0$, and $b_f \neq 0$. In this example, the width of constant parametric jerk is specified as $N_s = 2$.

$$a_0 = a_s, b_0 = b_s, a_N = a_f, b_N = b_f. \quad (7e)$$

The states a and b are smooth over the entire trajectory, while the control c and constraints can be discontinuous at the grid points. As shown in Fig. 3b, the constraints are discontinuous at u_k . For the 2nd-order constraints in Fig. 3a, the discretized constraint at u_k takes the following form:

$$\begin{bmatrix} n(u_k^-) \\ n(u_k^+) \end{bmatrix} a_k + \begin{bmatrix} m(u_k^-) \\ m(u_k^+) \end{bmatrix} b_k \leq \begin{bmatrix} g(u_k^-) \\ g(u_k^+) \end{bmatrix}, \quad (8)$$

which corresponds to (7b). Similarly, for the 3rd-order constraints in Fig. 3b, the discretized constraint at u_k should be (7c) and (7d) since $c(u)$ can also be discontinuous at u_k .

The state dynamic Eq. (6b) and the objective function (6a) will be discretized in Section 3.2.

Remark. If a toolpath and constraints of higher-order continuity are considered, then the constraint boundaries in Fig. 3, i.e., $g(u)$ and $f(u)$, are continuous. However, as will be shown in Section 3.2, the control $c(u)$ is allowed to be discontinuous at the grid points. Therefore, the discretized third-order constraints (7c) and (7d) remain necessary.

3.2. Real-time analytical interpolation

To achieve lossless discretization, it is a key step to determine the analytical interpolation of the planned trajectory in each interval. The proposed interpolation approach provides the analytical expressions of $u \mapsto t(u)$ and $t \mapsto u(t)$. Specifically, the map $u \mapsto t(u)$ is used to compute the time duration in each interval, while the inverse map $t \mapsto u(t)$ is used to interpolate the planned trajectory in the time domain. For the case where $a_k \neq 0$, in Section 3.2.1, the control $c(u)$ is stipulated to be constant in an interval. For the case where $a_k = 0$, in Section 3.2.2, $\ddot{u}$ is stipulated to be constant in an interval.

3.2.1. Zero-order hold of c at non-stationary points

For the case where $a_{k-1} \neq 0$ and $a_k \neq 0$, let $c(u) \equiv c_k$ be constant in the interval (u_{k-1}, u_k) , as shown in Fig. 4. Then, the continuous profile in the interval (u_{k-1}, u_k) is

$$a(u) = a_{k-1} + 2b_{k-1}(u - u_{k-1}) + c_k(u - u_{k-1})^2, \quad (9a)$$

$$b(u) = b_{k-1} + c_k(u - u_{k-1}), \quad (9b)$$

$$c(u) = c_k. \quad (9c)$$

Hence, the system dynamic Eq. (6b) is discretized as

$$a_k = a_{k-1} + 2b_{k-1}\Delta_k + c_k\Delta_k^2, \quad b_k = b_{k-1} + c_k\Delta_k, \quad (10)$$

where $\Delta_k = u_k - u_{k-1}$. The interpolation of the planned trajectory in the interval (u_{k-1}, u_k) can be analytically computed by the following proposition.

Proposition 1. Assume that $a_{k-1} \neq 0$ and $a_k \neq 0$. Let $c(u) \equiv c_k$ hold on (u_{k-1}, u_k) . Assume that $a(u) > 0$ holds for $u \in [u_{k-1}, u_k]$. The map $u \mapsto t(u)$ in $[u_{k-1}, u_k]$ is

$$t(u) = \Phi_k(u - u_{k-1}) - \Phi_k(0) + t_{k-1}, \quad (11)$$

where $\Phi_k(u)$ is defined as

$$\begin{cases} \frac{1}{\sqrt{c_k}} \ln \left| u + \frac{b_{k-1}}{c_k} + \sqrt{u^2 + \frac{2b_{k-1}}{c_k}u + \frac{a_{k-1}}{c_k}} \right|, & c_k > 0, \\ \frac{1}{\sqrt{-c_k}} \arcsin \frac{-c_k u - b_{k-1}}{\sqrt{b_{k-1}^2 - a_{k-1}c_k}}, & c_k < 0, \\ \frac{\sqrt{2b_{k-1}u + a_{k-1}}}{b_{k-1}}, & c_k = 0, b_{k-1} \neq 0, \\ \frac{u}{\sqrt{a_{k-1}}}, & c_k = b_{k-1} = 0. \end{cases} \quad (12)$$

Furthermore, the inverse map $t \mapsto u(t)$ in $[t_{k-1}, t_k]$ is

$$u(t) = \Phi_k^{-1}(t - t_{k-1} + \Phi_k(0)) + u_{k-1}, \quad (13)$$

where $\Phi_k^{-1}(t)$ is defined as

$$\begin{cases} -\frac{b_{k-1}}{c_k} \pm \frac{c_k^2 e^{\sqrt{c_k}t} + (b_{k-1}^2 - a_{k-1}c_k) e^{-\sqrt{c_k}t}}{2c_k^2}, & c_k > 0, \\ \frac{1}{-c_k} \left[b_{k-1} + \sqrt{b_{k-1}^2 - a_{k-1}c_k} \sin(\sqrt{-c_k}t) \right], & c_k < 0, \\ \frac{b_{k-1}^2 t^2 - a_{k-1}}{2b_{k-1}}, & c_k = 0, b_{k-1} \neq 0, \\ \sqrt{a_{k-1}}t, & c_k = b_{k-1} = 0. \end{cases} \quad (14)$$

Proof. For a given $u \in [u_{k-1}, u_k]$, (9a) implies that the time duration in the interval (u_{k-1}, u) can be expressed as $t(u) - t_{k-1} = \int_{u_{k-1}}^u \frac{dv}{\sqrt{a(v)}} = \int_0^{u-u_{k-1}} \frac{dv}{\sqrt{c_k v^2 + 2b_{k-1}v + a_{k-1}}}$. Through elementary calculus, it can be verified that Φ_k in (12) is an indefinite integral of $\frac{du}{\sqrt{c_k u^2 + 2b_{k-1}u + a_{k-1}}}$; hence, (11) holds.

For a given $t \in [t_{k-1}, t_k]$, (11) implies that $\Phi_k(u - u_{k-1}) = t - t_{k-1} + \Phi_k(0)$. Through elementary calculus, one can prove that $u = \Phi_k^{-1}(t)$ is equivalent to $t = \Phi_k^{-1}(u)$ where Φ_k^{-1} is defined in (14). Therefore, (13) holds. $\square$

Remark. The interpolation process in Proposition 1 is valid only if $a(u) > 0$ holds for $u \in [u_{k-1}, u_k]$. If $a = \dot{u}^2 < 0$ holds at some u , then the trajectory is ill-defined since the feedrate $\dot{u}$ would be imaginary. If $a(u)$ is tangent to zero at some u , then the zero-order hold of c leads to an infinite time duration in the interval, as will be further analyzed

in Section 3.2.2. The above two cases are called the *singularity at zero feedrate*, which should be avoided in TOTP.

For the case where $c_k < 0$, $b_{k-1}^2 - a_{k-1}c_k > 0$ holds since $a_{k-1} > 0$. For the case where $c_k = 0$ and $b_{k-1} \neq 0$, $2b_{k-1}u + a_{k-1} > 0$ holds since $a(u + u_{k-1})2b_{k-1}u + a_{k-1} > 0$. Therefore, the analytical interpolation Eqs. (11) and (13) are well-defined if no singularity at zero feedrate occurs along the trajectory.

Although Proposition 1 provides an analytical expression for the time duration, (11) is highly nonconvex in the optimization problem. Therefore, the objective function (5) is approximately discretized as

$$t_k - t_{k-1} = \int_{u_{k-1}}^{u_k} \frac{du}{\sqrt{a(u)}} \approx \frac{A_k}{2} \left(\frac{1}{\sqrt{a_{k-1}}} + \frac{1}{\sqrt{a_k}} \right), \quad (15)$$

which is convex w.r.t. (a_{k-1}, a_k) . Note that the interpolation process is still analytical conducted based on Proposition 1.

3.2.2. Zero-order hold of $\ddot{u}$ at stationary points

As will be guaranteed in Section 4, $a(u) > 0$ holds for $u \in (u_s, u_f)$. Therefore, this section considers the case where $a_s = 0$ or $a_f = 0$ which is specified by the boundary conditions (7e). Without loss of generality, assume that $a_s = 0$. To avoid reverse motion along the toolpath, it is required that $\dot{u} \geq 0$; hence, $b_s = 0$. If the zero-order of c is applied as in Section 3.2.1, i.e., $c(u) \equiv c_1$ in (u_s, u_1) , then (9a) implies that the time duration in (u_s, u_1) is

$$t_1 - t_s = \int_{u_s}^{u_1} \frac{du}{\sqrt{a(u)}} = \int_{u_s}^{u_1} \frac{du}{\sqrt{c_1}(u - u_s)} = \infty, \quad (16)$$

where the infinite time duration contradicts the time-optimality. Hence, the zero-order hold of c should be modified.

As shown in Fig. 4, let $\ddot{u}(t(u)) \equiv \ddot{u}_s$ be constant in (u_s, u_{N_s}) where $N_s \in \mathbb{N}^*$ is a positive integer specified by the user. Then, the interpolation of the planned trajectory in the interval (u_s, u_{N_s}) can be analytically computed by the following proposition.

Proposition 2. Assume that $a_s = b_s = 0$. Let $\ddot{u}(t(u)) \equiv \ddot{u}_s$ hold on (u_s, u_{N_s}) . The map $u \mapsto t(u)$ in $[u_s, u_{N_s}]$ is

$$t(u) = \sqrt[3]{\frac{6(u - u_s)}{\ddot{u}_s}} + t_s. \quad (17)$$

Furthermore, the inverse map $t \mapsto u(t)$ in $[t_s, t_{N_s}]$ is

$$u(t) = \frac{1}{6}\ddot{u}_s(t - t_s)^3 + u_s. \quad (18)$$

Proof. By $a_s = b_s = 0$, it holds that $\dot{u}(t_s) = \ddot{u}(t_s) = 0$. Then, $\ddot{u} \equiv \ddot{u}_s$ implies (18). Evidently, (17) holds. $\square$

Based on Proposition 2, the continuous profile in the interval (u_s, u_1) is derived as follows:

$$a(u) = \frac{3}{2}\sqrt[3]{6\ddot{u}_s^2(u - u_s)^4}, \quad (19a)$$

$$b(u) = \sqrt[3]{6\ddot{u}_s^2(u - u_s)}, \quad (19b)$$

$$c(u) = \frac{1}{3}\sqrt[3]{6\ddot{u}_s^2(u - u_s)^{-2}}. \quad (19c)$$

Therefore, the system dynamic Eq. (6b) is discretized as $\forall 1 \leq k \leq N_s$,

$$a_k = \left(\frac{u_k - u_s}{u_1 - u_s} \right)^{4/3} a_1, b_k = \frac{2a_k}{3(u_k - u_s)}, c_k = \frac{2a_k}{9(u_k - u_s)^2}. \quad (20)$$

By (17), the objective function (5) in the interval $[u_s, u_{N_s}]$ is analytically given as

$$t_{N_s} - t_s = \sqrt[3]{\frac{6(u_{N_s} - u_s)}{\ddot{u}_s}} = \frac{3(u_{N_s} - u_s)}{\sqrt{a_{N_s}}}. \quad (21)$$

Algorithm 1: Real-Time Analytical Interpolation

Input: $\{u_k\}_{k=0}^N, \{a_k\}_{k=0}^N, \{b_k\}_{k=0}^N, \{c_k\}_{k=1}^N, T_0, T_s, N_s, N_f$;

Output: Time durations $\{t_k\}_{k=0}^N$ and interpolation points

```

1:  $\{u(T_0 + iT_s)\}_{i=0}^M$ ;
2: if ( $a_0 = 0$  and  $b_0 \neq 0$ ) or ( $a_N = 0$  and  $b_N \neq 0$ ) then
3:   return an error;
4: end if
5: Let  $N_s \leftarrow 0$  if  $a_0 \neq 0$ ;
6: Let  $N_f \leftarrow 0$  if  $a_N \neq 0$ ;
7: Let  $i \leftarrow 0, t_0 \leftarrow 0$ ;
8: Compute  $\{t_k\}_{k=1}^{N_s}$  by Proposition 2;
9: while  $T_0 + iT_s \leq t_{N_s}$  do
10:   Compute  $u(T_0 + iT_s)$  by Proposition 2;
11:    $i \leftarrow i + 1$ ;
12: end while
13: for  $k \leftarrow N_s + 1, N_s + 2, \dots, N - N_f$  do
14:   Compute  $t_k$  by Proposition 1;
15:   while  $T_0 + iT_s \leq t_k$  do
16:     Compute  $u(T_0 + iT_s)$  by Proposition 1;
17:      $i \leftarrow i + 1$ ;
18:   end while
19: end for
20: Compute  $\{t_{N-k+1}\}_{k=1}^{N_f}$  by Proposition 2;
21: while  $T_0 + iT_s \leq t_N$  do
22:   Compute  $u(T_0 + iT_s)$  by Proposition 2;
23:    $i \leftarrow i + 1$ ;
24: end while
```

Different from the case in Section 3.2.1, the objective function (21) is convex w.r.t. a_{N_s} ; hence, the analytical expression is applied to both optimization and interpolation.

If a stationary terminal condition is specified, i.e., $a_f = b_f = 0$, then the zero-order hold of $\ddot{u}$ at u_f can be conducted with a width of N_f by similarly. The lossless discretization problem is provided as Eq. (22) in Box I. In this paper, let $N_s = 0$ if $a_s \neq 0$ and let $N_f = 0$ if $a_f \neq 0$.

Denote the interpolation period as T_s . The analytical interpolation of a planned trajectory is summarized in Algorithm 1. The position points can be directly obtained by $q(u(T_0 + iT_s))$.

Remark. In most existing interpolation methods for CNC machining [7,28], the planned trajectories in each interval are unknown and even not well-defined. In other words, these methods yield only the kinematic parameters $(\dot{u}, \ddot{u}, \ddot{u})$ at the grid points, without providing an analytical expression of the trajectory. Therefore, Taylor series expansion (TSE) is applied to interpolate the planned trajectory in real-time, which may introduce interpolation error. In contrast, the proposed real-time analytical interpolation approach can get rid of interpolation error compared to existing TSE-based works. Moreover, it avoids the requirement of using the arc-length parameterization which is computationally expensive.

Remark. The two stipulations in this section are interchangeable if the following conditions hold. (a) The stipulation within intervals should provide a template capable of fitting any feasible trajectories. (b) Similar to Propositions 1 and 2, the interpolation should take a closed-form to avoid TSE. (c) Similar to (10) and (20), the stipulation should provide a linear system dynamic equation in the discretization optimization problem. (d) Within each discretization interval, the constraint exceedance should be bounded, as will be proved in Theorem 1.

$$\min_{(a,b,c)} \frac{u_{N_s+1} + 5u_{N_s} - 6u_0}{\sqrt{a_{N_s}}} + \sum_{k=N_s+1}^{N-N_f-1} \frac{u_{k+1} - u_{k-1}}{\sqrt{a_k}} + \frac{6u_N - 5u_{N-N_f} - u_{N-N_f-1}}{\sqrt{a_{N-N_f}}}, \quad (22a)$$

$$\text{s.t. } a_k = \left(\frac{u_k - u_0}{u_1 - u_0} \right)^{4/3}, a_1, b_k = \frac{2a_k}{3(u_k - u_0)}, c_k = \frac{2a_k}{9(u_k - u_0)^2}, \forall 0 < k \leq N_s, \quad (22b)$$

$$a_k = a_{k-1} + 2b_{k-1}\Delta_k + c_k\Delta_k^2, b_k = b_{k-1} + c_k\Delta_k, \forall N_s < k \leq N - N_f, \quad (22c)$$

$$a_k = \left(\frac{u_k - u_N}{u_{N-1} - u_N} \right)^{4/3}, a_{N-1}, b_k = \frac{2a_k}{3(u_k - u_N)}, c_{k+1} = \frac{2a_k}{9(u_k - u_N)^2}, \forall N - N_f \leq k < N, \quad (22d)$$

$$0 \leq a_k \leq \bar{a}_k, n_k a_k + m_k b_k \leq g_k, \forall 0 < k < N, \quad (22e)$$

$$\left(r_k^+ + \frac{2s_k^+}{3(u_k - u_0)} + \frac{2t_k^+}{9(u_k - u_0)^2} \right) a_k^{3/2} + h_k^+ a_k^{1/2} \leq f_k^+, \forall 0 < k < N_s, \quad (22f)$$

$$\left(r_k^- + \frac{2s_k^-}{3(u_k - u_0)} + \frac{2t_k^-}{9(u_k - u_0)^2} \right) a_k^{3/2} + h_k^- a_k^{1/2} \leq f_k^-, \forall 0 < k \leq N_s, \quad (22g)$$

$$r_k^+ a_k + s_k^+ b_k + t_k^+ c_{k+1} + h_k^+ \leq f_k^+ a_k^{-1/2}, \forall N_s \leq k < N - N_f, \quad (22h)$$

$$r_k^- a_k + s_k^- b_k + t_k^- c_k + h_k^- \leq f_k^- a_k^{-1/2}, \forall N_s < k \leq N - N_f, \quad (22i)$$

$$\left(r_k^+ + \frac{2s_k^+}{3(u_k - u_N)} + \frac{2t_k^+}{9(u_k - u_N)^2} \right) a_k^{3/2} + h_k^+ a_k^{1/2} \leq f_k^+, \forall N - N_f \leq k < N, \quad (22j)$$

$$\left(r_k^- + \frac{2s_k^-}{3(u_k - u_N)} + \frac{2t_k^-}{9(u_k - u_N)^2} \right) a_k^{3/2} + h_k^- a_k^{1/2} \leq f_k^-, \forall N - N_f < k < N, \quad (22k)$$

$$a_0 = a_s, b_0 = b_s, a_N = a_f, b_N = b_f. (N_s = 0 \text{ if } a_s \neq 0; N_f = 0 \text{ if } a_f \neq 0.) \quad (22l)$$

Box I.

3.3. Discretization error analysis

As indicated in Section 3.1.2, the constraints in the discretized problem (6) are only enforced at the grid points, i.e., (7). However, as shown in Fig. 3, constraint exceedance may occur within the intervals between grid points. This section provides a theoretical analysis of such discretization error, guaranteeing that the constraints are satisfied within a bounded error in the interpolation process.

Lemma 1. Consider a C^2 -continuous function $f : [u_0, u_1] \mapsto \mathbb{R}$. If $f(u_0) \leq 0$ and $f(u_1) \leq 0$ hold, then $\forall u \in [u_0, u_1]$,

$$f(u) \leq \frac{1}{8} \|(-f'')^+\|_\infty (u_1 - u_0)^2, \quad (23)$$

$$\text{where } \|(-f'')^+\|_\infty \triangleq \max \{0, -\min_{u \in [u_0, u_1]} f''(u)\}.$$

Proof. f'' is bounded in the compact set $[u_0, u_1]$ due to the continuity. In other words, $\|(-f'')^+\|_\infty < \infty$ holds. Arbitrarily choose $u \in (u_0, u_1)$.

Let $g(v) = f(u)(v - u_0)(v - u_1) - f(v)(u - u_0)(u - u_1)$. Then, $g(u_0) = g(u) = g(u_1) = 0$ holds. By the Rolle's theorem, $\exists \xi \in (u_0, u_1)$, s.t. $g''(\xi) = 0$. Hence,

$$\begin{aligned} f(u) &\leq f(u) - \frac{u - u_0}{u_1 - u_0} f(u_1) - \frac{u_1 - u}{u_1 - u_0} f(u_0) \\ &= \frac{1}{2} f''(\xi) (u - u_0)(u - u_1) \\ &\leq \frac{1}{8} \|(-f'')^+\|_\infty (u_1 - u_0)^2. \end{aligned} \quad (24)$$

Therefore, (23) holds for all $u \in [u_0, u_1]$. $\square$

Theorem 1. Consider a feasible solution (a, b, c) of the lossless discretized problem (22). Applying the analytical interpolation in Algorithm 1, the

constraint violations in (1c), (1d), and (1e) do not exceed $C\Delta^2 = \mathcal{O}(\Delta^2)$. Specifically, $\Delta = \min_k \Delta_k > 0$ is the minimum width of the intervals, and $C > 0$ is a constant dependent on the constraints and the geometric path P .

Proof. According to Section 3.1.2, all constraints should be satisfied at the grid points. Without loss of generality, consider the 3rd-order constraints at (u_{k-1}, u_k) . The exceedance is

$$e(u) = \left[(ra + sb + tc + h)\sqrt{a} - f \right] \Big|_{u_0}^{u_k}. \quad (25)$$

Then, $e(u_{k-1}^+) \leq 0$ and $e(u_k^-) \leq 0$ hold. By Assumption 2, $e(u)$ is C^2 -continuous in (u_{k-1}, u_k) .

Consider the case where $c(u) \equiv c_{k-1}$ in Section 3.2.1. Then, $a(u) > 0$ holds for $u \in [u_{k-1}, u_k]$. Furthermore, Assumption 4 implies that $(a(u), b(u), c(u))$ is uniformly bounded in (u_{k-1}, u_k) . Note that $a'' = 2b'' = 2c' = 0$; hence, $e''(u)$ is uniformly bounded in (u_{k-1}, u_k) . By Lemma 1, $\exists C_k > 0$ independent of (a, b, c) , s.t. $e(u) \leq C_k \Delta_k^2 \mathbf{1}$ holds for all $u \in (u_{k-1}, u_k)$ where $\mathbf{1}$ denotes a column vector of ones with compatible dimension.

Consider the case where $a_s = b_s = 0$ and $\ddot{u} \equiv \ddot{u}_s$ in Section 3.2.2. Assumption 4 implies that $(a(u), b(u))$ is uniformly bounded in (u_{k-1}, u_k) . However, $c(u)$ is unbounded near u_s , as shown in Fig. 4. By (19), $c\sqrt{a} \equiv \ddot{u}_s$ is finite and constant in (u_s, u_{N_s}) . Therefore, $e(u)$ is C^2 -continuous in (u_s, u_{N_s}) . By Lemma 1, $\exists C_s > 0$ independent of (a, b, c) , s.t. $e(u) \leq C_s \Delta_k \mathbf{1}$ holds for all $u \in (u_{k-1}, u_k)$, $0 < k \leq N_s$. Specifically, $e(u) \leq C_s (u - u_s) \mathbf{1}$ for $u \in (u_s, u_1)$. Let $u \rightarrow u_s^+$, and it holds that $e(u) \leq 0$ near u_s . Therefore, $e(u) \leq C_s \Delta_k^2 \mathbf{1}$ holds for all $u \in (u_{k-1}, u_k)$, $0 < k \leq N_s$. The case with stationary terminal condition can be similarly proved.

A similar analysis can be conducted for the 1st-order and 2nd-order constraints. Therefore, the constraint violations do not exceed $C\Delta^2$ along the whole trajectories. $\square$

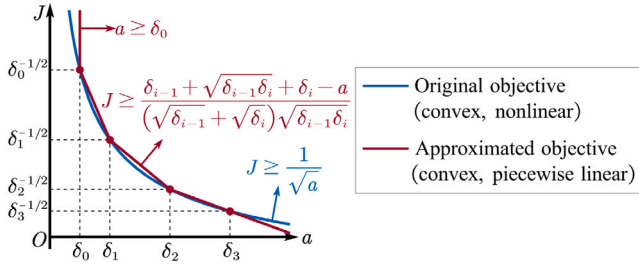


Fig. 5. Comparison of the original and approximated objective functions.

Based on Theorem 1, the discretization error is bounded by $\mathcal{O}(\Delta^2)$. Therefore, one can add more constraints in an interval to reduce the discretization error. For example, let

$$a \left(\frac{u_{k-1} + u_k}{2} \right) = a_{k-1} + b_{k-1} \Delta_k + \frac{1}{2} c_{k-1} \Delta_k^2 \leq \frac{\bar{a}_{k-1} + \bar{a}_k}{2}. \quad (26)$$

Then, the violation error of $a(u) \leq \bar{a}(u)$ can be reduced by approximately 75%. Similar inspiration can be applied to all other constraints. In this way, the feedrate oscillations are reduced. An example is shown in Section 6.3.1.

4. Time-optimal trajectory planning based on sequential piecewise linear programming

Assuming the planned trajectory is known, Section 3 provides the lossless discretized problem (22) and the real-time analytical interpolation in Algorithm 1. This section aims to solve problem (22) for a feasible trajectory with limited computational costs. Note that problem (22) is a nonconvex and high-dimensional optimization problem. To address this issue, a TOTP method based on sequential piecewise linear programming (SPLP) is proposed in Section 4.1. The feasibility of the initial solution is certifiably guaranteed in Section 4.2. Furthermore, Appendix B.2 presents a comparative analysis of the constraints and objectives between the proposed method and existing works.

4.1. TOTP based on SPLP

4.1.1. Objective function approximation

The objective function (22a) in problem (22) is convex but nonlinear. Theoretically speaking, one can apply sequential convex programming (SCP) [29] to solving problem (22) for a Karush-Kuhn-Tucker (KKT) solution [24]. However, a general SCP is computationally expensive and is not suitable for online trajectory planning, especially in CNC machining where the grid is dense due to the high accuracy requirement.

For each a_k , the original objective function takes the form $J_k = \frac{1}{\sqrt{a_k}}$. It is evident that J_k is monotonically decreasing w.r.t. a_k . Specifically, when a_k is large, J_k slightly encourages a_k to be larger. However, when a_k is small, J_k significantly penalizes a_k to be larger. Therefore, the objective function is highly sensitive to a_k when a_k is close to 0.

Inspired by the above observation, this paper approximates the objective function (22a) by a piecewise linear one to reduce the computational costs, as shown in Fig. 5. For each a_k , select sample points $0 < \delta_{k,0} < \delta_{k,1} < \dots < \delta_{k,p_k}$. The objective function J_k is approximate by the secant line passing through $(\delta_{k,i-1}, \delta_{k,i-1}^{-1/2})$ and $(\delta_{k,i}, \delta_{k,i}^{-1/2})$, i.e.,

$$J_k = \max_{1 \leq i \leq p_k} \frac{\delta_{k,i-1} + \sqrt{\delta_{k,i-1}\delta_{k,i}} + \delta_{k,i} - a_k}{(\sqrt{\delta_{k,i-1}} + \sqrt{\delta_{k,i}}) \sqrt{\delta_{k,i-1}\delta_{k,i}}}. \quad (27)$$

The approximated objective function (27) closely approximates the penalization behavior of the original formulation, as shown in Fig. 5.

In CNC machining, a full stop where $a_k = 0$ is prohibited except at the boundaries. From a numerical perspective, $a_k = 0$ leads to

an infinite time duration in the interpolation process due to (16). In practice, $a_k = 0$ might lead to numerical singularities, where an example is provided in Appendix B.2. To address this issue, an additional constraint is given as $a_k \geq \delta_{k,0}$ where $\delta_{k,0} > 0$ is a small positive constant. As shown in Fig. 5, the above approximation is equivalent to applying an infinitely large penalization to a_k as $a_k \rightarrow 0^+$. The following proposition guarantees the feasibility of the approximated problem.

Proposition 3. *The added constraints $\forall 0 < k < N$, $a_k \geq \delta_{k,0}$ do not affect the feasibility and optimality of problem (22) if $\delta_{k,0} > 0$ is small enough.*

Proof. Assume that problem (22) is feasible. Denote an optimal solution of problem (22) as (a^*, b^*, c^*) . To achieve a finitely large objective value (22a), it holds that $\forall N_s \leq k \leq N - N_f$, $a_k^* > 0$. By (22b) and (22d), it holds that $\forall 0 < k < N$, $a_k^* > 0$. Therefore, if $\forall 0 < k < N$, $0 < \delta_{k,0} < a_k^*$ holds, then (a^*, b^*, c^*) is also an optimal solution of the problem with the added constraints. Therefore, when $\delta_{k,0} > 0$ is small enough, the added constraints do not affect the feasibility and optimality of problem (22). $\square$

Remark. Some existing work approximated the objective function by a linear function. For example, in [8], the objective function $\min \int_{u_s}^{u_f} \frac{du}{\sqrt{a(u)}}$ is substituted by $\max \int_{u_s}^{u_f} a(u) du$, which can also achieve suboptimal solutions in some cases. However, when encountering sharp corners or other regions requiring deceleration, the linear approximation may suffer substantial losses in time optimality, where $a_k = 0$ will be encountered. In fact, once an a_k approaches zero, then the profile $a(u)$ might become negative in an interval since the constraint $a_k \geq 0$ is only introduced at grid points. As a result, a singularity at zero feedrate occurs. Compared to these existing works, the proposed piecewise linear approximation offers the following two advantages. On the one hand, the proposed approximation can achieve a similar penalization to the original objective function, resulting in near-optimal solutions. On the other hand, the proposed approximation can avoid the singularity at zero feedrate, which is essential due to the stable requirement of CNC machining. A comparative example is provided in Appendix B.2.

Remark. The proposed approximation (27) is highly flexible. For example, tangent approximations can be applied instead of using secant lines. The selected secant lines are allowed not to pass through the graph of $\frac{1}{\sqrt{a}}$, as long as the convexity and monotonicity of the approximated objective function are maintained. Furthermore, the number of sample points P_k can be adaptively adjusted based on the trade-off between computational cost and accuracy.

Remark. As a lower bound safeguard, the constraints $a_k \geq \delta_{0,k}$ are designed to remain inactive by choosing small enough $\delta_{0,k}$. In this way, the penalization on a becomes effective when a is close to 0. For the above case, one can also choose a smaller $\delta_{0,k}$ which has little influence on the final solution since the constraints are inactive.

4.1.2. Addressing nonconvex constraints

According to Assumption 3, $f_k^\pm > 0$ holds for all k . Therefore, in problem (22), the 3rd-order constraints (22f)–(22k) are concave, which poses challenges for the practical implementation of the developed TOTP method.

Consider the 3rd-order constraints where $\ddot{u} \equiv \text{const}$, i.e., (22f), (22g), (22j), and (22k). Note that each of these constraints only explicitly contains one variable a_k . Therefore, the feasible region of a_k can be obtained by solving some cubic equations. In other words, the 3rd-order constraints where $\ddot{u} \equiv \text{const}$ can be transformed into a upper bounds on a_k , denoted by $\bar{a}_{k,\text{jerk}}$. The transformed constraints are

$$a_k \leq \bar{a}_{k,\text{jerk}}, \quad \forall 0 < k \leq N_s \text{ or } N - N_f \leq k < N. \quad (28)$$

$$\min_{(a,b,c,J)} \left(u_{N_s+1} + 5u_{N_s} - 6u_0 \right) J_{N_s} + \sum_{k=N_s+1}^{N-N_f-1} (u_{k+1} - u_{k-1}) J_k + \left(6u_N - 5u_{N-N_f} - u_{N-N_f-1} \right) J_{N-N_f}, \quad (29a)$$

$$\text{s.t. } a_k \leq \bar{a}_{k,\text{jerk}}, \forall 0 < k \leq N_s \text{ or } N - N_f \leq k < N, \quad (29b)$$

$$a_k \geq \delta_{k,0}, \forall N_s \leq k \leq N - N_f, \quad (29c)$$

$$J_k \geq \frac{\delta_{k,i-1} + \sqrt{\delta_{k,i-1}\delta_{k,i}} + \delta_{k,i} - a_k}{(\sqrt{\delta_{k,i-1}} + \sqrt{\delta_{k,i}}) \sqrt{\delta_{k,i-1}\delta_{k,i}}}, \forall N_s \leq k \leq N - N_f, 1 \leq i \leq P_k, \quad (29d)$$

$$r_k^+ a_k + s_k^+ b_k + t_k^+ c_{k+1} + h_k^+ \leq \frac{1}{2} a_k^{(p-1)-3/2} f_k^+ \left(3a_k^{(p-1)} - a_k \right), \forall N_s \leq k < N - N_f, \quad (29e)$$

$$r_k^- a_k + s_k^- b_k + t_k^- c_k + h_k^- \leq \frac{1}{2} a_k^{(p-1)-3/2} f_k^- \left(3a_k^{(p-1)} - a_k \right), \forall N_s < k \leq N - N_f, \quad (29f)$$

$$\text{Linear constraints (22b), (22c), (22d), (22e), and (22l) hold.} \quad (29g)$$

Box II.

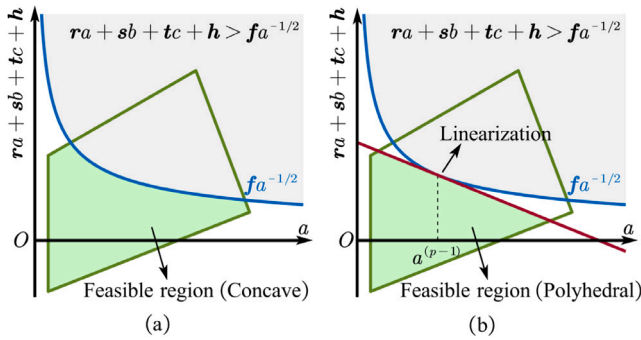


Fig. 6. The feasible region in the TOTP-SPLP process. (a) The 3rd-order constraints are not linearized; hence, the original feasible region is nonconvex. (b) The 3rd-order constraints are linearized at $a^{(p-1)}$, resulting in a polyhedral feasible region which is contained within the original one. The linearized feasible region is updated iteratively in the SPLP process.

Consider the 3rd-order constraints where $c \equiv \text{const}$, i.e., (22h) and (22i). An SCP is applied to handle these constraints in this paper. Since $f_k^{\pm} > 0$, the term $a_k^{-1/2}$ should be locally linearized, resulting in a sequential linear programming (SLP). To emphasize the piecewise linear approximation of the objective function in Section 4.1.1, the SLP is named as sequential piecewise linear programming (SPLP) in this paper.

The SPLP is iteratively conducted as follows. For the p th iteration, the problem with approximated objective function (27) and the transformed constraints is solved, resulting in $a^{(p)}$. Specifically, constraints (22h) and (22i) are linearized into (29e) and (29f), respectively. The vector $a^{(p-1)}$ is called the *linearization point* in the p th iteration. The feasible regions before and after the linearization are shown in Fig. 6. Evidently, the original feasible region is concave, and the linearized feasible region is a linear subset of the original one. The linear programming (LP) problem in the p th iteration is given in Eq. (29) in Box II. As will be introduced in Section 4.2, an initial guess $a^{(0)}$, called the TOTP seed, is not required to be feasible in the original problem (22), but it should guarantee the feasibility of the LP problem in the first iteration, i.e., $p = 1$.

The above iterative process is repeated until one of the following stopping criterion is satisfied:

$$\left\| t_f^{(p)} - t_f^{(p-1)} \right\| < \varepsilon_t \text{ or } \left\| a^{(p)} - a^{(p-1)} \right\| < \varepsilon_a \text{ or } p \geq N_{\text{iter}}, \quad (30)$$

where ε_t and ε_a are predefined tolerances and N_{iter} is the maximum number of iterations. In practice, a fully converged solution is usually not required; instead, a feasible and near-optimal solution is sufficient.

Algorithm 2: TOTP-SPLP.

Input: Information in (22), $a^{(0)}$, $\{\delta_{k,i}\}$, ε_t , ε_a , N_{iter} ;
Output: A feasible near-optimal solution (a, b, c, J) ;
1: Let $p \leftarrow 0$;
2: **repeat**
3: Let $p \leftarrow p + 1$;
4: Linearize problem (22) with $\{\delta_{k,i}\}$ and $a^{(p-1)}$ and obtain PLP problem (29);
5: Solve problem (29) for $(a^{(p)}, b^{(p)}, c^{(p)}, J^{(p)})$;
6: **until** Stopping criterion (30) is satisfied;
7: Let $(a, b, c, J) \leftarrow (a^{(p)}, b^{(p)}, c^{(p)}, J^{(p)})$;

The following theorem certifiably guarantees the feasibility and convergence of the proposed SPLP.

Theorem 2. Given a seed $a^{(0)}$, assume that problem (29) is feasible for $p = 1$. Then, the proposed SPLP guarantees the feasibility and convergence to a KKT solution of problem (22) with the approximated objective function (27) and the linearized constraints.

Proof. The proposition is proved by induction. Since the seed $a^{(0)}$ guarantees the feasibility of problem (29) for $p = 1$, a feasible initial solution $a^{(1)}$ can be obtained. Consider the p th iteration, $p > 1$. Note that $a^{(p-1)}$ serves as a feasible solution. According to Proposition 3, the added constraints $a_k \geq \delta_{k,0}$ do not affect the feasibility and optimality of problem (22) with the approximated objective function (27). Note that the linearized constraints (29e) and (29f) holds for $a^{(p-1)}$; hence, the problem is feasible in the p th iteration. Therefore, the SPLP process can be conducted with feasible guarantee. The convergence to a KKT solution is guaranteed by properties of SCP [29]. $\square$

Remark. For a general SCP problem, artificial unboundness and artificial infeasibility are the two key issues that hinder solving the problem [30]. In this paper, the artificial unboundness is avoided since the piecewise linear objective function is fixed. Conditioned by a feasible initial solution $a^{(1)}$, the artificial infeasibility is avoided by Theorem 2.

Based on Theorem 2, the proposed TOTP-SPLP method is summarized in Algorithm 2. The implementation procedure of TOTP-SPLP is provided in Appendix A.1.1. An example is shown in Fig. 7a. The seed $a^{(0)}(u)$ is infeasible in problem (22); however, problem (29) linearized

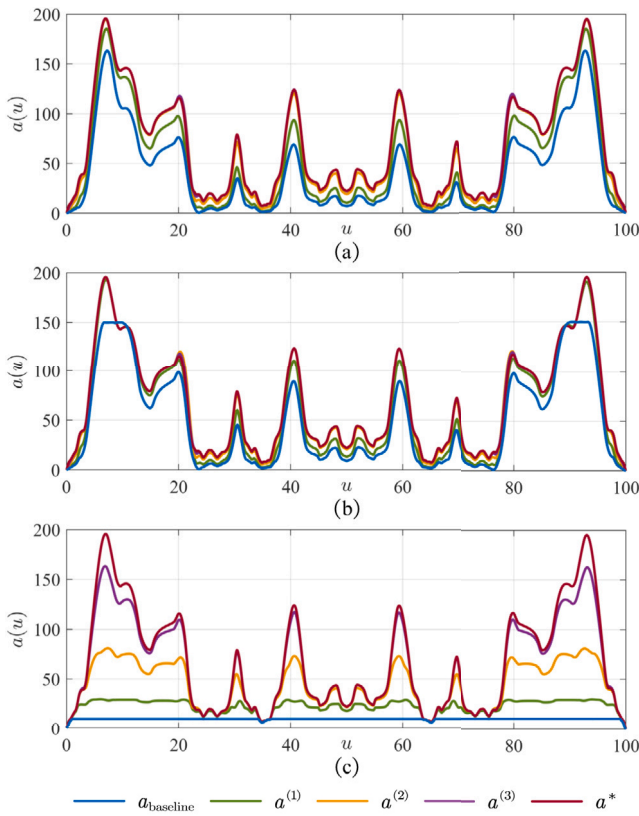


Fig. 7. An example of the TOTP-SPLP process with different seeds. (a) $a^{(0)} \equiv 250$. (b) $a^{(0)} \equiv 150$. (c) $a^{(0)} \equiv 10$. In this example, all seeds are infeasible in problem (22), even without considering 3rd-order constraints. Although the seed $a^{(0)}$ significantly affects initial iterations, the influence on the final time-optimality is relatively minor.

at $a^{(0)}(u)$ is feasible. During the SPLP process, the overall feedrate of feasible trajectories $a^{(p)}$ increases, and the motion duration $t_f - t_s$ decreases. In this example, the motion durations at the first 4 iterations are 25.159, 18.647, 16.803, and 16.466, respectively.

4.2. TOTP seed for the feasible initial solution

For a general SCP problem, it is challenging to obtain a feasible initial solution, let alone solving a near-optimal solution. According to Theorem 2, once a TOTP seed $a^{(0)}$ is obtained that guarantees the feasibility of problem (29) for $p = 1$, the developed TOTP-SPLP can plan a feasible near-optimal trajectory beginning from the initial feasible solution $a^{(1)}$.

The key issue in the proposed method is to conservatively linearize the feasible region of problem (22), as shown in Fig. 8a. If the linearized feasible region is non-empty, then the feasible region of the original problem is non-empty. Specifically, the concave constraints (22h) and (22i) should be linearized. Generally, consider a scalar component of the above concave constraint as follows:

$$r_\gamma a_{k_\gamma} + s_\gamma b_{k_\gamma} + t_\gamma c_{k_\gamma} + h_\gamma \leq f_\gamma a_{k_\gamma}^{-1/2}. \quad (31)$$

Note that $\forall a_{k_\gamma}^{(0)} > 0$, $a_{k_\gamma}^{-1/2} \geq \frac{3a_{k_\gamma}^{(0)} - a_{k_\gamma}}{2a_{k_\gamma}^{(0)3/2}}$ holds. Hence, for each constraint in (22h) and (22i), one can arbitrarily choose an $a_{k_\gamma}^{(0)} > 0$ and linearize the constraint as follows:

$$r_\gamma a_{k_\gamma} + s_\gamma b_{k_\gamma} + t_\gamma c_{k_\gamma} + h_\gamma \leq f_\gamma \frac{3a_{k_\gamma}^{(0)} - a_{k_\gamma}}{2a_{k_\gamma}^{(0)3/2}}. \quad (32)$$

Therefore, if the problem under linearized constraints (32) is feasible, then the original problem under concave constraints (31) is feasible.

Remark. In most existing work [8,31], the feasible region is linearized as shown in Fig. 8b. The concave term $a_k^{-1/2}$ in (22h) and (22i) is directly substituted by a constant $a_k^{(0)-1/2}$ where $a_k^{(0)}$ is sufficiently large. For example, $a_k^{(0)}$ is set as the optimal solution of 2nd-order problem. For these methods, the jerk constraint is called the pseudo-jerk constraint. However, the above linearization might be overly conservative, leading to an empty feasible region. The proposed method can achieve a larger feasible region, as shown in Fig. 8a. Furthermore, the selection of $a_{k_\gamma}^{(0)}$ is flexible, noting that one can choose $a_{k_\gamma}^{(0)}$ independently for each component of different constraints.

In contrast to [8,31], the proposed approach does not require $a_{k_\gamma}^{(0)}$ to form a trajectory. For example, in Fig. 7a, the seed $a^{(0)}(u)$ is infeasible. In contrast, the linearized feasible region is non-empty, leading to a feasible initial solution $a^{(1)}(u)$. The larger feasible region shown in Fig. 8a contributes to a shorter motion duration for the initial solution. Specifically, the motion durations of $a^{(1)}(u)$ and $a_{\text{baseline}}(u)$ are 25.159 and 30.976, respectively. In the context, $a^{(1)}(u)$ is the feasible initial trajectory planned by the proposed Algorithm 2, and $a_{\text{baseline}}(u)$ is the trajectory planned by the baseline method in [8,31]. The two trajectories are both solved through a single optimization process based on the same seed $a^{(0)}(u)$.

Furthermore, the proposed method is relatively robust to the choice of seeds. As shown in Fig. 7, three different seeds are applied. In all three cases, the final converged solutions a^* are similar. The differences arise in the iteration process. For example, in Fig. 7c, the seed $a^{(0)} \equiv 10$ is close to the lower bound, leading to a small initial solution. In contrast, for the baseline [8], an extra constraint $a \leq a^{(0)}$ is introduced to guarantee the conservative linearization of the pseudo-jerk constraint. Hence, the final solution is significantly affected by the seed.

A more detailed example is provided in Appendix B.2 to compare the constraints between the proposed method and existing works. The example also shows that the seed might affect the final solution in the SPLP process, which is similar to the general SCP.

Remark. Consider non-stationary boundary conditions, i.e., $a_s \neq 0$ or $a_f \neq 0$. In this case, the trajectory $a(u) \equiv 0$ is infeasible due to the non-zero boundary conditions. Despite the non-empty nonconvex feasible region, the conservative linearization might lead to infeasibility, as shown in Fig. 9a. The above issue exists in not only this paper but also existing work [8,31]. Furthermore, non-stationary boundary conditions are inevitable in online TOTP. Note that a proper seed always exists to obtain a non-empty linearized feasible region, as shown in Fig. 9b. Although it is challenging to find such a seed for a general SCP, the structure of online TOTP is utilized and a seed with feasibility guarantee is constructed in Section 5.

5. Online hierarchical look-ahead windowing

In high-speed, high-precision CNC machining, the G-code for a single part may contain millions of tool pose points, with the number of corresponding discrete grid intervals reaching up to hundreds of millions. In terms of computer memory and computational time, a one-time offline calculation is intolerable, especially under the limitations of the CNC system's computational resources. To address the above issue, an online hierarchical look-ahead windowing (HLAW) framework is proposed in Section 5.1. The feasibility of the online TOTP is certifiably guaranteed in Section 5.2, thereby ensuring robustness and reliability in CNC machining applications. Then, a comparison between hierarchical and non-hierarchical windowing is discussed in Section 5.3, with examples provided in Appendix B.3. Finally, the complexity of TOTP-SPLP and HLAW is analyzed in Section 5.4.

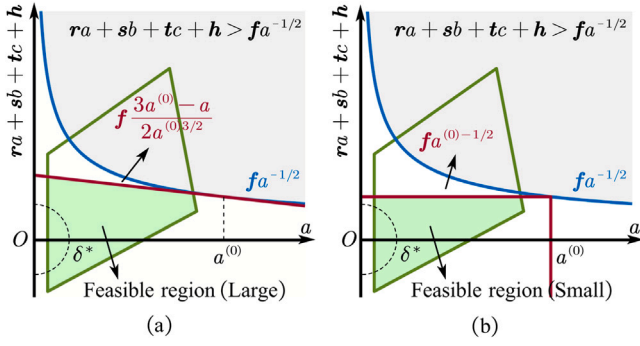


Fig. 8. Conservatively linearize the feasible region. (a) The proposed method. (b) The widely-used method in existing work [8,31]. The linear feasible set obtained by (b) is a subset of that obtained by (a).

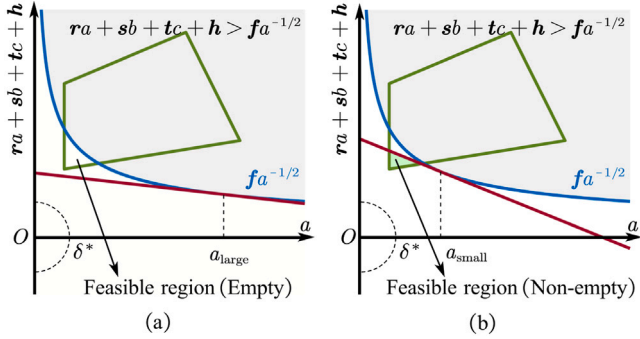


Fig. 9. Feasibility analysis with different seeds $a^{(0)}$ with non-stationary boundary conditions. (a) An improper seed leads to an empty linearized feasible region. (b) A proper seed contributes to a non-empty linearized feasible region.

5.1. HLAW framework

Assume that the long toolpath is divided into N_{win} windows, where window # j covers the interval $[u_{k_j}^1, u_{k_j}^r]$. Specifically, the following monotonicity holds:

$$0 = k_1^1 < k_2^1 < k_1^r < k_3^1 < \dots < k_{N_{\text{win}}-1}^r < k_{N_{\text{win}}}^r = N. \quad (33)$$

In other words, $(u_{k_j}^1, u_{k_j}^r) \cap (u_{k_{j+1}}^1, u_{k_{j+1}}^r) \neq \emptyset$ holds, which is called the overlap between two adjacent windows. The window $[u_{k_j}^1, u_{k_j}^r]$ moves forward as j increases.

An overview of the developed HLAW framework is shown in Fig. 10. The HLAW framework contains three windows moving forward in the long toolpath, namely the seed window, the feasibility window, and the optimality window. During the online process, the seed window always precedes the feasibility window, which in turn precedes the optimality window. In other words, it always holds that

$$k_{\text{seed}}^1 < k_{\text{seed}}^r < k_{\text{fea}}^1 < k_{\text{fea}}^r < k_{\text{opt}}^1 < k_{\text{opt}}^r, \quad (34)$$

where $[u_{k_{\text{seed}}^1}^1, u_{k_{\text{seed}}^r}^r]$, $[u_{k_{\text{fea}}^1}^1, u_{k_{\text{fea}}^r}^r]$, and $[u_{k_{\text{opt}}^1}^1, u_{k_{\text{opt}}^r}^r]$ are the seed, feasibility, and optimality windows, respectively. A frame of the HLAW framework is presented in Fig. 10c.

5.1.1. Seed window

The seed window generates TOTP seeds $a^{(0)}(u)$ for the feasible initial solution. As pointed out in Section 4.2, $a^{(0)}(u)$ is not required to form a feasible trajectory of problem (6). In fact, an arbitrarily chosen positive

$a^{(0)}(u)$ is sufficient to guarantee the feasibility of the HLAW process, as will be proved in Section 5.2.

Inspired by [8], $a^{(0)}(u)$ is chosen as the optimal solution of the 2nd-order problem in this paper. For the window $[u_{k^1}, u_{k^r}]$, assume that the historical seed windows provide the seed $\{a_k^{(0)}\}_{k \leq k_{\text{before}}^r}$ where $k^1 < k_{\text{before}}^r < k^r$. Let the trajectory in the current seed window start from the state $a_{k^1}^{(0)}$ and end at the state $a_{k^r}^{(0)} = 0$. Similar to the developed lossless discretization in Section 3.2.1, assume that $b(u) \equiv b_k$ for $u \in (u_{k-1}, u_k)$. The 2nd-order problem is given as follows:

$$\max_{(a,b)} \sum_{k=k^1+1}^{k^r-1} (u_{k+1} - u_{k-1}) a_k, \quad (35a)$$

$$\text{s.t. } a_k = a_{k-1} + 2b_{k-1} \Delta_k, \forall k^1 < k \leq k^r, \quad (35b)$$

$$0 \leq a_k \leq \bar{a}_k, \forall k^1 < k < k^r, \quad (35c)$$

$$n_k^+ a_k + m_k^+ b_{k+1} \leq g_k^+, \forall k^1 \leq k < k^r, \quad (35d)$$

$$n_k^- a_k + m_k^- b_k \leq g_k^-, \forall k^1 < k \leq k^r, \quad (35e)$$

$$a_{k^1} = a_{k^1}^{(0)}, a_{k^r} = 0. \quad (35f)$$

Specifically, constraints (35d) and (35e) are derived from the continuous constraints (6e) at $u_k^\pm$, respectively. Evidently, the 2nd-order problem (35) is an LP. Furthermore, $a^{(0)}(u)$ provides a uniform upper bound for the final 3rd-order trajectory $a^*(u)$, as shown in Fig. 8e.

Some examples are shown in Fig. 10a–c, where seeds $a^{(0)}$ planned by the historical windows and the current window are represented by dashed and solid lines, respectively.

5.1.2. Feasibility window

The feasibility window generates feasible initial solutions $(a^{(1)}, b^{(1)}, c^{(1)}, J^{(1)})$ for the TOTP-SPLP. To satisfy the requirement on the seed $a^{(0)}(u)$, the feasibility window should always remain behind the seed window, i.e., (34).

For the current window $[u_{k^1}, u_{k^r}]$, assume that the historical feasibility window provides the feasible initial solution $\{(a_k^{(1)}, b_k^{(1)})\}_{k \leq k_{\text{before}}^r}$ where $k^1 < k_{\text{before}}^r < k^r$. Then, the feasible initial solution in the current window $\{(a_k^{(1)}, b_k^{(1)})\}_{k=k^1}^{k^r}$ is solved by problem (29) with $p = 1$, where the seed $a^{(0)}$ is provided by the seed window. The boundary conditions are $a_s = a_{k^1}^{(1)}$, $b_s = b_{k^1}^{(1)}$, $a_f = 0$, and $b_f = 0$.

Some examples are shown in Fig. 10b–d, where seeds $a^{(1)}$ planned by the historical windows and the current window are represented by dashed and solid lines, respectively.

5.1.3. Optimality window

The optimality window aims to solve the SPLP problem (29) with $p \geq 2$ to obtain the near-optimal solution, denoted by (a^*, b^*, c^*, J^*) . Similar to the feasibility window, the optimality window should always remain behind the feasibility window, i.e., (34).

For the current window $[u_{k^1}, u_{k^r}]$, assume that the historical optimality window provides the near-optimal solution $\{(a_k^*, b_k^*, c_k^*, J_k^*)\}_{k \leq k_{\text{before}}^r}$ where $k^1 < k_{\text{before}}^r < k^r$. Then, problem (29) with $p \geq 2$ is iteratively solved to obtain the near-optimal solution in the current window $\{(a_k^*, b_k^*, c_k^*, J_k^*)\}_{k=k^1}^{k^r}$. The boundary conditions are $a_s = a_{k^1}^*$, $b_s = b_{k^1}^*$, $a_f = a_{k^r}^{(1)}$, and $b_f = b_{k^r}^{(1)}$.

Some examples are shown in Fig. 10c,d. Specifically, it can be observed from Fig. 10d that $a^*(u) \leq a^{(0)}(u)$ holds uniformly.

The developed online HLAW framework is summarized in Algorithm 3, and the implementation procedure of HLAW is provided in Appendix A.1.2. As will be shown in Section 6.3.2, the developed HLAW framework can be directly applied to both TOTP-SPLP and other optimization-based TOTP methods, where the feasibility can be certifiably guaranteed in Section 5.2.

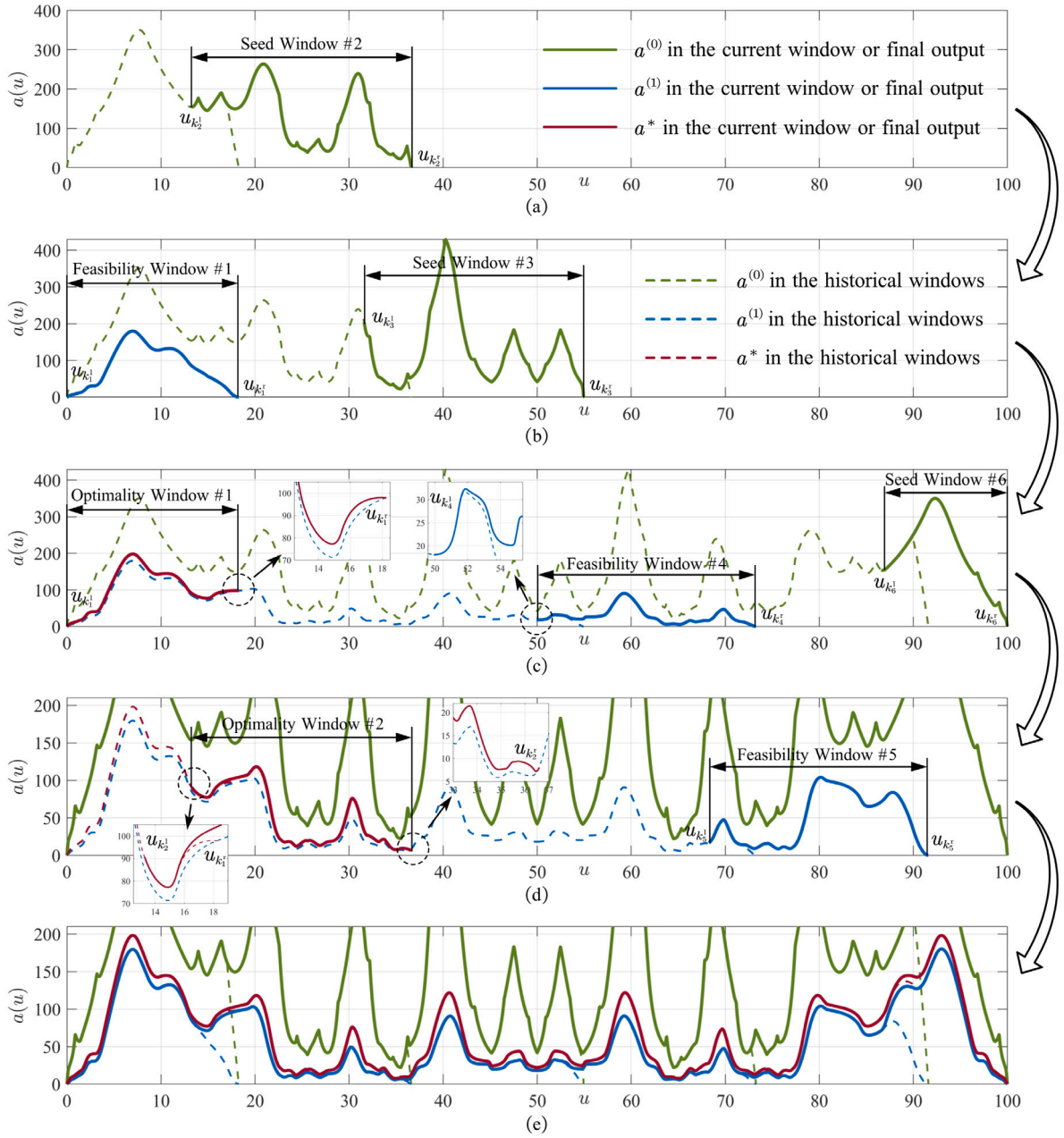


Fig. 10. An overview of the developed online hierarchical look-ahead windowing framework. (a)–(d) Different stages of the online HLAW process. (e) Final solutions of the seed, feasibility, and optimality windows. Profiles $a(u)$ planned by the historical windows and the current window are represented by dashed and solid lines, respectively.

Remark. Similar to existing windowing methods [8,26], both the window width and the overlap between adjacent windows can affect the computational efficiency and optimality of the final solution. If the width of windows is too small, then the distance is insufficient for acceleration in each window, which hinders time-optimality. Similarly, if the overlap between windows is too small, then the deceleration caused by the stationary terminal conditions in each window cannot be eliminated in the next window, which also hinders time-optimality. In contrast, if the width of windows or the overlap is too large, then the computational costs increase, which is detrimental to the online process.

As will be proved in Section 5.2, the feasibility of the HLAW framework is guaranteed regardless of the width and overlap of windows. However, for the sake of time-optimality and computational efficiency, it remains essential to tune the two parameters properly. In practice,

the parameters can be tuned once for a family of toolpaths with similar constraints and discretization. The tuning process is discussed in Appendix A.3.

5.2. Feasibility guarantee of the HLAW framework

The most significant advantage of the developed HLAW framework is the certifiable feasibility guarantee of the TOTP-SPLP problem in the whole trajectory, which is proved in the following theorem.

Theorem 3. Consider a stationary boundary condition $a_0 = b_0 = a_N = b_N = 0$ and apply the HLAW framework. Then, optimization problems in all windows, i.e., the seed, feasibility, and optimality windows, are feasible. For each optimality window, the solution converges to a KKT point of problem (29) in the corresponding interval.

Proof. Firstly, consider the feasibility of the 2nd-order problem (35) in the seed window $[u_{k_j^l}, u_{k_j^r}]$. Under Assumption 3, the stationary trajectory $(a, b, c) \equiv \mathbf{0}$ for $u \in [u_{k_{j-1}^r}, u_{k_j^r}]$ is feasible for the 3rd-order problem (22). Therefore, $(a, b) \equiv \mathbf{0}$ for $u \in [u_{k_{j-1}^r}, u_{k_j^r}]$ is feasible for the 2nd-order problem (35) since the feasible region of problem (22) is a subset of that of problem (35). Note that in the previous seed window, the terminal condition is $a_{k_{j-1}^r}^{(0)} = 0$, where $(a^{(0)}(u), b^{(0)}(u))$ is the seed generated by historical seed windows. Hence, the trajectory

$$(a, b)|_u = \begin{cases} (a^{(0)}, b^{(0)})|_u, & \text{if } u \in (u_{k_j^l}, u_{k_{j-1}^r}), \\ (0, 0), & \text{if } u \in (u_{k_{j-1}^r}, u_{k_j^r}), \end{cases} \quad (36)$$

is feasible for problem (35), where k_0^r is conventionally defined as 0. In other words, the 2nd-order problem (35) is feasible in each seed window $[u_{k_j^l}, u_{k_j^r}]$.

Secondly, consider the feasibility of the linearized 3rd-order problem (29) with $p = 1$ in the feasibility window $[u_{k_j^l}, u_{k_j^r}]$. Denote the trajectory planned by historical feasibility windows as $(a^{(1)}(u), b^{(1)}(u), c^{(1)}(u))$. Evidently, $(a^{(1)}, b^{(1)}, c^{(1)})$ satisfies the constraints linearized at $a^{(0)}$ in problem (29). Note that the terminal condition of the previous feasibility window is $a_{k_{j-1}^r}^{(1)} = b_{k_{j-1}^r}^{(1)} = 0$. By Assumption 3, the trajectory

$$(a, b, c)|_u = \begin{cases} (a^{(1)}, b^{(1)}, c^{(1)})|_u, & \text{if } u \in (u_{k_j^l}, u_{k_{j-1}^r}), \\ (0, 0, 0), & \text{if } u \in (u_{k_{j-1}^r}, u_{k_j^r}), \end{cases} \quad (37)$$

is feasible for problem (29) linearized at $a^{(0)}$ with $p = 1$. In other words, the linearized 3rd-order problem (29) with $p = 1$ is feasible in each feasibility window $[u_{k_j^l}, u_{k_j^r}]$.

Finally, the feasibility of the 3rd-order problem (29) with $p \geq 2$ in the optimality window $[u_{k_j^l}, u_{k_j^r}]$ is guaranteed by Theorem 2.

Therefore, the feasibility of the HLAW framework in each window is proved. Furthermore, Theorem 2 guarantees that the solution converges to a KKT point in each optimality window. $\square$

Remark. Theorem 3 requires zero boundary velocities and accelerations of the entire trajectory. Among existing literature [32], it remains challenging to solve the problem with non-stationary boundary conditions even in a single window, let alone considering an online process with multiple windows. Compared to existing works, the conditions of feasibility are generalized, i.e., the initial states of the first window and the terminal states of the last window are all zero.

Note that the stationary boundary conditions serve as sufficient but not necessary conditions for the feasibility of the HLAW framework. In practice, some problems under non-stationary boundary conditions can also be handled by the developed HLAW framework, especially when the optimization problems in the first and last feasibility windows are feasible. For extreme boundary conditions, one can employ the incremental linearization method proposed in [32] to carefully handle the first and last feasibility windows.

5.3. Comparison with non-hierarchical windowing

This section compares the developed HLAW framework with two non-hierarchical windowing methods, i.e., look-ahead windowing (LAW) and parallel windowing (PW) [8]. The fundamental procedures of the three frameworks are illustrated in Fig. 11. The dependency

Algorithm 3: Online HLAW framework.

Input: Information in (29), windows $\{(k_j^l, k_j^r)\}_{j=1}^{N_{\text{win}}}$, T_s ;

Output: Near-optimal trajectory (a^*, b^*, c^*) , interpolation points $\{u(T_0 + iT_s)\}_{i=0}^M$;

```

1: Initialize  $a^{(0)} \leftarrow [a_s, \mathbf{0}_{1 \times (N-2)}, a_f]^T$ ,  $a^* \leftarrow [a_s, \mathbf{0}_{1 \times (N-2)}, a_f]^T$ , and
    $b^* \leftarrow [b_s, \mathbf{0}_{1 \times (N-2)}, b_f]^T$ ;
2: for  $j \leftarrow 1, 2, \dots, N_{\text{win}} + 4$  do
3:   // Seed Window
4:   if  $j \leq N_{\text{win}}$  then
5:     Solve problem (35) in  $[u_{k_j^l}, u_{k_j^r}]$  to obtain  $\hat{a}^{(0)}$  with  $a_{k_j^l} \leftarrow a_{k_j^l}^{(0)}$ 
       and  $a_{k_j^r} \leftarrow a_{k_j^r}^{(0)}$ ;
6:     Update  $a_k^{(0)} \leftarrow \hat{a}_k^{(0)}$  for  $k \leftarrow k_j^l, k_j^l + 1, \dots, k_j^r$ ;
7:   end if
8:   // Feasibility Window
9:   if  $1 \leq j - 2 \leq N_{\text{win}}$  then
10:    Linearize problem (29) in  $[u_{k_{j-2}^l}, u_{k_{j-2}^r}]$  at  $a^{(0)}$ ;
11:    Solve problem (29) to obtain  $(\hat{a}^{(1)}, \hat{b}^{(1)})$  with  $a_{k_{j-2}^l} \leftarrow a_{k_{j-2}^l}^*$ ,
        $b_{k_{j-2}^l} \leftarrow b_{k_{j-2}^l}^*$ ,  $a_{k_{j-2}^r} \leftarrow a_{k_{j-2}^r}^*$ , and  $b_{k_{j-2}^r} \leftarrow b_{k_{j-2}^r}^*$ ;
12:    Update  $a_k^* \leftarrow \hat{a}_k^{(1)}$ ,  $b_k^* \leftarrow \hat{b}_k^{(1)}$  for  $k \leftarrow k_{j-2}^l, k_{j-2}^l + 1, \dots, k_{j-2}^r$ ;
13:   end if
14:   // Optimality Window
15:   if  $j \geq 5$  then
16:     Set the initial solution as  $(a_k^*)_{k=k_{j-4}^l}^{k_{j-4}^r}$ ;
17:     Set the boundary conditions as  $a_{k_{j-4}^l} \leftarrow a_{k_{j-4}^l}^*$ ,  $b_{k_{j-4}^l} \leftarrow b_{k_{j-4}^l}^*$ ,
        $a_{k_{j-4}^r} \leftarrow a_{k_{j-4}^r}^*$ , and  $b_{k_{j-4}^r} \leftarrow b_{k_{j-4}^r}^*$ ;
18:     Apply Algorithm 2 to solve problem (29) in  $[u_{k_{j-4}^l}, u_{k_{j-4}^r}]$  and
       obtain  $(\hat{a}^*, \hat{b}^*)$ ;
19:     Update  $a_k^* \leftarrow \hat{a}_k^*$ ,  $b_k^* \leftarrow \hat{b}_k^*$  for  $k \leftarrow k_{j-4}^l, k_{j-4}^l + 1, \dots, k_{j-4}^r$ ;
20:   end if
21:   // Interpolation
22:   if  $6 \leq j \leq N_{\text{win}} + 3$  then
23:     Apply Algorithm 1 to  $(a^*, b^*)$  in  $[u_{k_{j-5}^l}, u_{k_{j-5}^r}]$ ;
24:   else if  $j = N_{\text{win}} + 4$  then
25:     Apply Algorithm 1 to  $(a^*, b^*)$  in  $[u_{k_{j-5}^l}, u_{k_{j-5}^r}]$ ;
26:   end if
27: end for

```

relations indicated by the arrows between windows determine the order of solving optimization problems. Adjacent windows overlap in the u domain.

As shown in Fig. 11a, the window dependencies between different levels in the developed HLAW framework are unidirectional. For example, the solution of seed windows is independent to that of feasibility and optimality windows. Moreover, a feasibility window can be solved only when it is fully covered in the u domain by the solved seed windows. Similar analysis can be applied between feasibility and optimality windows. Therefore, the HLAW framework solves optimization problems *level-by-level*.

In contrast, the two non-hierarchical framework solve optimization problems *window-by-window*, as shown in Fig. 11b,c. Taking the LAW framework in Fig. 11b as an example, windows $\#(i+1)$ at different levels depends on the optimality window $\#i$. Specifically, the $(i+1)$ th seed, feasibility, and optimality windows share the boundary conditions derived from the solution of optimality window $\#i$.

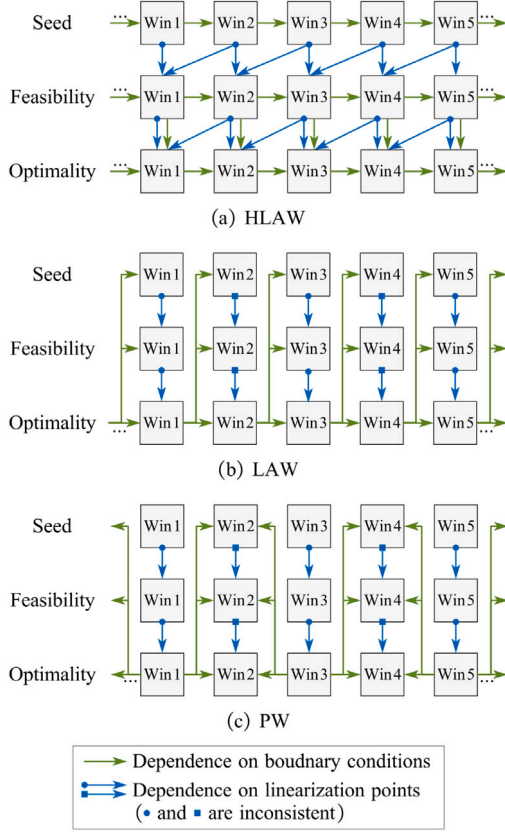


Fig. 11. Comparison of different online windowing frameworks. (a) HLAW. (b) LAW. (c) PW [8]. An arrow from window $\mathcal{W}_i$ to $\mathcal{W}_j$ means that the optimization problem in $\mathcal{W}_j$ depends on the solution of $\mathcal{W}_i$. A green arrow denotes that the optimization problem in $\mathcal{W}_j$ uses the solution in $\mathcal{W}_i$ as boundary conditions. A blue arrow means that the nonconvex constraints in $\mathcal{W}_j$ are linearized at the solution of $\mathcal{W}_i$.

Although the LAW framework appears intuitively reasonable, it inherently suffers from infeasibility issues caused by the *inconsistency* of linearization points among different windows. Denote the solution in the i th seed, feasibility, and optimality windows as $a_i^{(0)}(u)$, $(a_i^{(1)}(u), b_i^{(1)}(u))$, and $(a_i^*(u), b_i^*(u))$, $u \in [u_{k_i}^l, u_{k_i}^r]$, respectively. For window $\#i$, the boundary conditions are given as $a_s = a_{i-1}^*(u_{k_i}^l)$, $b_s = b_{i-1}^*(u_{k_i}^l)$, $a_f = 0$, and $b_f = 0$ for all levels. Although $(a_i^*(u), b_i^*(u))$ is feasible in $[u_{k_{i+1}}^l, u_{k_i}^r]$, it might be infeasible when considering the constraints linearized at $a_i^{(0)}(u)$. Note that the feasibility of $(a_i^*(u), b_i^*(u))$ w.r.t. the nonconvex constraints (29e) and (29f) is equivalent to that w.r.t. the constraints linearized at $a_i^*(u)$ itself, while $a_i^*(u) \neq a_{i+1}^{(0)}(u)$ in $[u_{k_{i+1}}^l, u_{k_i}^r]$. As a result, the feasibility guarantee established in Theorem 3 no longer holds for either LAW or PW.

Two examples regarding the infeasibility issue of LAW and PW are provided in Appendix B.3. The examples show that the infeasibility issue of non-hierarchical windowing frameworks arises from inconsistency of linearization points among different windows, which might lead to a contradiction between the conservative jerk constraint and the requirement of rapid deceleration. In contrast, the developed HLAW framework eliminates such inconsistency by solving optimization problems level-by-level.

Remark. Some existing works [8] did not introduce the optimality window, i.e., the initial solution is not further optimized. Such feasible solutions can also be applied in real-world applications, though they

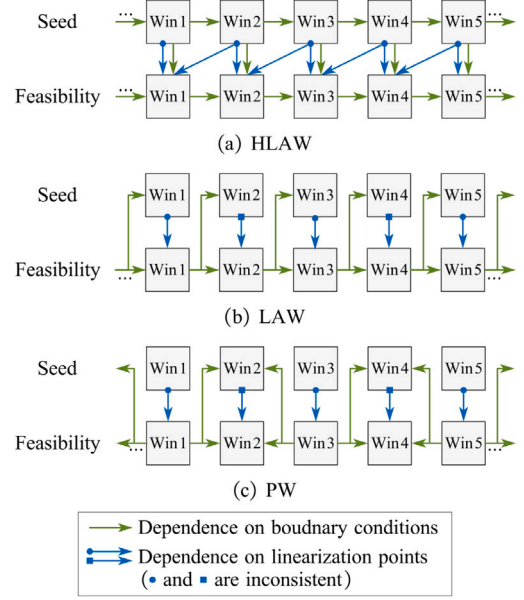


Fig. 12. Comparison of different simplified online windowing frameworks without optimality windows. (a) HLAW. (b) LAW. (c) PW [8]. Notations are the same as those in Fig. 11.

require longer manufacturing time. The original PW framework in [8] is provided in Fig. 12c. Similarly, the HLAW and LAW frameworks without optimality windows are provided in Fig. 12a,b. However, the infeasibility issue of non-hierarchical windowing frameworks still exists without optimality windows. An example is provided in Appendix B.3.1.

Some recent works [26] utilized the LAW framework without optimality windows in Fig. 12b to generate an initial solution, which is further optimized with NLP processes. However, the infeasibility issue of the LAW framework still exists due to the LAW structure.

5.4. Complexity analysis

Consider the complexity of the TOTP-SPLP method in a single window. For an LP with n variables and L constraints, the randomized simplex method [33] has an expected time complexity of $\mathcal{O}(n^3 L)$. For a window with N_{win} grid intervals, the number of variables is $\mathcal{O}(N_{\text{win}})$, and the number of constraints is also typically $\mathcal{O}(\beta_L N_{\text{win}})$, where β_L denotes the average number of constraints per grid interval. Therefore, solving the LP in each iteration of the SPLP process has an expected time complexity of $\mathcal{O}(\beta_L N_{\text{win}}^4)$.

Assume that the width of the overlap between adjacent windows is $N_{\text{ov}} < \frac{1}{2} N_{\text{win}}$, and the number of intervals in the long toolpath is $N \gg N_{\text{win}}$. Then, the number of windows is $\mathcal{O}\left(\frac{N}{N_{\text{win}} - N_{\text{ov}}}\right) = \mathcal{O}\left(\frac{N}{N_{\text{win}}}\right)$.

In each seed, feasibility, and optimality window, the expected time complexity is $\mathcal{O}(\beta_L N_{\text{win}}^4)$, $\mathcal{O}(\beta_L N_{\text{win}}^4)$, and $\mathcal{O}(K \beta_L N_{\text{win}}^4)$, respectively, where K is the number of iterations of the SPLP process in each optimality window. Therefore, the expected time complexity of the HLAW framework is $\mathcal{O}(K \beta_L N_{\text{win}}^3 N)$. If the long toolpath is directly solved by the TOTP-SPLP method in a single window, then the complexity is $\mathcal{O}(K \beta_L N^4) \gg \mathcal{O}(K \beta_L N_{\text{win}}^3 N)$, which demonstrates the necessity of online windowing.

Furthermore, the time complexity for each discretization interval is $\mathcal{O}(K \beta_L N_{\text{win}}^3)$, which is independent of the number of intervals in the long toolpath. Therefore, one can tune the iteration number K in the optimality window and the width of the window N_{win} to achieve

a desired computational efficiency for the online process. If computational resources are limited, then a simpler problem with a smaller β_L , i.e., fewer constraints, or a rougher grid with a smaller N_{win} can be chosen to improve computational efficiency.

Remark. The improvement of the objective function and the iterative procedure of the SPLP process increase the computational cost of the TOTP task compared to existing LP-based TOTP methods [8]. In [8], only one iteration, i.e., $K = 1$, is performed, which is fewer than that of the TOTP-SPLP method. Furthermore, TOTP-SPLP introduces auxiliary variables J and linear constraints (29d), which also increase β_L .

Nevertheless, TOTP-SPLP can utilize computational resources more efficiently than the baselines. As shown in the experiment of Section 6.3.1, TOTP-SPLP achieves better time-optimality within fewer iterations than some baselines, which results in shorter computational time. These results demonstrate the practicality of TOTP-SPLP despite the increased complexity per iteration.

6. Experiments

6.1. Experimental setup

6.1.1. Baselines

The baselines of TOTP methods are as follows:

- **TOTP-SK** [13]: A TOTP method that connects keypoints with S-shaped curves using look-ahead methods. The keypoints of a toolpath are defined as the points of maximum curvature.
- **TOTP-FIR** [1]: A one-step global smoothing and interpolation method based on two low-pass finite impulse response (FIR) filters in series. Pre-discretization in [22] is applied to the input G-code for higher efficiency.
- **TOTP-SQP** [24]: A TOTP method based on sequential quadratic programming (SQP) whose objective is to minimize $\int_{u_s}^{u_f} \frac{1}{\sqrt{a(u)}} du$.
- **TOTP-LP** [8]: An LP-based TOTP method whose objective is to maximize $\int_{u_s}^{u_f} a(u) du$.
- **TOTP-SLP**: To verify the performance of the approximated objective function in Section 4.1.1, let TOTP-SLP conduct the similar procedure of the proposed TOTP-SPLP. During the iterative process, the objective function is the same as that in TOTP-LP. The idea of TOTP-SLP is similar to the two-stage method in [26].
- **TOTP-SLP***: Based on TOTP-SLP, the iteration is further conducted until the computational time exceeds that of TOTP-SPLP. TOTP-SLP* allows for a fair comparison of time-optimality with TOTP-SPLP while conservatively eliminating the influence of computational time.
- **TOTP-SPLP**: The proposed method in Algorithm 2. Let $\delta_{k,l} = 10^{l-4} a_k^{(0)}$ for $l = 0, 1, 2, 3$ in (27), where $a_k^{(0)}$ is the solution generated by the seed window. The iteration steps are the same as those in TOTP-SLP.
- **TOTP-SPLP¹**: To verify the effectiveness of TOTP-SPLP when the computational resources are limited, let TOTP-SPLP¹ be the feasible initial solution $a^{(1)}$. Other settings are the same as those in TOTP-SPLP. Note that TOTP-SPLP¹ represents the configuration of our method with the least computational effort.

For the five optimization-based methods, i.e., TOTP-SQP, TOTP-LP, TOTP-SLP, TOTP-SLP*, and TOTP-SPLP, the seed $a^{(0)}$ is generated by solving problem (35), which is consistent with existing works [8,26,32]. Specifically, the initial solution $a^{(1)}$ of TOTP-SLP is the same as the final solution of TOTP-LP, where the computational time of solving TOTP-LP optimization problems is included in that of TOTP-SLP.

The proposed interpolation method, i.e., Algorithm 1, is applied to the five optimization-based methods. The interpolation methods used in TOTP-SK and TOTP-FIR are adopted as described in the original

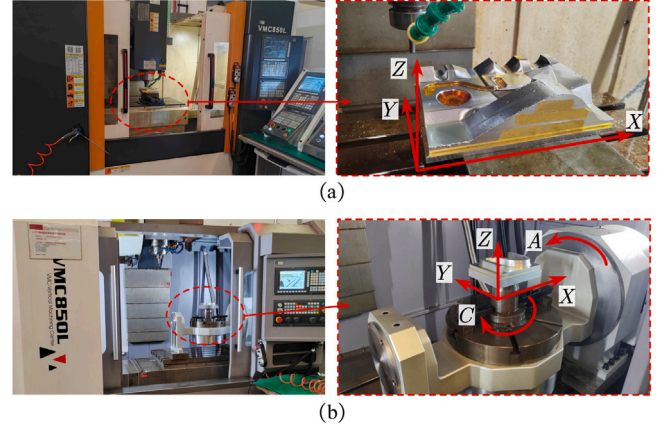


Fig. 13. Machine tools used in experiments and the corresponding coordinate systems. (a) The 3-axis machine tool. (b) The 5-axis machine tool.

references. For G01 inputs, the toolpaths are smoothed by quintic B-splines [3] except for the one-step method TOTP-FIR.

For long toolpaths, the baseline online planning frameworks are as follows:

- **PW** [8]: In the parallel windowing (PW) method, two adjacent windows intersect at an endpoint. Connecting trajectories are planned to avoid full stops at the endpoints. The procedure is shown in Fig. 11c.
- **LAW**: The non-hierarchical look-ahead windowing (LAW) method is similar to the developed HLAW framework, while the three windows are combined into a single window. The procedure is shown in Fig. 11b. The width of each window is 1000 and the overlap between adjacent windows is 100.
- **HLAW**: The proposed online hierarchical look-ahead windowing (HLAW) framework in Algorithm 3. The procedure is shown in Fig. 11a. The parameters are the same as those in LAW.

6.1.2. Metrics

- **The computational time T_c** : All algorithms are implemented in MATLAB 2024b on a computer equipped with an Intel® Core™ Ultra 9 Processor 285K. Optimization problems are solved by Gurobi [34]. The computational time includes the time for discretizing the toolpath, generating constraints, solving optimization problems and interpolation for each TOTP methods.
- **The terminal time T_f** : The terminal time is defined as the motion duration with stationary boundary conditions.
- **The exceedance ratio R_v** [14]: Consider v as a performance metric, such as the feedrate, the axial acceleration, the jerk, and so on. The upper and lower bounds are v^+ and v^- , respectively. Then, the maximal exceedance rate is defined as follows:

$$E_v(t) = \frac{\max \{v(t) - v^+, 0, v^- - v(t)\}}{v^+ - v^-}, \quad (38a)$$

$$R_v = \max_{0 \leq t \leq T_f} E_v(t). \quad (38b)$$

Then, $R_v = 0$ if and only if $v^- \leq v(t) \leq v^+$ holds a.e. The larger R_v is, the greater the violation of the constraint $v^- \leq v \leq v^+$ is.

- **The exceedance duration ratio D_v** : Apply the symbols in R_v . Then, D_v is defined as the ratio of the duration when $v(t)$ exceeds the bounds $v^\pm$ by more than 1% of the range $v^+ - v^-$, i.e.,

$$D_v = \frac{1}{T_f} \int_0^{T_f} \mathbb{I}(E_v(t) > 0.01) dt, \quad (39)$$

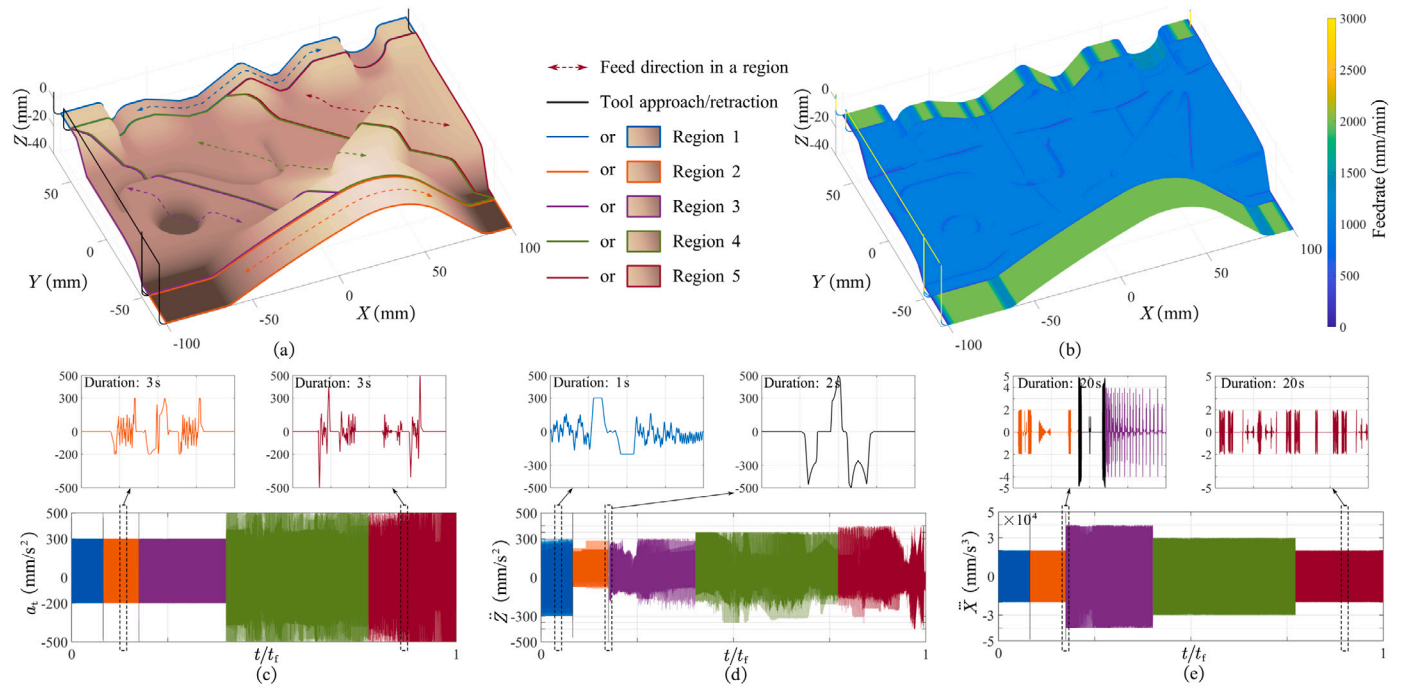


Fig. 14. Results of the 3-axis point milling experiment with asymmetric and parameter-varying constraints. (a) The toolpaths and the divided regions. (b) The distribution of the planned feedrate. (c) Tangential acceleration. (d) Z-axis acceleration. (e) X-axis jerk. All constraints are satisfied in this experiment.

where $\mathbb{I}(\cdot)$ is the indicator function. Evidently, $D_v = 0$ if and only if $v^- \leq v(t) \leq v^+$ holds a.e. The larger D_v is, the longer the duration of violating the constraint $v^- \leq v \leq v^+$ is.

In this section, the control frequency is set to 1 kHz. In other words, the duration between two interpolation points is 1 ms. The metrics R_v and D_v are calculated based on the interpolation points. Note that the information between two interpolation points is not sent to the servo controller in practice; hence, the metrics based on the interpolation points are sufficient to evaluate a TOTP method's ability to satisfy the constraints.

6.1.3. Machine tools and cutters

The machine tools and the corresponding coordinate systems are shown in Fig. 13. The 3-axis experiments are conducted on a VMC850L machining center equipped with an R4 ball-end milling cutter. For 5-axis experiments, a VMC850L machining center equipped with a 5° tapered R1.5 ball-end milling cutter is used.

6.2. Experiments for validating constraint flexibility

In this section, two simulation experiments are conducted to validate the constraint flexibility of the proposed method. In Section 6.2.1, a 3-axis point milling simulation is conducted with asymmetric and parameter-varying constraints, where the shape of profiles is discussed. Then, a 5-axis flank milling simulation is conducted with constraints in different coordinate systems in Section 6.2.2. Owing to such flexibility, the proposed method, as well as some existing optimization-based methods, can serve as full-process modules for feedrate planning in CNC systems.

6.2.1. Asymmetric and parameter-varying constraints

Consider a Mercedes-Benz mold for point milling in Fig. 13a. The feed directions of the input G01 zigzag toolpaths are shown in Fig. 14a. The upper bound of contour error is set as 1 μm in the toolpath smoothing process. The constraints are set as follows:

- Feedrate v_t in (41a): In tool approach or retraction, $v_t \leq 3000$ mm/min. In Regions 1 and 2, $v_t \leq 2000$ mm/min. In other regions, $v_t \leq 1000$ mm/min.
- Tangential acceleration a_t in (41b): In Regions 1, 2, and 3, $-200 \text{ mm/s}^2 \leq a_t \leq 300 \text{ mm/s}^2$. In other regions, $|a_t| \leq 500 \text{ mm/s}^2$.
- Axial acceleration $\ddot{q}$ in (40b): In tool approach or retraction, $\|\ddot{q}\|_\infty \leq 500 \text{ mm/s}^2$. In Regions 4 and 5, $\|\ddot{q}\|_\infty \leq 350$ and 400 mm/s^2 , respectively. In other regions, $\|\ddot{q}\|_\infty \leq 300 \text{ mm/s}^2$.
- Axial jerk $\ddot{\dot{q}}$ in (40c): In tool approach or retraction, $\|\ddot{\dot{q}}\|_\infty \leq 50,000 \text{ mm/s}^3$. In Regions 3 and 4, $\|\ddot{\dot{q}}\|_\infty \leq 40,000$ and $30,000 \text{ mm/s}^3$, respectively. In other regions, $\|\ddot{\dot{q}}\|_\infty \leq 20,000 \text{ mm/s}^3$.

Representative results are shown in Fig. 14b–e, with a motion time of $t_f \approx 1.266 \times 10^4$ s. It can be verified that all specified constraints, which are asymmetric and parameter-varying, are satisfied. The profile planned by the proposed TOTP-SPLP is fundamentally different from the traditional S-shaped curve which is widely adopted in CNC systems. Specifically, the S-shaped curve exhibits piecewise-linear tangential acceleration and cannot handle axial constraints. In contrast, the tangential acceleration planned by TOTP-SPLP exhibits a more complex profile due to axial constraints, as shown in Fig. 14c. For example, TOTP-SPLP can maintain a constant axial acceleration over a period, as shown in Fig. 14d, which is a behavior unattainable by S-shaped curves. In summary, the proposed TOTP-SPLP, along with other optimization-based methods, can plan more complex trajectories to satisfy different types of constraints compared to non-optimization-based methods like TOTP-SK.

6.2.2. Constraints in different coordinate systems

Consider an S-shaped sample for 5-axis flank milling in Fig. 13b. The upper bounds of contour error in position and orientation are set as 1 μm and 1 mdeg, respectively. Denote $\bullet^w$ and $\bullet^t$ as the variable in the workpiece and tool coordinate systems, respectively. The feedrate constraints are set as $\|(\dot{x}^w, \dot{y}^w, \dot{z}^w)\|_2 \leq 1000 \text{ mm/min}$ and $\|(\dot{i}^w, \dot{j}^w, \dot{k}^w)\|_2 \leq 10 \text{ rad/min}$. To validate the constraint flexibility of TOTP-SPLP, the following constraints in different coordinate systems are set:

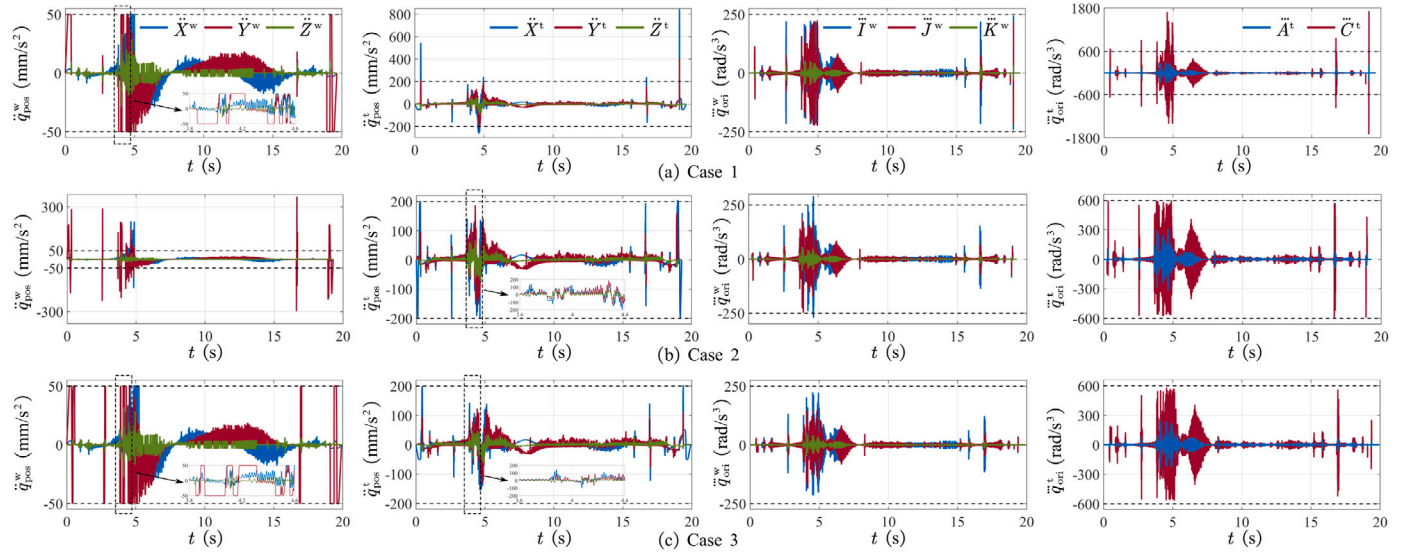


Fig. 15. Results of the 5-axis flank milling experiment with constraints in different coordinate systems. (a)–(c) The planned axial acceleration and jerk for Cases 1, 2, and 3, respectively. The first two columns correspond to axial position acceleration in workpiece and tool coordinate systems, respectively. The last two columns correspond to axial orientation jerk in workpiece and tool coordinate systems, respectively. The inequivalence of constraints in different coordinate systems is verified in this experiment.

- **Case 1:** Consider the workpiece coordinate system. The acceleration satisfies $|\ddot{X}^w|, |\ddot{Y}^w|, |\ddot{Z}^w| \leq 50 \text{ mm/s}^2$, and $|\ddot{I}^w|, |\ddot{J}^w|, |\ddot{K}^w| \leq 2 \text{ rad/s}^2$. The jerk satisfies $|\dddot{X}^w|, |\dddot{Y}^w|, |\dddot{Z}^w| \leq 20,000 \text{ mm/s}^3$, and $|\dddot{I}^w|, |\dddot{J}^w|, |\dddot{K}^w| \leq 250 \text{ rad/s}^3$.
- **Case 2:** Consider the tool coordinate system. The acceleration satisfies $|\ddot{X}^t|, |\ddot{Y}^t|, |\ddot{Z}^t| \leq 200 \text{ mm/s}^2$, and $|\ddot{A}^t|, |\ddot{C}^t| \leq 5 \text{ rad/s}^2$. The jerk satisfies $|\dddot{X}^t|, |\dddot{Y}^t|, |\dddot{Z}^t| \leq 30,000 \text{ mm/s}^3$, and $|\dddot{A}^t|, |\dddot{C}^t| \leq 600 \text{ rad/s}^3$.
- **Case 3:** Consider both workpiece and tool coordinate systems. Constraints in Cases 1 and 2 are both applied.

Representative results are provided in Fig. 15. As shown in Fig. 15a, for Case 1, the axial constraints in the workpiece coordinate system are satisfied, while those in the tool coordinate system are violated. The opposite holds for Case 2, as shown in Fig. 15b. Furthermore, all constraints in both coordinate systems are satisfied in Case 3, as shown in Fig. 15c. These results verify the inequivalence of constraints across different coordinate systems. However, most non-optimization-based TOTP methods can only handle constraints in the workpiece coordinate system, as will be validated in Sections 6.3.1 and 6.4.2. In contrast, the proposed TOTP-SPLP, along with other optimization-based methods, can simultaneously handle constraints in different coordinate systems. Therefore, these findings suggest that optimization-based approaches, including TOTP-SPLP, offer a higher upper bound in constraint-handling capability and thus deserve consideration in practical CNC implementations. As will be discussed in Section 6.3, existing optimization-based methods fail to achieve stable online planning, which further highlights the significance of the proposed method.

6.3. Simulation experiments

In this section, the proposed TOTP-SPLP method and HLA framework are compared with the baselines in Sections 6.3.1 and 6.3.2, respectively.

6.3.1. Comparison of TOTP methods

To compare the performance of the proposed TOTP-SPLP method and TOTP baselines, 100 random 3-axis and 100 random 5-axis 5th-order Bézier toolpaths are generated and all TOTP methods are applied

to these toolpaths. The control points are sampled in the box space with $|X| \leq 60 \text{ mm}$, $|Y| \leq 50 \text{ mm}$, $|Z| \leq 20 \text{ mm}$, $|I| \leq 0.4$, $|J| \leq 0.5$, and $K = \sqrt{1 - I^2 - J^2} > 0$. Then, 50 tool points with different random distances are sampled from each toolpath as the input to TOTP methods. Positional constraints are set as follows: the feedrate $\|\dot{q}_{\text{pos}}\|_2 \triangleq \|(\dot{X}, \dot{Y}, \dot{Z})\|_2 \leq 2000 \text{ mm/min}$, the axial acceleration $|\ddot{X}|, |\ddot{Y}|, |\ddot{Z}| \leq 1000 \text{ mm/s}^2$, and the axial jerk $|\dddot{X}|, |\dddot{Y}|, |\dddot{Z}| \leq 20,000 \text{ mm/s}^3$. The orientational constraints are set as follows: the velocity $\|\dot{q}_{\text{ori}}\|_2 \triangleq \|(\dot{I}, \dot{J}, \dot{K})\|_2 \leq 30 \text{ rad/min}$, the axial acceleration $|\ddot{I}|, |\ddot{J}|, |\ddot{K}| \leq 15 \text{ rad/s}^2$, and the axial jerk $|\dddot{I}|, |\dddot{J}|, |\dddot{K}| \leq 300 \text{ rad/s}^3$. The contour errors of position and orientation are set as $1 \mu\text{m}$ and 5 mrad , respectively.

The distributions of the computational time T_c and the terminal time T_t are shown in Fig. 16 for all TOTP methods. It is evident that the proposed TOTP-SPLP achieves one of the shortest T_t , whereas the computational time T_c is significantly shorter than the motion duration T_f . The quantitative results are shown in Table 1. In the 3-axis experiment, TOTP-SPLP improves machining efficiency by 25.6%, 56.0%, 31.1%, and 15.6% compared to TOTP-SK, TOTP-FIR, TOTP-LP, and TOTP-SLP, respectively, on average. Although TOTP-SPLP requires 1.7% longer terminal time T_t than the offline method TOTP-SQP, the computational time is reduced by 90.6%. Note that TOTP-SQP can only be executed offline, as its computational time is 73.2% longer than the terminal time. In contrast, TOTP-SPLP can be executed online with its computational time for only 83.9% of the terminal time. Similar conclusions can be drawn from the 5-axis experiment. Therefore, the proposed TOTP-SPLP achieves one of the best time-optimality with limited computational cost among all TOTP methods.

Although TOTP-SPLP requires more computational time than some existing TOTP methods, it achieves the highest overall practicality among all optimization-based approaches. In the following, the 3-axis experiment is taken as an example, where TOTP-SQP is neglected due to its expensive computational cost. First, for the case with sufficient computational resources, most methods except TOTP-SLP can only generate a trajectory without the ability of further optimization; hence, the computational cost is underutilized, leading to longer terminal time. When TOTP-SLP is iterated until convergence, denoted by TOTP-SLP*, its terminal time remains 9.5% longer than that of TOTP-SPLP, even with more than twice the computational time. In the Pareto sense, TOTP-SLP* is dominated by the proposed TOTP-SPLP, which indicates

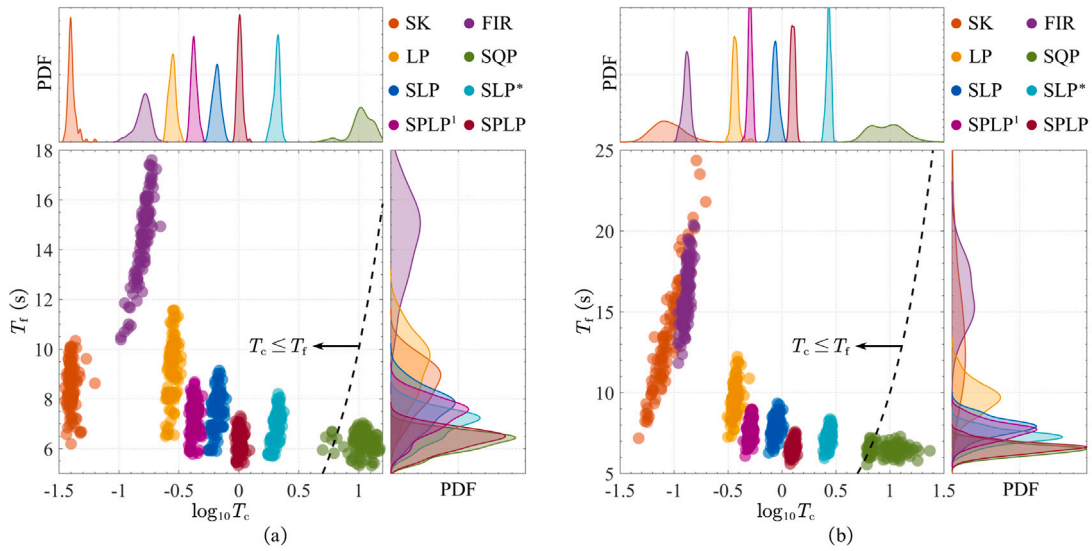


Fig. 16. Probability density functions (PDFs) of the computational time T_c and the terminal time T_f for different TOTP methods. (a) 3-axis. (b) 5-axis.

Table 1

Quantitative results of TOTP methods with 100 random toolpaths. Values are presented as mean $\pm$ standard deviation ($n = 100$).

(a) 3-axis experiment. R_{acc} and D_{acc} of all TOTP methods are all $<10^{-3}$.

Method	T_f (s)	T_c (s)	R_{vel}	R_{jerk}	D_{vel}	D_{jerk}
SK	8.543 ± 0.957	0.041 ± 0.004	$<10^{-3}$	0.712 ± 0.186	$<10^{-3}$	0.010 ± 0.004
FIR	14.461 ± 1.726	0.160 ± 0.022	$<10^{-3}$	0.001 ± 0.003	$<10^{-3}$	$<10^{-3}$
LP	9.222 ± 1.270	0.280 ± 0.019	0.019 ± 0.002	0.009 ± 0.002	0.029 ± 0.013	$<10^{-3}$
SLP	7.536 ± 0.798	0.654 ± 0.050	0.017 ± 0.004	0.060 ± 0.035	0.020 ± 0.015	0.001 ± 0.001
SLP*	6.961 ± 0.601	2.067 ± 0.127	0.018 ± 0.003	0.029 ± 0.038	0.024 ± 0.017	$<10^{-3}$
SQP	6.254 ± 0.410	10.831 ± 2.144	0.015 ± 0.003	0.006 ± 0.002	0.010 ± 0.010	$<10^{-3}$
SPLP ¹ (Ours)	7.310 ± 0.683	0.426 ± 0.025	$<10^{-3}$	0.002 ± 0.003	$<10^{-3}$	$<10^{-3}$
SPLP (Ours)	6.357 ± 0.448	1.021 ± 0.050	$<10^{-3}$	0.004 ± 0.004	$<10^{-3}$	$<10^{-3}$

(b) 5-axis experiment. R_{acc}^{pos} , R_{ori}^{pos} , D_{acc}^{pos} and D_{ori}^{pos} of all TOTP methods are all $<10^{-3}$.

Method	T_f (s)	T_c (s)	R_{vel}^{pos}	R_{jerk}^{pos}	D_{vel}^{pos}	D_{jerk}^{pos}	R_{ori}^{pos}	R_{jerk}^{pos}	D_{ori}^{pos}	D_{jerk}^{pos}
SK	13.667 ± 3.546	0.092 ± 0.028	$<10^{-3}$	3.539 ± 0.840	$<10^{-3}$	0.011 ± 0.006	$<10^{-3}$	3.586 ± 1.604	$<10^{-3}$	0.015 ± 0.009
FIR	16.099 ± 1.857	0.131 ± 0.010	$<10^{-3}$	0.001 ± 0.006	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$
LP	9.676 ± 1.020	0.370 ± 0.030	0.019 ± 0.002	0.009 ± 0.002	0.024 ± 0.012	$<10^{-3}$	0.009 ± 0.015	0.003 ± 0.005	0.002 ± 0.004	$<10^{-3}$
SLP	7.877 ± 0.623	0.875 ± 0.057	0.017 ± 0.003	0.057 ± 0.039	0.018 ± 0.013	$<10^{-3}$	0.009 ± 0.015	0.033 ± 0.034	0.003 ± 0.007	$<10^{-3}$
SLP*	7.250 ± 0.480	2.709 ± 0.123	0.018 ± 0.003	0.019 ± 0.030	0.022 ± 0.015	$<10^{-3}$	0.009 ± 0.015	0.017 ± 0.029	0.003 ± 0.007	$<10^{-3}$
SQP	6.478 ± 0.355	9.824 ± 3.350	0.016 ± 0.003	0.006 ± 0.002	0.011 ± 0.009	$<10^{-3}$	0.008 ± 0.014	0.005 ± 0.006	0.003 ± 0.007	$<10^{-3}$
SPLP ¹ (Ours)	7.634 ± 0.585	0.506 ± 0.022	$<10^{-3}$	0.003 ± 0.003	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$	0.001 ± 0.002	$<10^{-3}$	$<10^{-3}$
SPLP (Ours)	6.598 ± 0.385	1.259 ± 0.056	$<10^{-3}$	0.004 ± 0.004	$<10^{-3}$	$<10^{-3}$	$<10^{-3}$	0.003 ± 0.004	$<10^{-3}$	$<10^{-3}$

that TOTP-SPLP inherently formulates and solves a better optimization problem. Second, if computational resources are limited, the initial solution of TOTP-SPLP, i.e., TOTP-SPLP¹, can also be employed to balance time-optimality and computational cost. Similarly, TOTP-SPLP¹ also dominates TOTP-SLP in the Pareto sense. The computational time of TOTP-SPLP¹ is comparable to that of TOTP-LP, whereas its terminal time is 20.7% shorter. Third, if the computational resources are extremely limited such that even TOTP-SPLP¹ cannot be executed online, then no optimization-based TOTP methods, including TOTP-LP, would be practically applicable. Note that the threshold of computational resources for online TOTP is not strict in practice. Therefore, non-optimization-based methods are more recommended in such extreme cases. In summary, the proposed TOTP-SPLP achieves the best overall practicality among optimization-based TOTP methods in terms of time-optimality and computational cost, except under extremely limited computational resources where no optimization-based methods could be executed online.

As shown in Table 1, the proposed TOTP-SPLP strictly adheres to all constraints, with the exceedance ratio and the exceedance duration ratio for each constraint less than 0.1%. Trajectories following a 5-axis toolpath are shown in Fig. 17 as an example. As shown in Fig.

17a, TOTP-SK fails to meet the axial jerk constraint, because it can only deal with tangential constraints and static normal constraints. As a result, the axial jerk significantly exceeds the constraint at the sharp corners of the toolpath. For most optimization-based TOTP methods, including TOTP-LP, TOTP-SLP, TOTP-SLP*, and TOTP-SQP, the feedrate typically exhibit oscillations at the upper limitation due to the discretization of the problem, as shown in Fig. 17c-e. Quantitatively, feedrates planned by the above three baselines exceed the upper bound by approximately 2% which is dependent on the intervals' length. The exceedance duration ratios w.r.t. feedrates are more than 1%. To avoid feedrate oscillations, the feedrate constraints are densely sampled in the proposed TOTP-SPLP like (26). As shown in Fig. 17f, the feedrate planned by TOTP-SPLP follows the specified constraints with little oscillations with less than 0.1% exceedance ratio as well as exceedance duration ratio.

In this experiment, for all methods except TOTP-FIR, the interpolation points lie precisely on the smoothed toolpaths generated by [3]; hence, no contour error is theoretically introduced in the TOTP process. Without dependence on toolpath smoothing, the one-step method TOTP-FIR introduces the contour error in the interpolation process. Fig. 17 provides the contour error between the final interpolation points and

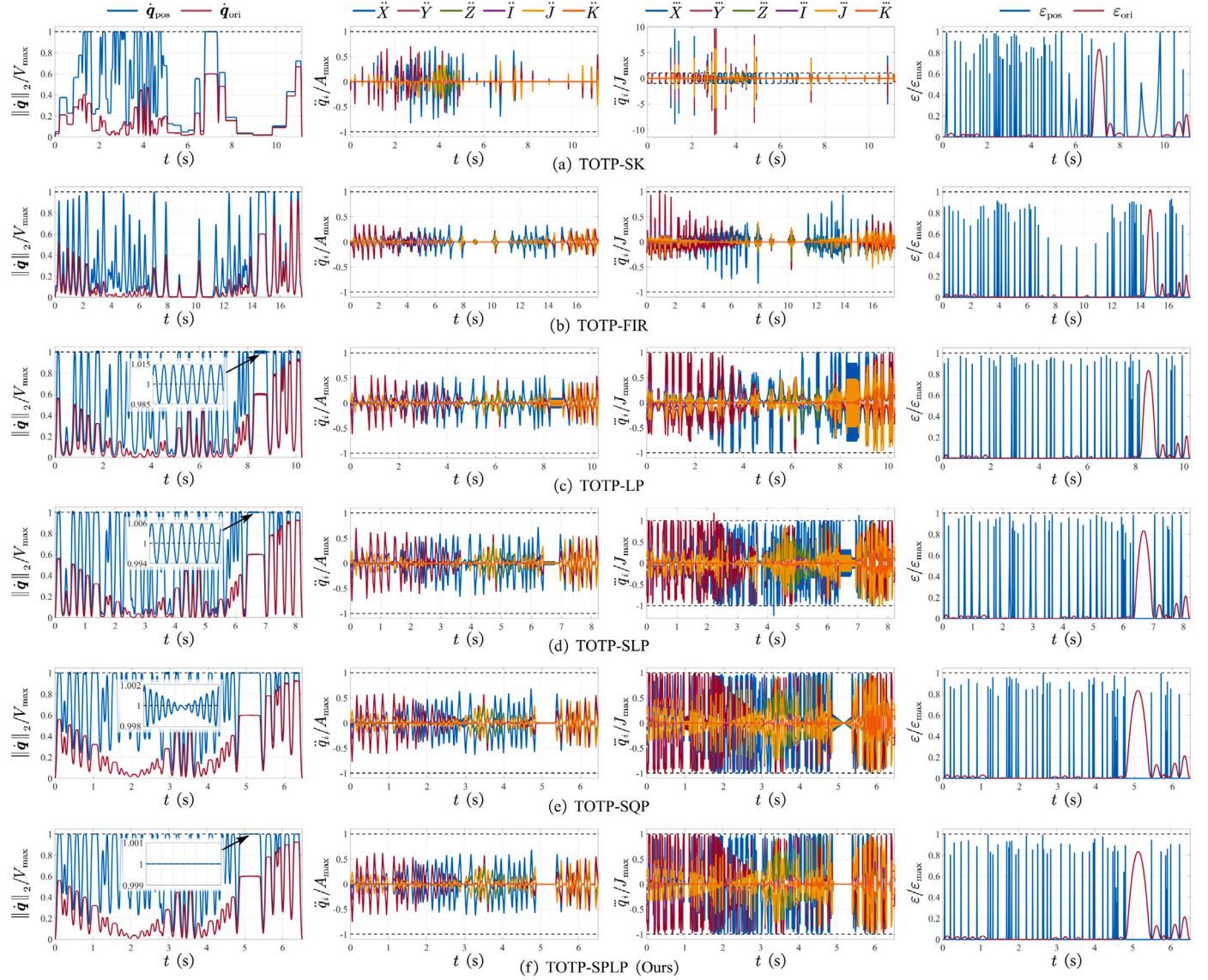


Fig. 17. Trajectories planned by different TOTP methods along the same 5-axis toolpath under the same constraints. (a) TOTP-SK. (b) TOTP-FIR. (c) TOTP-LP. (d) TOTP-SLP. (e) TOTP-SQP. (f) TOTP-SPLP. In this figure, the bounds of variables are normalized to ± 1 and represented by black dashed lines. The four columns from left to right represent the tangential velocity, axial acceleration, axial jerk, and contour error, respectively. The proposed TOTP-SPLP achieves one of the shortest terminal time while strictly adhering to all constraints without feedrate oscillations.

the original G01 points before smoothing. It is evident that all methods follow the contour error constraints.

To further compare the performance of optimization-based TOTP methods under different constraints, the upper bounds of velocity, acceleration, and jerk are individually varied. The results are shown in Fig. 18, where the 5-axis toolpaths and the default constraints are the same as those in Table 1b. Compared to the default constraints, the upper bounds of kinematic constraints on position and orientation of the same order are scaled proportionally. As shown in Fig. 18a–c, under different constraints, our TOTP-SPLP achieves time-optimality that is only slightly inferior to, and closely approaches TOTP-SQP. Specifically, TOTP-SPLP and TOTP-SQP can fully utilize the velocity limits. The computational time of most methods, except TOTP-SQP, keeps nearly unchanged under different constraints, as shown in Fig. 18d–f. Consistent with the results in Table 1, TOTP-SPLP achieves one of the best time-optimality with limited computational cost among all online TOTP methods. Specifically, the dominance relationships in

computational time and terminal time between TOTP-SPLP and TOTP-SLP*, as well as between TOTP-SPLP¹ and TOTP-SLP, remain consistent under different constraint settings.

In summary, the proposed TOTP-SPLP achieves the best time-optimality and constraint satisfaction among all online TOTP methods. Under different constraints and with different computational resource limitations, TOTP-SPLP or the simplified version, i.e., TOTP-SPLP¹, demonstrate the most practicality among all optimization-based TOTP methods.

6.3.2. Comparison of online windowing frameworks

To compare the performance of the proposed HLA framework with the baselines, the Mercedes-Benz mold in Section 6.2.1 is used as the testing sample. In this example, 280,921 G01 blocks are smoothed and discretized into 4,835,927 intervals for TOTP. The kinematic constraints and the contour error are the same as those in Section 6.3.1. Three optimization-based TOTP methods, i.e., TOTP-LP, TOTP-SLP, and the proposed TOTP-SPLP, are integrated with three windowing frameworks, i.e., PW, LAW, and the developed HLA. During an online

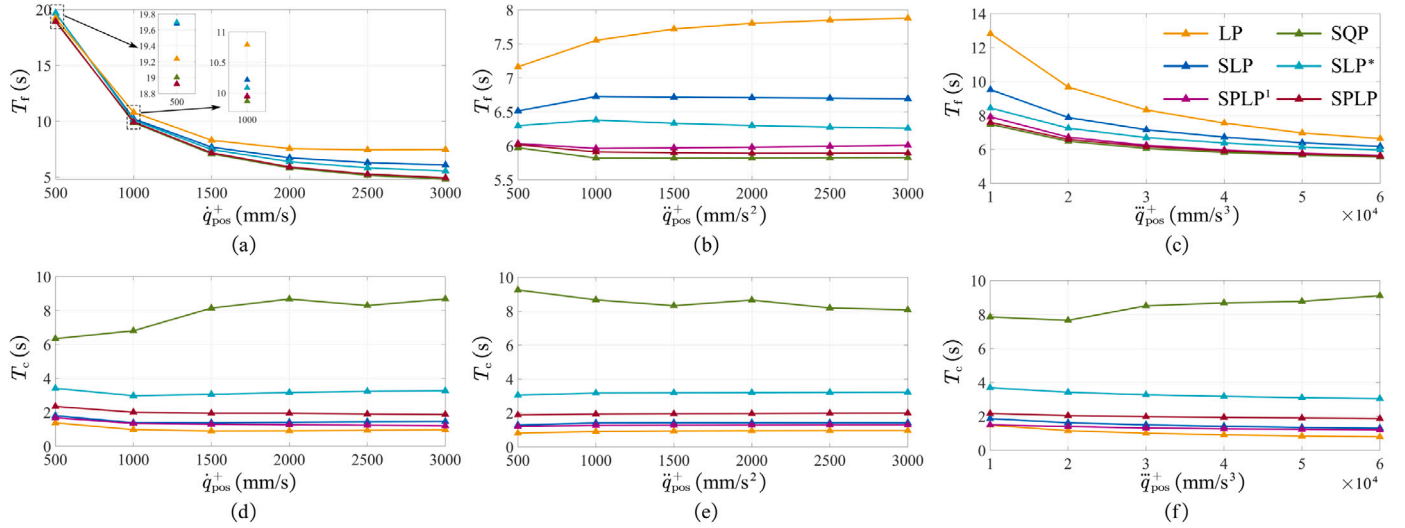


Fig. 18. Results under different constraints. (a)–(c) Terminal time T_f under different velocity, acceleration, and jerk constraints, respectively. (d)–(f) Computational time T_c under different velocity, acceleration, and jerk constraints, respectively. The relative magnitudes of computation time and terminal time among different TOTP methods remain almost unchanged under different constraints.

Table 2

Quantitative results of online windowing TOTP frameworks with the Mercedes-Benz mold.

TOTP	Windowing	T_f (h)	T_c (h)	Infeasibility
LP	PW	3.063	0.885	66
	LAW	3.064	0.599	40
	HLAW (Ours)	3.061	0.581	0
SLP	PW	2.872	1.239	1905
	LAW	2.709	1.232	2870
	HLAW (Ours)	2.637	1.075	0
SPLP (Ours)	PW	2.517	1.512	2436
	LAW	2.499	1.535	1314
	HLAW (Ours)	2.474	1.383	0

process, if an optimization problem is infeasible, then a full stop is introduced and the next window is initiated. In this way, the online process can be guaranteed for the entire part.

The quantitative results are shown in Table 2. Evidently, no infeasibility occurs in the proposed HLAW framework, regardless of the TOTP method used. In contrast, infeasibility occurs thousands of times in the PW and LAW frameworks when applying TOTP-SLP and TOTP-SPLP. Specifically, the infeasibility issue for SLP-PW is shown in Fig. 1a. As for TOTP-LP, only dozens of infeasibilities occur in the PW and LAW frameworks, which is still intolerable for real-world machining. The feasibility guarantee of HLAW can be attributed to the consistency of linearization points among different windows, where simple examples can be found in Appendix B.3. In this experiment, the proposed HLAW framework also achieves the best time-optimality among all online windowing frameworks since numerical infeasibilities and full stops are avoided. Therefore, the developed HLAW framework is numerically robust and stable for online TOTP.

Another advantage of the developed HLAW framework is the computational cost. HLAW saves 32.1%, 13.2%, and 8.5% computational time T_c than PW integrated with TOTP-LP, TOTP-SLP, and TOTP-SPLP, respectively. Compared to LAW, the developed HLAW also significantly reduces computational cost with TOTP-SLP and TOTP-SPLP. The lower computational cost contributes to the online process, where $T_c < T_f$ is required. As shown in Fig. 19, the computational time is always less than the motion duration during the online process, regardless of the TOTP and windowing methods used. In this case, HLAW can conserve computation resources for other tasks, such as supporting

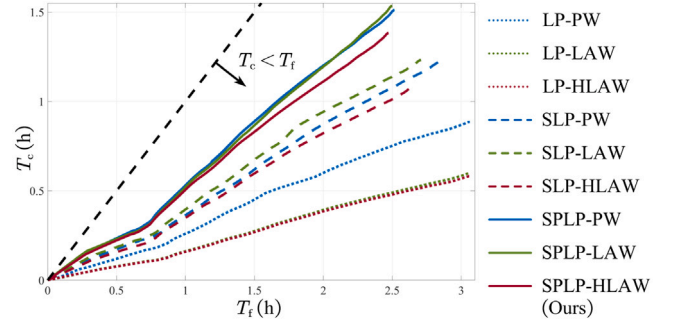


Fig. 19. Computational time T_c versus motion duration T_f for different online windowing frameworks and TOTP methods. All methods achieve shorter computational time than motion duration. For each TOTP method, the proposed HLAW framework consistently achieves the shortest computational time and terminal time among all windowing frameworks.

complex optimization algorithms like TOTP-SPLP or handling unforeseen situations. Moreover, the data corresponding to HLAW in Table 2 forms a Pareto front relative to alternatives. In other words, regardless of the TOTP method used, the HLAW framework remains the most competitive alternative regarding numerical stability, computational cost, and machining efficiency.

6.4. Real-world experiments

Two real-world machining experiments are conducted in this section. In Section 6.4.1, we compare the surface quality under the same machining time, as well as the required machining time to attain a comparable surface quality. In Section 6.4.2, we compare the constraint satisfaction of the actual feedback trajectories under the same constraints, along with the corresponding machining time. Optimization-based baselines are not included in this section, as the differences mainly lie in computational efficiency and manufacturing time, rather than constraint satisfaction.

6.4.1. 3-axis machining

In this section, TOTP-SK, TOTP-FIR, and the proposed TOTP-SPLP integrated with the HLAW framework are applied to machining Mercedes-Benz molds, where the feed directions are the same as

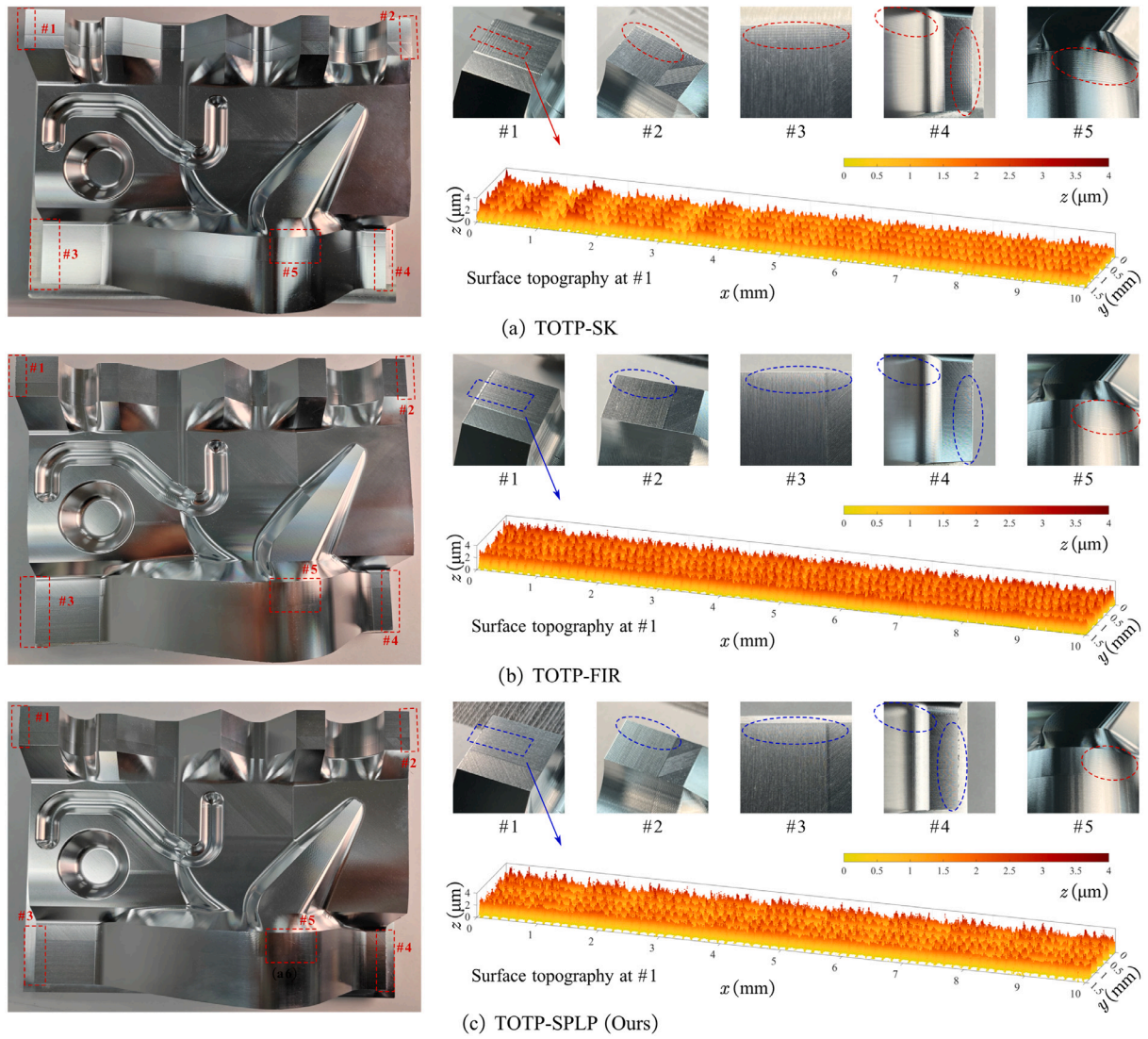


Fig. 20. Results of real-world 3-axis machining experiments. (a) TOTP-SK. (b) TOTP-FIR. (c) TOTP-SPLP. In #1–#5 for each method, vibration streaks are marked by red dashed ellipses, while the places without vibration streaks are marked in blue. Both TOTP-FIR and the proposed TOTP-SPLP eliminate most vibration streaks compared to TOTP-SK, whereas TOTP-SPLP achieves shorter machining time than TOTP-FIR.

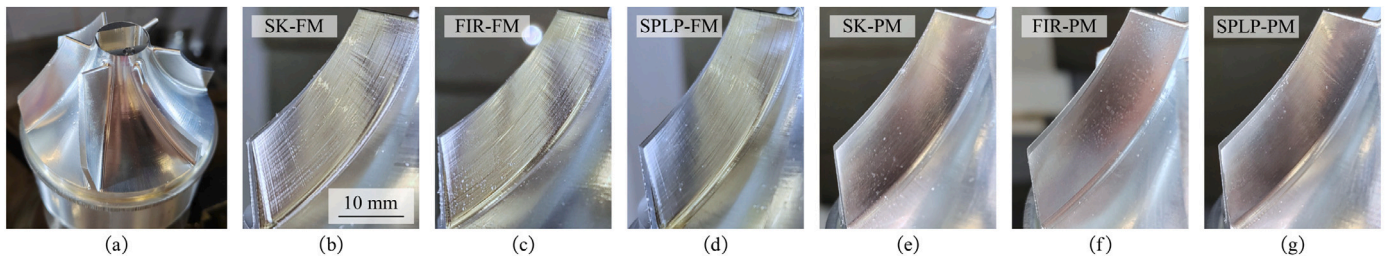


Fig. 21. Results of real-world 5-axis machining experiments. (a) The machined impeller. (b)–(d) The three blades machined using flank milling (FM). (e)–(g) The three blades machined using point milling (PM). For each machining process, different TOTP methods achieve comparable surface quality, while the proposed TOTP-SPLP achieves the shortest machining time and the strictest constraint satisfaction.

those in Fig. 14a. The surface topographies are measured by a Bruker ContourGT-X white-light interferometer. To avoid collisions between the interferometer's lens and the Mercedes-Benz molds, this section selects planar measurement points at #1 of Fig. 20a for white-light interferometry, where acceleration and deceleration occur due to the toolpath's corner. For comparison, the motion durations of TOTP-SPLP and TOTP-SK are adjusted to be close, where the feedrate constraints

are the same. All algorithms are executed online. Specifically, the motion durations of TOTP-SK, TOTP-FIR, and TOTP-SPLP are 4 h 3 m 19 s, 5 h 6 m 36 s, and 4 h 6 m 50 s, respectively.

It can be observed from Fig. 20 that all methods achieve an acceptable surface quality overall. However, as shown in Fig. 20a, some streaks perpendicular to the feed direction, namely vibration streaks, exist in the surface machined by TOTP-SK. The surface topography of

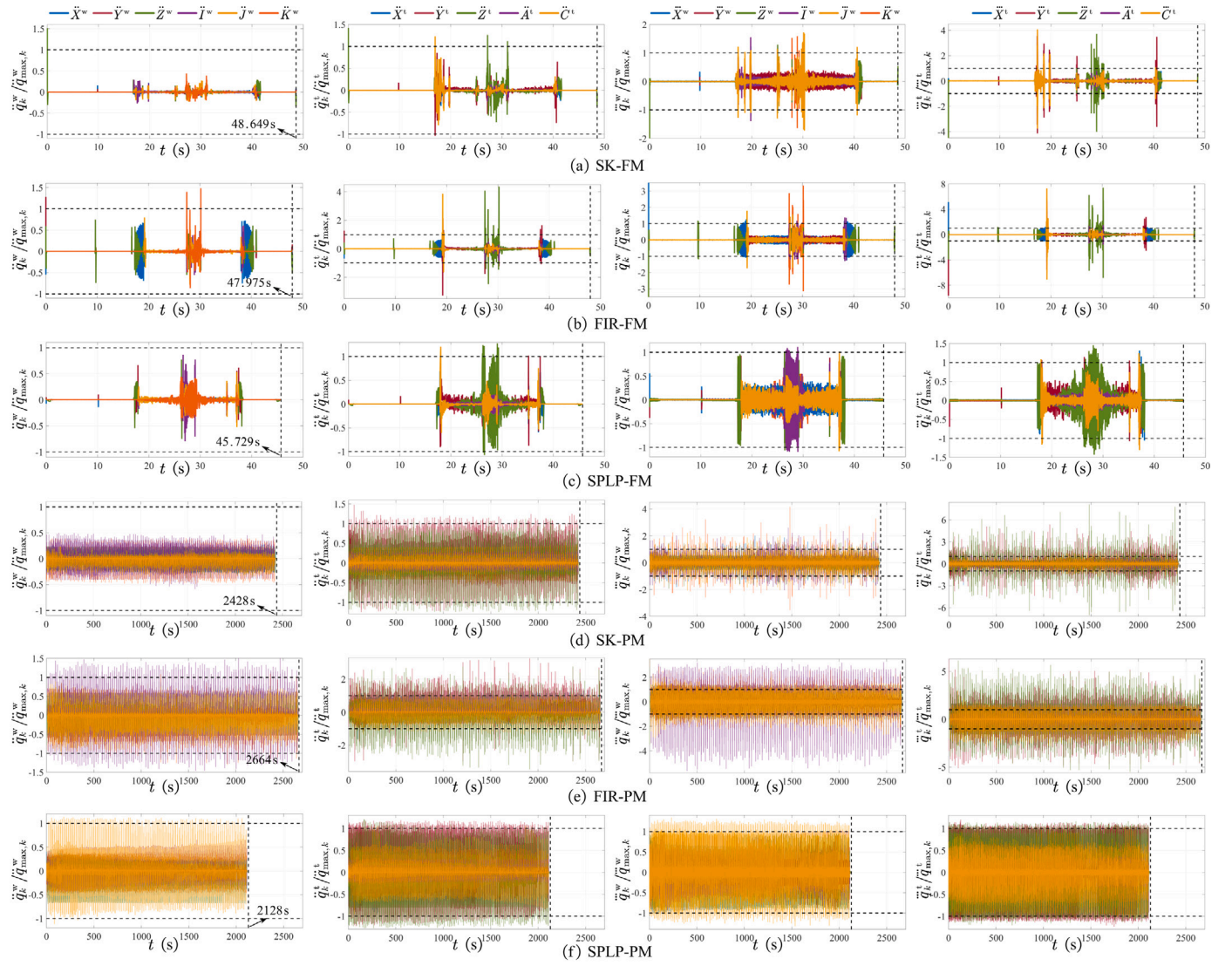


Fig. 22. The profiles of acceleration and jerk in workpiece and tool coordinate systems during the 5-axis machining of the impeller. (a) SK-FM. (b) FIR-FM. (c) SPLP-FM. (d) SK-PM. (e) FIR-PM. (f) SPLP-PM. The first and second columns are the acceleration in the workpiece and tool coordinate systems, respectively. The third and fourth columns are the jerk in the workpiece and tool coordinate systems, respectively. In this figure, the bounds of variables are normalized to ± 1 and represented by black dashed lines. The position information is obtained from the encoder feedback of the machine tool. The trajectories planned by TOTP-SPLP satisfy all constraints in simulation. However, due to tracking errors and measurement noise, the feedback trajectories of TOTP-SPLP exceed the constraints in this experiment, especially for jerk constraints, which are highly sensitive to noise.

#1 demonstrates that variations in geometric error and residual height along the z direction can be observed as x varies. Note that accelerations and declarations exist at the boundary of the surface, where the feed direction changes. Therefore, the streaks can be explained by vibrations in the x direction, which are caused by the exceedance of axial jerk constraints. The vibrations attenuate as the position moves away from the boundary of the surface. Specifically, when $x > 5$ mm, vibration streaks almost disappear, which can be observed from both macroscopic visual inspection and surface morphology of #1. Similarly, vibration streaks exist when the feed direction changes in #2–#5.

In contrast to TOTP-SK, the surface qualities of TOTP-FIR and TOTP-SPLP are of higher quality, as shown in Fig. 20b,c. Specifically, vibration streaks at #1–#4 are significantly reduced and almost unobservable at the macroscopic level, noting that the acceleration and jerk constraints are strictly satisfied by TOTP-FIR and TOTP-SPLP. For #5 of the two methods, vibration streaks are significantly reduced in TOTP-FIR and TOTP-SPLP compared to TOTP-SK. TOTP-SPLP and TOTP-FIR significantly improves surface quality by reducing the vibration streaks compared to TOTP-SK. Furthermore, TOTP-SPLP achieves

the same surface quality as TOTP-FIR, while saving 19.5% machining time. In summary, TOTP-SPLP simultaneously achieves high machining efficiency and superior surface quality.

6.4.2. 5-axis machining

As shown in Fig. 21a, an impeller is machined using a 5-axis machine tool. Three TOTP methods, i.e., TOTP-SK, TOTP-FIR, and the proposed TOTP-SPLP, are applied to plan the trajectories, where three blades are machined using flank milling (FM) and the other three blades are machined using point milling (PM), as shown in Fig. 21b–g. For comparison, the input toolpaths of different blades differ only by a rotation of the C -axis during semi-finishing and finishing stages. Specifically, the finishing stage is conducted using the above three TOTP methods. Due to the limitations of the tilt table's performance, the feedrate is set relatively low in this experiment. As a result, the surface quality of blades machined by the three TOTP methods are similar. Therefore, this section focuses on the actual constraint satisfaction of each method, as shown in Fig. 22.

The constraints are set as follows. In the workpiece coordinate system, the feedrate $\|\dot{q}_{\text{pos}}^w\|_2 \leq 300 \text{ mm/min}$, $\|\dot{q}_{\text{ori}}^w\|_2 \leq 18 \text{ rad/min}$, the axial acceleration $|\ddot{x}^w|, |\ddot{y}^w|, |\ddot{z}^w| \leq 300 \text{ mm/s}^2$, $|\ddot{x}^w|, |\ddot{y}^w|, |\ddot{z}^w| \leq 5 \text{ rad/s}^2$, and the axial jerk $|\dddot{x}^w|, |\dddot{y}^w|, |\dddot{z}^w| \leq 20,000 \text{ mm/s}^3$, $|\dddot{x}^w|, |\dddot{y}^w|, |\dddot{z}^w| \leq 250 \text{ rad/s}^3$. In the tool coordinate system, the axial acceleration $|\ddot{x}^t|, |\ddot{y}^t|, |\ddot{z}^t| \leq 300 \text{ mm/s}^2$, $|\ddot{x}^t|, |\ddot{y}^t|, |\ddot{z}^t| \leq 500 \text{ rad/s}^2$, and the axial jerk $|\dddot{x}^t|, |\dddot{y}^t|, |\dddot{z}^t| \leq 20,000 \text{ mm/s}^3$, $|\dddot{x}^t|, |\dddot{y}^t|, |\dddot{z}^t| \leq 30,000 \text{ rad/s}^3$. The contour errors of position and orientation are set as $1 \mu\text{m}$ and 5 mrad , respectively. Note that TOTP-SK and TOTP-FIR cannot handle constraints in the tool coordinate system; hence, the constraints in the tool coordinate system are only applied to the proposed TOTP-SPLP. The established HLA framework is used to guarantee the feasibility for the three TOTP methods in this experiment.

The feedback positions of each axis are sampled at a frequency of 1 kHz and then filtered by a 4th-order Butterworth low-pass filter with a cutoff frequency of 10 Hz . The resulting acceleration and jerk in the workpiece and tool coordinate systems are shown in Fig. 22. It can be observed from Fig. 22c,f that the feedback acceleration and jerk of the proposed TOTP-SPLP remain within 10% and 50% bounds, respectively, even under the influence of tracking errors and measurement noise. It should be clarified that the planned trajectories, i.e., the command profiles, strictly satisfy the given constraints for TOTP-SPLP. Evidently, TOTP-SPLP can handle the both coordinate systems and the both milling processes. However, both TOTP-SK and TOTP-FIR significantly violate the axial jerk constraints, especially in the tool coordinate system.

Another observation is that the proposed TOTP-SPLP achieves shorter terminal times than TOTP-SK and TOTP-FIR. In the FM process, the TOTP-SPLP saves 6.0% and 4.7% machining times compared to TOTP-SK and TOTP-FIR, respectively. In the PM process, TOTP-SPLP also saves 12.4% and 20.1% machining times compared to TOTP-SK and TOTP-FIR, respectively.

In summary, the proposed TOTP-SPLP can handle complex constraints in both workpiece and tool coordinate systems, achieving higher machining efficiency and stricter axial constraint satisfaction than TOTP-SK and TOTP-FIR.

7. Conclusion

This paper has developed an online time-optimal toolpath planning (TOTP) method, namely TOTP-SPLP, for CNC machining with strict constraint satisfaction and certifiable feasibility guarantee. TOTP-SPLP can strictly satisfy complex constraints, including parameter-varying and asymmetric constraints across different coordinate systems. It also accounts for axial, tangential, and normal kinematic and dynamic constraints in both 3- and 5-axis machining. Owing to the piecewise linear approximation of the time objective and the sequential procedure, TOTP-SPLP reduces terminal time by 15.6%–56.0% compared with baselines of lower computational cost, and still outperforms baselines by 9.5% even when they consume twice the computational time. Comparable time-optimality to TOTP-SPLP is achieved by offline baselines only when consuming tenfold computational time. An analytical interpolation was proposed, contributing to the lossless discretization and the reduction of feedrate oscillations. An online hierarchical look-ahead windowing (HLAW) framework was established to certifiably guarantee the feasibility of all optimization problems among the entire online process. Compared to existing online frameworks, the developed HLA framework eliminates 100% infeasibilities and achieves a Pareto front regarding numerical stability, computational cost, and machining efficiency in the experiment. In hour-level real-world experiments on Mercedes-Benz molds and impellers, the proposed methods significantly reduced vibrations and improved surface quality with high manufacturing efficiency.

In summary, this work addresses several longstanding challenges of optimization-based TOTP approaches regarding online stability, time-optimality, and computational efficiency, thereby unlocking the benefits of strict constraint satisfaction in real-world machining. The proposed approach can serve as a reliable and practical online optimization-based TOTP foundation in CNC machining.

CRedit authorship contribution statement

Yunan Wang: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chuxiong Hu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition. **Yuanshenglong Li:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis, Data curation. **Jichuan Yu:** Writing – review & editing, Visualization, Validation, Methodology. **Jizhou Yan:** Writing – review & editing, Validation, Resources, Investigation, Funding acquisition. **Yixuan Liang:** Writing – review & editing, Validation, Methodology, Formal analysis. **Zhao Jin:** Writing – review & editing, Validation, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Procedure for implementing SPLP-HLAW

A.1. Step-by-step implementation procedure

This section summarizes the implementation procedure of the proposed TOTP-SPLP integrated with the HLA framework. For a short toolpath, TOTP-SPLP without windowing can be directly applied, which is introduced in Appendix A.1.1. For a long toolpath, the HLA framework should be applied to save computational cost and guarantee feasibility of the entire online process, as introduced in Appendix A.1.2. The flowchart of the entire SPLP-HLAW is shown in Fig. 2.

A.1.1. Non-windowing version

Step 1. Generating the smooth toolpath. In CNC machining, the input to the CNC system is usually G01 blocks, i.e., a sequence of tool pose points. The toolpath should be generated based on corner smoothing, piecewise fitting, or other methods. The resulting toolpath should be at least C^2 - and piecewise C^3 -continuous. As the output of Step 1, the toolpath is typically represented by finite parameters, including control points, knot vectors, and so on.

Step 2. Generating a discretization grid. Denote the toolpath by $q = q(u)$, $u \in [u_s, u_f]$. A discretization grid $\{u_k\}_{k=0}^{N_f}$ is generated for $[u_s, u_f]$. The grid is allowed to be non-uniform. The number of intervals should be tuned to balance the computational cost and the discretization error, as discussed in Theorem 1. The positions and derivatives at grid points, i.e., $q(u_k)$, $q'(u_k)$, $q''(u_k)$, and $q'''(u_k)$, are calculated as the output of Step 2.

Step 3. Formulating the nonconvex optimization problem (22). The user-specified constraints can be expressed in the form of problem (1), where some examples are provided in Appendix A.2. Based on the state transformation (3), the optimization problem in the u domain (22) is formulated. Details can be found in Section 3.

Step 4. Choosing the seed $a^{(0)}$. As a typical value in the literature, the seed $a^{(0)}$ is recommended to be the optimal solution of the 2nd-order problem. Compared to problem (22), the 2nd-order problem neglects the 3rd-order constraints and modifies the objective function as $\min \sum_{k=1}^{N_f-1} (u_{k+1} - u_{k-1}) a_k$. The 2nd-order problem can be directly solved as an LP. If further prior knowledge is available, alternative seeds may also be used.

Step 5. Formulating the linear optimization problem (29). For the piecewise linear objective function (27), secant points $\delta_{k,i}$ should be chosen to approximate the original objective function, which is highly flexible. If $\ddot{u} \equiv \text{const}$ holds in some boundary intervals, then the corresponding 3rd-order constraints (22f), (22g), (22j), and (22k) can be transformed into linear constraints w.r.t. a . As a result, problem (29) is formulated, where $a^{(p-1)}$ is temporarily considered as the input. The details can be found in Section 4.

Step 6. Solving the 3rd-order problem through TOTP-SPLP. Starting from the seed $a^{(0)}$, the 3rd-order problem (29) is iteratively solved. In the p th iteration, the 3rd-order constraints (22h) and (22i) are linearized at $a^{(p-1)}$ into (29e) and (29f). The optimization problem in the p th iteration is solved as an LP, and the solution is denoted by $a^{(p)}$. Iterations continue until the stopping criteria (30) are satisfied. In practice, exact convergence is usually unnecessary. Instead, one can set a small N_{iter} to limit the computational cost and obtain a near-optimal feasible solution.

Step 7. Interpolation. Based on the solution a^* , the interpolation points $\{u(kT_s)\}_{k=1}^M$ are calculated by the analytical interpolation method in Section 3.2, where T_s is the sampling period of the CNC system. Then, the interpolation tool pose points $\{q(u(kT_s))\}_{k=1}^M$ are obtained based on Step 1 and sent to the servo controller as the command reference.

A.1.2. Online windowing version

Further details of the HLA framework are provided in Section 5.1. The width of windows and their overlap can be tuned according to Appendix A.3. Toolpath information is streamed online during planning and execution.

Step 1. Seed window. The 2nd-order problem is solved in the seed window to obtain the seed $a^{(0)}$. The procedure is the same as that in Steps 1–4 in Appendix A.1.1.

Step 2. Feasibility window. Denote the current seed window and the next feasibility window by $[u_1, u_2]$ and $[u_3, u_4]$, respectively. Then, the next feasibility window can be computed only if $u_4 < u_1$. In the feasibility window, the initial solution $a^{(1)}$ of the 3rd-order problem is solved based on the seed $a^{(0)}$, where the procedure is the same as that in Steps 5–6 in Appendix A.1.1.

Step 3. Optimality window. Denote the current feasibility window and the next optimality window by $[u_1, u_2]$ and $[u_3, u_4]$, respectively. Then, the next optimality window can be computed only if $u_4 < u_1$. Starting from the initial solution $a^{(1)}$, the optimal solution a^* of the 3rd-order problem is solved in the optimality window, where the procedure is the same as that in Steps 5–6 in Appendix A.1.1.

Step 4. Interpolation. Denote the current optimality window by $[u_1, u_2]$. Then, the interpolation process can be conducted for $u \leq u_1$. The interpolation procedure is the same as that in Step 7 in Appendix A.1.1.

A.2. Examples of constraint formulations

A.2.1. Axial kinematic constraints

The axial velocity, acceleration, and jerk are the most common constraints in manufacturing, which contribute to improving tracking accuracy and avoiding chatter [35,36]. Axial constraints are expressed as follows:

$$\dot{q}^-(u) \leq \dot{q}(u) = \dot{q}'\dot{u} \leq \dot{q}^+(u), \quad (40a)$$

$$\ddot{q}^-(u) \leq \ddot{q}(u) = \ddot{q}'\dot{u}^2 + \ddot{q}'\ddot{u} \leq \ddot{q}^+(u), \quad (40b)$$

$$\ddot{q}^-(u) \leq \ddot{q}(u) = \ddot{q}''\dot{u}^3 + 3\ddot{q}'\ddot{u}\dot{u} + \ddot{q}'''\ddot{u} \leq \ddot{q}^+(u). \quad (40c)$$

Constraints (40a), (40b), and (40c) can be evidently formulated as (1c), (1d), and (1e), respectively.

Remark. The upper bounds $\dot{q}^+(u)$, $\ddot{q}^+(u)$, and $\ddot{q}^+(u)$ are allowed to be asymmetric and dependent on the path parameter u in this paper, which also hold for other constraints in the following. An example is provided in Section 6.2.1.

Remark. The toolpath $q = q(u)$ can be considered in both the workpiece and the tool coordinate systems. For example, the feedrate constraint can be introduced in the workpiece coordinate system for the consistency of the residual height [5], while constraints on axial acceleration and jerk can be introduced in the tool coordinate system to avoid vibrations of the machine tool [12]. It is necessary to simultaneously handle constraints in the both coordinate systems when the toolpath is close to singular configurations of the machine tool, as the nonlinearity and sensitivity of the mapping between the two coordinate systems significantly increase. An example is provided in Section 6.2.2.

A.2.2. Tangential and normal kinematic constraints

The tangential velocity, i.e., feedrate, significantly affects the cutting force, tool wear, and surface quality [5,37]. Denote the position vector as $\bar{q} = (X, Y, Z)$. The analysis of the kinematic constraints in the tool orientation is similar.

The tangential velocity v_t , acceleration a_t , and jerk j_t are formulated as follows:

$$v_t = \|\dot{\bar{q}}\|_2 = \|\dot{\bar{q}}'\|_2 \dot{u}, \quad (41a)$$

$$a_t = \dot{v}_t = \frac{\dot{\bar{q}}'^T \ddot{\bar{q}}''}{\|\dot{\bar{q}}'\|_2} \dot{u}^2 + \|\dot{\bar{q}}'\|_2 \ddot{u}, \quad (41b)$$

$$j_t = \dot{a}_t = \left(\frac{\dot{\bar{q}}'^T \ddot{\bar{q}}''}{\|\dot{\bar{q}}'\|_2} \right)' \dot{u}^3 + \frac{3\dot{\bar{q}}'^T \ddot{\bar{q}}''}{\|\dot{\bar{q}}'\|_2} \ddot{u}\dot{u} + \|\dot{\bar{q}}'\|_2 \ddot{u}. \quad (41c)$$

The normal acceleration a_n , and jerk j_n are formulated by:

$$a_n = \frac{\|\dot{\bar{q}} \times \ddot{\bar{q}}\|_2}{\|\dot{\bar{q}}\|_2} = \frac{\|\dot{\bar{q}}' \times \ddot{\bar{q}}''\|_2}{\|\dot{\bar{q}}'\|_2} \dot{u}^2 \quad (42a)$$

$$j_n = \dot{a}_n = \left(\frac{\|\dot{\bar{q}}' \times \ddot{\bar{q}}''\|_2}{\|\dot{\bar{q}}'\|_2} \right)' \dot{u}^3 + \frac{2\|\dot{\bar{q}}' \times \ddot{\bar{q}}''\|_2}{\|\dot{\bar{q}}'\|_2} \ddot{u}\dot{u}. \quad (42b)$$

Evidently, the tangential and normal constraints are expressed as (1c), (1d), and (1e). Note that normal constraints are not approximated by centripetal ones, which are usually adopted in existing keypoint-based methods [13,17].

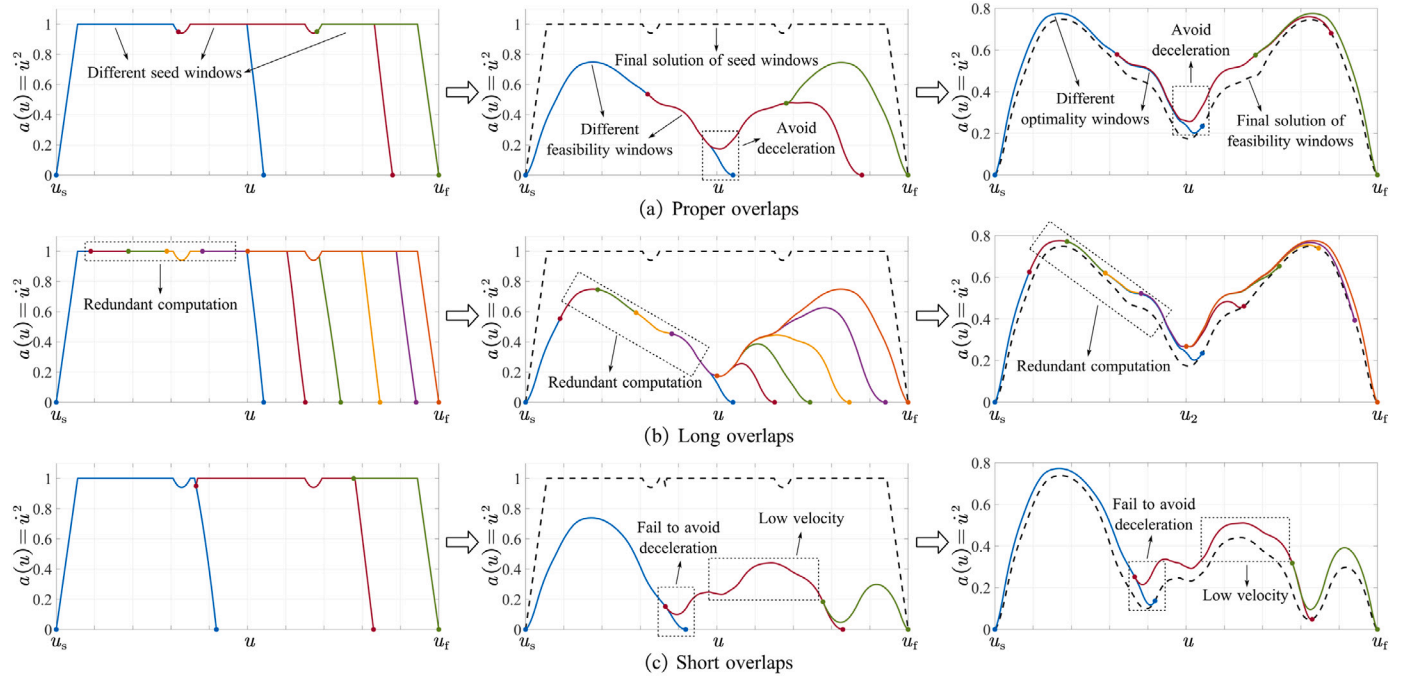


Fig. 23. HLAW process with different parameters. (a) Proper parameter setting. (b) Long overlap. (c) Short overlap. Columns from left to right correspond to the seed, feasibility, and optimality windows, respectively. A proper overlap setting can save computational cost without sacrificing machining efficiency.

A.2.3. Dynamic constraints

Constraints on the torque τ can help to avoid overloading and improve the machined surface quality [38]. The torque is expressed as follows:

$$\tau = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) + \hat{\mathbf{F}}(\mathbf{q}, \dot{\mathbf{q}}). \quad (43)$$

In (43), $\mathbf{M}(\mathbf{q}) \in \mathbb{R}^{n \times n}$ is the mass matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{n \times n}$ is the Coriolis matrix linear in $\dot{\mathbf{q}}$, and $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^n$ is the gravity vector. The external force $\hat{\mathbf{F}}$ can be ignored or approximated by a standard form $\hat{\mathbf{F}}(\mathbf{q}, \dot{\mathbf{q}}) = \hat{\mathbf{F}}_0(\mathbf{q}) + \dot{\mathbf{q}}^T \hat{\mathbf{F}}_1(\mathbf{q})\dot{\mathbf{q}}$, which can formulate Coulomb friction or approximated cutting force. Then, the torque constraints are expressed as follows:

$$\tau^-(u) \leq \tau = \mathbf{n}_{\text{tor}}(u)\dot{u}^2 + \mathbf{m}_{\text{tor}}(u)\ddot{u} + \mathbf{g}_{\text{tor}}(u) \leq \tau^+(u). \quad (44)$$

Evidently, the torque constraints (44) are expressed as (1d).

If the cutting force, the friction force, and other disturbances are not considered in the TOTP process, i.e., $\hat{\mathbf{F}} = \mathbf{0}$, then the torque rate $\dot{\tau}$ can also be constrained. By (44), the torque rate is expressed as follows:

$$\dot{\tau} = \mathbf{n}'_{\text{tor}}\dot{u}^3 + (2\mathbf{n}_{\text{tor}} + \mathbf{m}'_{\text{tor}})\dot{u}\ddot{u} + \mathbf{m}_{\text{tor}}\ddot{u} + \mathbf{g}'_{\text{tor}}\dot{u}. \quad (45)$$

Therefore, the torque rate constraints $\dot{\tau}^-(u) \leq \dot{\tau}(u) \leq \dot{\tau}^+(u)$ are expressed as (1e).

A.2.4. Other static constraints

Some static constraints are considered in existing work on toolpath smoothing and TOTP [13,18]. Given the sample time T_s in interpolation and the upper bound of the chord error $\delta_{\text{ch}}(u)$, the constraint on the chord error [16] is expressed by:

$$v_t(u) \leq \frac{2}{T_s} \sqrt{\frac{2\delta_{\text{ch}}(u) \|\bar{\mathbf{q}}'(u)\|_2^3}{\|\bar{\mathbf{q}}'(u) \times \bar{\mathbf{q}}''(u)\|_2}} - \delta_{\text{ch}}(u)^2. \quad (46)$$

Similarly, the constraint on the control error [17] is given by

$$v_t(u) \leq \rho\omega_n \sqrt{1 - 2\zeta^2 + \sqrt{(1 - 2\zeta^2)^2 - \frac{1 - 2\rho/\delta_{\text{co}}}{(1 - \rho/\delta_{\text{co}})^2}}}, \quad (47)$$

where ω_n and ζ are the undamped natural frequency and the damping ratio of the second-order feed system, respectively. $\rho(u)$ is the curvature radius, and $\delta_{\text{co}}(u)$ is the upper bound of the contour error. Evidently, the constraints on the chord error and the contour error can both be formulated as (1c).

A.3. Parameter tuning for HLAW

As analyzed in Section 5, the widths and overlaps of windows in the HLAW framework do not affect the feasibility guarantee. However, these parameters would affect the computational cost and time-optimality if the tuning is extremely improper. This section provides some examples and practical guidelines for parameter tuning.

Consider a 2-axis toolpath of C^3 -continuity:

$$q_2 = \begin{cases} q_1(q_1^2 - 1)^4 - 0.1(100q_1^2 - 1)^4, & |q_1| \leq 0.1, \\ q_1(q_1^2 - 1)^4, & 0.1 < |q_1| \leq 1. \end{cases} \quad (48)$$

Let u be the arc-length parameter of the toolpath, where $u \in [u_s, u_f]$, $x(u_s) = -1$, $x(u_f) = 1$. The constraints are set as $\|\dot{\mathbf{q}}\|_2 \leq 1$, $\|\ddot{\mathbf{q}}\|_\infty \leq 4$, and $\|\ddot{\mathbf{q}}\|_\infty \leq 8$. Apply different overlaps in the HLAW framework. The results are shown in Fig. 23.

As illustrated in Section 5.1, the terminal state in each window is stationary, which results in a full stop. The overlap between two adjacent windows is introduced to avoid deceleration caused by the full stop at the connection of windows. As shown in Fig. 23a, when the overlap is relatively proper, the long trajectory can be planned with a limited number of windows, and deceleration is effectively avoided.

If the overlap is too long, as shown in Fig. 23b, then the computational cost increases due to the large number of windows and redundant computations across windows. However, the time-optimality is not significantly affected, as reflected by the similar velocity profiles in Fig. 23a,b.

If the overlap is too short, as shown in Fig. 23c, then deceleration at connections of windows is not fully avoided. As a result, the velocity is low near the connections of windows, which can also affect the overall time-optimality. Similarly, if the width of windows itself is too

short, then the system lacks sufficient distance to accelerate to the maximum velocity before decelerating to reach the stationary terminal state within each window, which further affects the time-optimality.

Therefore, one can tune the width and overlap of windows as follows. Given some example toolpaths, one can first test the terminal time and computational time under different parameters. The width and overlap is long enough when the terminal time no longer significantly decreases with the increase of these parameters. Then, one can select proper parameters within acceptable computational cost for online implementation. Fortunately, the online implementation is robust relatively long overlap and window width, unless these parameters are chosen extremely large. For example, when the overlap is shorter than the half of the window width, then the number of windows increases by no more than a factor of two compared to zero overlap. Therefore, the parameter tuning process is generally straightforward.

Appendix B. Some examples

B.1. A toy example of problem (1)

To illustrate the developed TOTP model (1), a toy example is provided as follows. Consider a 2-axis toolpath $q = q(u)$, $u \in [-1, 1]$, where

$$q_1(u) = \begin{cases} u, & u \in [-1, 0], \\ u - u^3, & u \in (0, 1], \end{cases} \quad (49a)$$

$$q_2(u) = \begin{cases} 0, & u \in [-1, 0], \\ u^4, & u \in (0, 1]. \end{cases} \quad (49b)$$

Evidently, $q(u)$ is piecewise C^3 -continuous in $[-1, 2]$. The constraints are specified as follows and formulated in the TOTP model (1).

Tangential velocity: $0 \leq v_t \leq 2 - u^2$ for $u \in (-1, 1)$. By (41a), it is required that

$$\dot{u} \leq 2 - u^2, \quad u \in (-1, 0), \quad (50a)$$

$$\dot{u} \leq \frac{2 - u^2}{\sqrt{(1 - 3u^2)^2 + 16u^6}}, \quad u \in (0, 1). \quad (50b)$$

The constraint at $u = 0$ can be equivalently neglected due to the continuity of $\dot{u}$. Evidently, denote $\mathcal{A} = \{\alpha_1, \alpha_2\}$. Let $v_{\alpha_1}(u) = 2 - u^2$, $u_{\alpha_1 1} = -1$, $u_{\alpha_1 2} = 0$, $v_{\alpha_2}(u) = \frac{2 - u^2}{\sqrt{(1 - 3u^2)^2 + 16u^6}}$, $u_{\alpha_2 1} = 0$, and $u_{\alpha_2 2} = 1$. Then, the tangential velocity constraints (50) can be formulated in the form of (1c).

Axial acceleration: $-2 \leq \dot{q}_1 \leq 1$ for $u \in (0, 0.6)$, and $-1 \leq \dot{q}_2 \leq 2$ for $u \in (0.4, 1)$. By (40b), the constraints can be rewritten as

$$\begin{bmatrix} -6u \\ 6u \end{bmatrix} \ddot{u} + \begin{bmatrix} 1 - 3u^2 \\ -1 + 3u^2 \end{bmatrix} \dot{u}^2 \leq \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad u \in (0, 0.6), \quad (51a)$$

$$\begin{bmatrix} 12u^2 \\ -12u^2 \end{bmatrix} \ddot{u} + \begin{bmatrix} 4u^3 \\ -4u^3 \end{bmatrix} \dot{u}^2 \leq \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad u \in (0.4, 1), \quad (51b)$$

Similarly, denote $\mathcal{B} = \{\beta_1, \beta_2\}$. Axial acceleration constraints (51) can be formulated in the form of (1d).

Axial jerk: $|\ddot{q}_k| \leq 1$ for $k \in \{1, 2\}$, $u \in (-1, 1)$. For the same reason, let $\mathcal{G} = \{\gamma_1, \gamma_2, \gamma_3, \gamma_4\}$, and the axial jerk constraints can be formulated in the form of (1e). For example, the constraint $|\ddot{q}_2| \leq 1$, $u \in (0, 1)$ can be rewritten as $r_{\gamma_4}(u)\dot{u}^3 + s_{\gamma_4}(u)\ddot{u} + t_{\gamma_4}(u)\ddot{u} + h_{\gamma_4}(u)\ddot{u} \leq f_{\gamma_4}(u)$, $u \in (u_{\gamma_4 1}, u_{\gamma_4 2})$, where

$$\begin{aligned} u_{\gamma_4 1} = 0, u_{\gamma_4 2} = 1, r_{\gamma_4}(u) &= \begin{bmatrix} 24u \\ -24u \end{bmatrix}, h_{\gamma_4} = \mathbf{0}, \\ s_{\gamma_4}(u) &= \begin{bmatrix} 36u^2 \\ -36u^2 \end{bmatrix}, t_{\gamma_4} = \begin{bmatrix} 4u^3 \\ -4u^3 \end{bmatrix}, f_{\gamma_4}(u) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}. \end{aligned} \quad (52)$$

Consider the discretization grid in Section 3.1.2. To ensure continuity of constraints in each interval, the points $u = -1, 0, 1, 0.4, 0.6, 1$ should be included in the discretization grid $\{u_k\}_{k=0}^N$. Then, the discretized constraints can be formulated in the form of (7).

B.2. Comparison on constraints and objectives

This section compares the constraints and objectives of different TOTP methods in Section 6.3.1, where an example of singularity where $a(u) = \dot{u}^2 < 0$ is provided. Consider tool pose points $\{(0, 0), (0, 1.5), (0.2, 2)\}$. A 2-axis toolpath is generated based on the corner smoothing algorithm in [3], where the contour error is 0.002. The constraints are set as $\|\dot{q}\|_2 \leq 1$, $\|\ddot{q}\|_\infty \leq 2$, and $\|\ddot{q}\|_\infty \leq \ddot{q}^+ = 4$.

Consider the jerk constraint as follows. Denote

$$J(u, a, b, c) = \|\ddot{q}'''(u)a(u) + 3\ddot{q}''(u)b(u) + \ddot{q}'(u)c(u)\|_\infty. \quad (53)$$

Then, the original jerk constraint can be rewritten as

$$J(u, a, b, c) \leq \frac{\ddot{q}^+}{\sqrt{a(u)}}. \quad (54)$$

In [8], the nonconvex jerk constraint (54) is conservatively approximated by the following pseudo-jerk constraint:

$$J(u, a, b, c) \leq \frac{\ddot{q}^+}{\sqrt{a^{(0)}(u)}}. \quad (55)$$

where $a^{(0)}(u)$ is the solution of the 2nd-order problem. In this paper, the jerk constraint is conservatively approximated by the following linearized jerk constraint:

$$J(u, a, b, c) \leq \ddot{q}^+ \frac{3a(u) - a^{(0)}(u)}{2(a^{(0)}(u))^{3/2}}. \quad (56)$$

In this section, (55) and (56) are called LP- and PLP-constraints, respectively.

Consider the objective function. The original objective function $\int_{u_s}^{u_f} \frac{1}{\sqrt{a(u)}} du$ is called QP-objective [24]. The linearized objective function $\int_{u_s}^{u_f} a(u) du$ is called LP-objective [8,26]. The sequential linear objective function (27) in this paper is called PLP-objective.

Five different combinations of constraints and objectives are compared in this section. For initial solutions, C1 and C2 utilize LP-constraints, while C3, C4, and C5 utilize PLP-constraints. As for objectives, C1, C2, C3, C4, and C5 use LP-, PLP-, LP-, PLP-, and QP-objectives, respectively. Starting from the initial solutions, all methods converge to KKT solutions after several iterations, where the objectives are the same as those of the initial solutions, and the constraints are all PLP-constraints. The results are shown in Fig. 24.

The objectives are compared for the initial solutions as follows. As shown in Fig. 24a, the feedrate profiles $a(u)$ of C3, C4, and C5 intersect pairwise; hence, none of them uniformly dominates the others. The underlying reason is that the three optimization objectives all inherently tend to maximize $a(u)$. Specifically, the profiles of C4 and C5 are similar, implying that our proposed PLP-objective can approximate the original QP-objective well. Since QP- and PLP-objectives provide sufficiently large penalization as $a(u) \rightarrow 0$, the profiles of C4 and C5 tend to prevent $a(u)$ from approaching zero. In contrast, C3 allocates the potential for velocity enhancement in a relatively uniform manner over the entire toolpath. The terminal times of C3, C4, and C5 are 4.805, 4.526, and 4.365, respectively. Compared to the LP-objective, the proposed PLP-objective emphasizes time minimization instead of velocity maximization.

The above phenomenon is more evident when comparing C1 and C2. Specifically, C1 with LP-objective even allows zero feedrate at a discretization point, as shown in the third column of Fig. 24a. Based on the dynamic equations within each interval in Section 3.2, $a(u) = \dot{u}^2 < 0$ occurs in an interval of C1, resulting in an ill-defined duration. The above phenomenon, i.e., singularity, is avoided in C2 whose PLP-objective prevents $a(u)$ from approaching zero.

The constraints are compared for the initial solutions as follows. As shown in Fig. 24a, the profile $a(u)$ of C4 is uniformly larger than that of C2, as the PLP-constraint is less conservative than the LP-constraint. A similar analysis can be applied to C3 and C1.

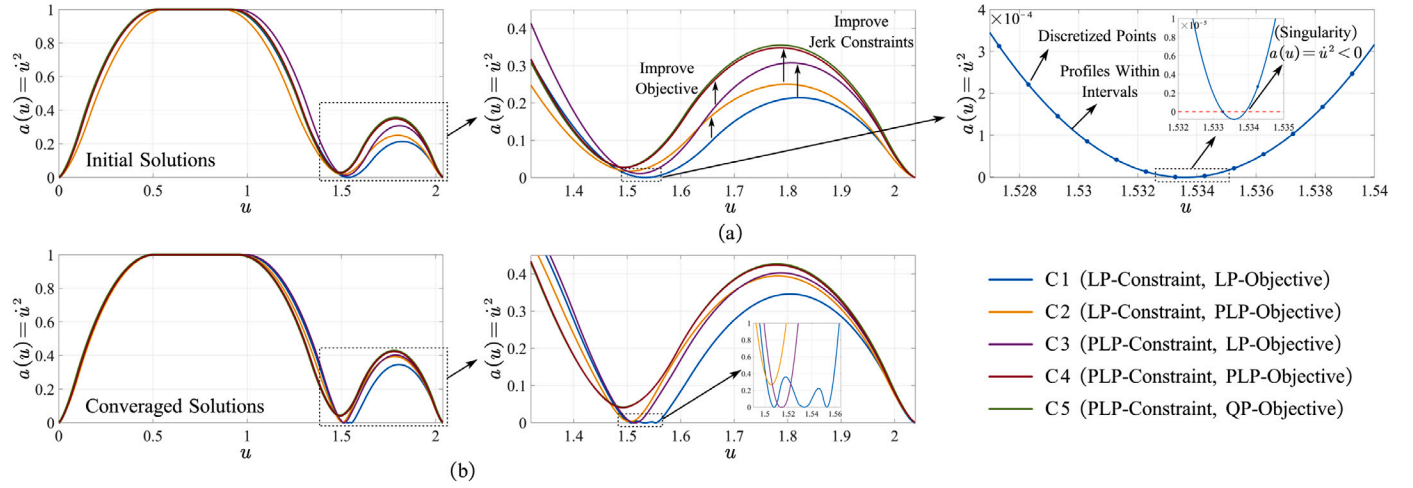


Fig. 24. Comparison on constraints and objectives. (a) Initial solutions of different methods. Specifically, the planned $a(u)$ in C1 crosses zero, which results in a numerical singularity and failure in interpolation. (b) Converged solutions of different methods.

As shown in Fig. 24b, the converged solution of each method improves the feedrate compared to its initial solution, which demonstrates the effectiveness of the sequential optimization framework. However, the singularities in C1 and C3 become even more serious due to the LP-objective. Furthermore, the difference among the converged solutions of C2 and C4 implies that the initial solution also affects the final solution. The initial solutions of C2 and C4 require motion times of 5.058 and 4.526, respectively. The converged solutions of C2 and C4 require motion times of 4.215 and 3.842, respectively. Note that during the iteration of C2 and C4, the PLP-objective and the PLP-constraint are utilized.

In summary, the proposed PLP-objective approximates the original QP-objective better than LP-objective, where singularity is inherently avoided. The proposed PLP-constraint is less conservative than LP-constraint, which improves the machining efficiency.

B.3. Infeasible examples for non-hierarchical windowing

This section provides representative infeasible examples for non-hierarchical windowing frameworks to illustrate the advantages of the developed HLAW framework. All methods can be found in Section 6.1.1. Consider the 2-axis toolpath (48) in Appendix A.3. The constraints are set as $\|\ddot{q}\|_2 \leq 1$, $\|\ddot{q}\|_\infty \leq 2$, and $\|\ddot{q}\|_\infty \leq \ddot{q}^+$.

B.3.1. Case 1: Comparison of LP-PW and SPLP-HLAW

In Case 1, the jerk constraint is set to $\ddot{q}^+ = 8$. As shown in the first column of Fig. 25a, the TOTP-LP itself can feasibly handle the entire toolpath without windowing, which is equivalent to LP-PW with a single window. The seed $a_{\text{tol}}^{(0)}$ is generated by the 2nd-order problem, and the final solution $(a_{\text{tol}}^{(1)}, b_{\text{tol}}^{(1)}, c_{\text{tol}}^{(1)})$ satisfies the pseudo-jerk constraint:

$$\mathcal{J}(u, a_{\text{tol}}^{(1)}, b_{\text{tol}}^{(1)}, c_{\text{tol}}^{(1)}) \leq \frac{\ddot{q}^+}{\sqrt{a_{\text{tol}}^{(0)}(u)}}. \quad (57)$$

However, with the introduction of non-hierarchical windowing like PW, TOTP-LP may instead encounter infeasibility. For example, two windows, i.e., $[u_s, u_2]$ and $[u_2, u_f]$, are introduced in the second column of Fig. 25a, each of which is feasible. Denote the seed and the final solution in the two windows by $a_{\text{win}}^{(0)}$ and $(a_{\text{win}}^{(1)}, b_{\text{win}}^{(1)}, c_{\text{win}}^{(1)})$, respectively. It holds that

$$\mathcal{J}(u, a_{\text{win}}^{(1)}, b_{\text{win}}^{(1)}, c_{\text{win}}^{(1)}) \leq \frac{\ddot{q}^+}{\sqrt{a_{\text{win}}^{(0)}(u)}}. \quad (58)$$

Then, as shown in the third column of Fig. 25a, a connected window $[u_1, u_3]$ is introduced to further optimize the trajectory. For the

connected window, u_1 and u_3 are the local minimum of $a_{\text{win}}^{(1)}$, where $u_1 < u_2 < u_3$. The seed and the final solution are denoted by $a_{\text{con}}^{(0)}$ and $(a_{\text{con}}^{(1)}, b_{\text{con}}^{(1)}, c_{\text{con}}^{(1)})$, respectively. Then, the pseudo-jerk constraint

$$\mathcal{J}(u, a_{\text{con}}^{(1)}, b_{\text{con}}^{(1)}, c_{\text{con}}^{(1)}) \leq \frac{\ddot{q}^+}{\sqrt{a_{\text{con}}^{(0)}(u)}} \quad (59)$$

should be satisfied under the boundary condition of $(a_{\text{win}}^{(1)}(u_1), b_{\text{win}}^{(1)}(u_1), c_{\text{win}}^{(1)}(u_1), a_{\text{win}}^{(1)}(u_3), b_{\text{win}}^{(1)}(u_3), c_{\text{win}}^{(1)}(u_3))$. However, TOTP-LP encounters infeasibility in the connected window, which can be intuitively explained as follows. In Windows #1 and #2, the optimality in TOTP-LP results in the velocities at u_1 and u_3 being maximized under the pseudo-jerk constraint (58). Then in the connected window, $a_{\text{con}}^{(0)} \geq a_{\text{win}}^{(0)}$ implies that the pseudo-jerk bound in (59) is more conservative than that in (58). As a result, the velocity fails to be decreased to a sufficiently safe value from u_1, u_3 to u_2 in the connected window, as the deceleration distance is short.

On the other hand, the relatively high boundary velocities at u_1 and u_3 require a high jerk for deceleration within a relatively short distance. Therefore, the two aspects lead to the infeasibility of TOTP-LP in the connected window.

The above issue is inherently resolved in the proposed HLAW framework, as shown in Fig. 25b. Firstly, the seed window generates the solution $a^{(0)}$ through two adjacent windows $[u_s, u_2]$ and $[u_1, u_f]$, the overlap of which helps to avoid full stops. Then, feasibility window #1 generates the solution $(a_{\text{win1}}^{(1)}, b_{\text{win1}}^{(1)}, c_{\text{win1}}^{(1)})$ conditioned on the solution $a^{(0)}$, where the linearized jerk constraint is formulated as

$$\mathcal{J}(u, a_{\text{win1}}^{(1)}, b_{\text{win1}}^{(1)}, c_{\text{win1}}^{(1)}) \leq \ddot{q}^+ \frac{3a^{(0)} - a_{\text{win1}}^{(1)}}{2a^{(0)3/2}}. \quad (60)$$

The above optimization problem is feasible, i.e., the feasibility region is non-empty since it contains a special solution $a \equiv 0$. Let $a_{\text{win1}}^{(1)}(u) \equiv 0$ for $u \geq u_1$. With the same seed $a^{(0)}$, feasibility window #2 generates a feasible solution $(a_{\text{win2}}^{(1)}, b_{\text{win2}}^{(1)}, c_{\text{win2}}^{(1)})$ under the linearized jerk constraint

$$\mathcal{J}(u, a_{\text{win2}}^{(1)}, b_{\text{win2}}^{(1)}, c_{\text{win2}}^{(1)}) \leq \ddot{q}^+ \frac{3a^{(0)} - a_{\text{win2}}^{(1)}}{2a^{(0)3/2}} \quad (61)$$

and other constraints including the boundary condition of $(a_{\text{win1}}^{(1)}(u_1), b_{\text{win1}}^{(1)}(u_1), c_{\text{win1}}^{(1)}(u_1), a_{\text{win1}}^{(1)}(u_2), b_{\text{win1}}^{(1)}(u_2), c_{\text{win1}}^{(1)}(u_2))$. Similarly, the optimization problem in feasibility window #2 is also feasible since $a_{\text{win1}}^{(0)}|_{u \in [u_1, u_f]}$ is also in the feasible region. Finally, optimization problems in the optimality window are guaranteed feasible for the same reason.

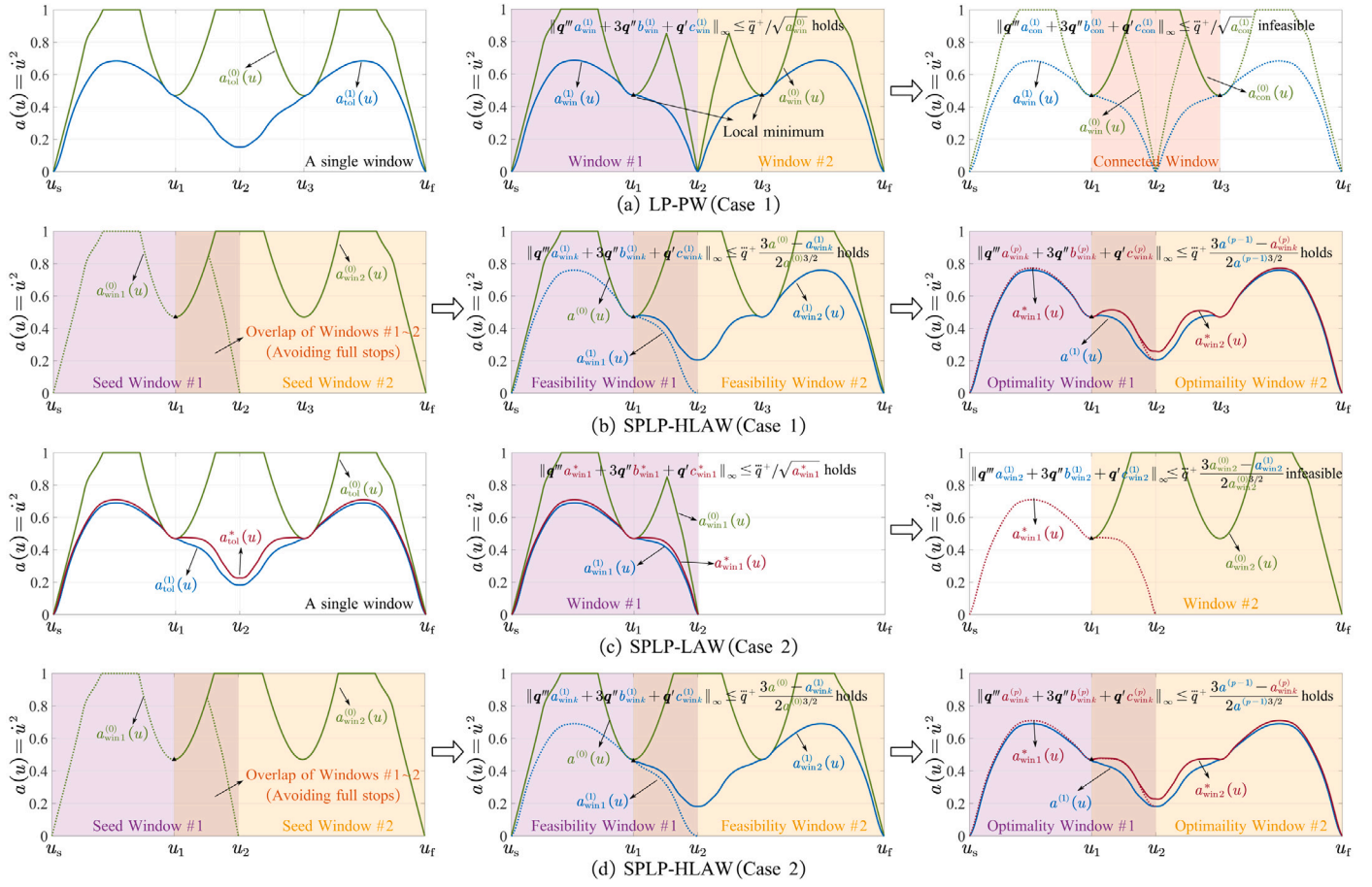


Fig. 25. Infeasible examples for windowing frameworks. (a)–(b) LP-PW and SPLP-HLAW are applied to Case 1, respectively. (c)–(d) SPLP-LAW and SPLP-HLAW are applied to Case 2, respectively. In this figure, the first columns of (a) and (c) have a single window, while all other subfigures have two windows. $(u_s, u_1, u_2, u_3, u_f) \approx (0, 0.731, 1.115, 1.499, 2.229)$ are fixed. Only the developed HLAW guarantees feasibility, while others encounter infeasibility.

This example illustrates the following two advantages of the proposed SPLP-HLAW over LP-PW [8]:

- **TOTP method:** TOTP-SPLP outperforms TOTP-LP in terms of time-optimality, which can be observed from Fig. 25a,b. On the one hand, the linearized jerk constraint like (60) in TOTP-SPLP is less conservative than the pseudo-jerk constraint like (58) in TOTP-LP. On the other hand, the optimality windows in TOTP-SPLP further improve the time-optimality.
- **Windowing framework:** The proposed HLAW framework certifiably guarantees the feasibility of all optimization problems. The primary reason is that all feasible windows share the same seed $a^{(0)}$. In contrast, non-hierarchical windowing frameworks like PW and LAW may encounter infeasibility due to the inconsistency of linearization points among windows.

B.3.2. Case 2: Comparison of SPLP-LAW and SPLP-HLAW

In Case 2, the jerk constraint is set to $\bar{q}^+ = 7$. The discussion is similar to that in Case 1. As shown in the first column of Fig. 25c, the TOTP-SPLP itself can feasibly handle the entire toolpath without windowing. Consider window #1 of SPLP-LAW in the second column of Fig. 25c. Distinct from LP-PW in Case 1, the solution $(a_{win1}^*, b_{win1}^*, c_{win1}^*)$ in window #1 satisfies (59) due to the convergence, i.e.,

$$\mathcal{J}(u, a_{win1}^*, b_{win1}^*, c_{win1}^*) \leq \bar{q}^+ \frac{3a_{win1}^* - a_{win1}^{(0)}}{2a_{win1}^{*3/2}}. \quad (62)$$

In other words, the solution a_{win1}^* is feasible when considering the seed as $a_{win1}^{(0)}$ itself. Similar to LP-PW in Case 1, $a_{win2}^{(0)} \geq a_{win1}^*$ leads to the

infeasibility of TOTP-SPLP in window #2, as shown in the third column of Fig. 25c. In other words, the TOTP-SPLP fails to find a feasible solution a_{win2}^* , which also can be explained by Fig. 9a.

For the same reason as in Case 1, SPLP-HLAW guarantees the feasibility of all optimization problems in this example. The difference between LAW and the developed HLAW is the consistency of linearization points among windows.

Data availability

Data will be made available on request.

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